



NEPHROLOGY · DIALYSIS · INTERNAL MEDICINE

SK Nefro

**Proven,
probable,
open**

*Nephrology and internal medicine
as of October 2026*

Chapter editorials to SK Nefro Báza 1 · 387 articles · English
edition

Ľubomír Polaščin, MD

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Nephrology and internal medicine as of October 2026: chapter editorials to SK Nefro Bába 1

English edition

Articles from April 25, 2026 to October 2, 2026

Ľubomír Polaščín, MD

Bratislava 2026

SK Nefro: Proven, Probable, Open

Nephrology and internal medicine as of October 2026: chapter editorials to SK Nefro Bába 1

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Author: Lubomír Polaščín, MD

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Scope: 16 chapter editorials and an editorial to the appendix, covering 387 specialist articles. Full text of the articles: SK Nefro Bába 1, English edition (2026); abstracts and key points: SK Nefro Bába 1 Compendium, English edition (2026).

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Information on drugs, indications, dosing, and reimbursement reflects the situation as of the publication date of each article and refers primarily to Slovakia; it may differ in other countries and at the time of reading. It should be verified in primary sources before use, in particular in the current SmPC or prescribing information, in the database of the national medicines agency (in Slovakia, the State Institute for Drug Control, ŠÚKL), and in the guidelines of professional societies.

Online version of the articles (Slovak): nefro.polascin.net

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Introduction

SK Nefro Báza 1 brought together 387 specialist articles from the nefro.polasci.net portal published between April 25, 2026 and October 2, 2026. The *Compendium* condensed them into abstracts and key points and opened each chapter with a short synthesis. This third publication goes in the opposite direction: it seeks not the shortest form but the widest view. For each of the 16 chapters and for the appendix, it offers an extended editorial that tries to answer a single question: where does knowledge in the field stand as of October 2026, judging by what the articles of the past months have brought?

The editorials are not summaries of the articles. They link individual papers into thematic lines, compare where the evidence agrees and where it conflicts, and explicitly separate randomized trials and meta-analyses from observational cohorts, conference abstracts, preprints, case reports, and hypotheses. The title of the publication is also its method. Each editorial closes with three sections: what is proven, what is probable, and what remains open; implications for practice in a Slovak outpatient clinic, on the ward, or in the dialysis center; and what to watch in the coming period. Every factual statement refers, by a number in parentheses, to the article it comes from. The numbering is identical to that of the full edition and of the *Compendium*, so the full text and the primary sources are always within reach.

I deliberately kept to the rule that the editorials draw only on the content of the articles. They add no trials or guidelines that are not in the book. They therefore present a picture of knowledge as captured by one portal over half a year, not a systematic review of the specialty. Topics the portal covered less are represented less in the editorials, and vice versa. Readers should keep this selection in mind.

Summaries of product characteristics, reimbursement conditions, and the recommendations of professional societies change continuously. Before use in practice, they must be verified in current primary sources. The evaluative comments in the editorials are the author's opinion, supported by the data they cite, and are meant to prompt the reader's own reading, not to replace it.

This English edition is a translation of the Slovak original. References to Slovak practice, legislation, and institutions have been kept. The online versions of the articles linked at the end of each chapter are in Slovak; the English full text is available in SK Nefro Báza 1, English edition.

Bratislava, October 6, 2026

Lubomír Polaščin, MD

How to use this publication

Organization. The publication follows the structure of the full edition of *SK Nefro Báza 1*: 16 thematic chapters and an appendix with cheat sheets, checklists, and algorithms. Each chapter contains one extended editorial with its own subheadings and, at the end, a list of the chapter's articles with their publication dates.

References to articles. Numbers in parentheses, for example (3.12) or (A.4), refer to articles in the full edition and in the Compendium. In the electronic editions, each number links directly to the article's entry in the list at the end of the chapter, and from there to the full text on the nefro.polasci.net portal.

Structure of an editorial. An untitled introduction states what the chapter is about and which type of evidence predominates in it. The thematic sections link the articles into lines of argument. The three closing sections have the same titles in every chapter: What is proven, what is probable, and what remains open; Implications for practice; What to watch next.

Strength of evidence. The texts consistently distinguish randomized trials and meta-analyses from observational data, conference abstracts, preprints, case reports, and expert opinion. Where an association is reported, it is not evidence of a causal relationship.

Terminology and units. Abbreviations are explained in the list at the end. Values are given in SI units with a decimal point; numerical ranges are written with "to". Spelling follows US English.

What you will not find here. Abstracts and key points of the individual articles are in the Compendium; full texts, tables, figures, and reference lists are in the full edition and on the portal.

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Chronic kidney disease: diagnosis, risk, and kidney protection

Estimating and measuring glomerular filtration rate, albuminuria, risk stratification, and layered kidney protection. The chapter follows how the treatment of chronic kidney disease is moving from a single drug to the concurrent use of several pillars. Number of articles: 29.

The chapter on chronic kidney disease brings together 29 articles from April to October 2026 and centers on three areas: how to stratify risk more precisely, how many patients escape detection, and how to layer kidney protection. The evidence base is uneven. At one end stand an individual participant data meta-analysis of 46 randomized trials of blood pressure lowering (1.1) and outcome RCTs of finerenone, including the prespecified FIND-CKD subgroup (1.11, 1.29); at the other end are conference posters (1.6, 1.27), an uncontrolled observation without a complete design (1.10), cross-sectional associations (1.12, 1.25), and a mouse model (1.21). The main message is twofold. The assessment of CKD is shifting from the eGFR category to individual risk estimation, in which albuminuria is an indispensable variable (1.3, 1.28), and the therapeutic framework has settled on four pillars that have outcome evidence individually but not as a whole (1.19, 1.24). At the same time, observational data show that the largest gap in practice is not a shortage of drugs but an unrecorded diagnosis and an unreferred patient (1.6, 1.27).

From stage to risk and the limits of eGFR

KDIGO 2024 builds on the four-variable KFRE with age, sex, eGFR, and UACR: a 5-year risk of kidney failure of 3% to 5% is the criterion for referral to a nephrologist, a 2-year risk above 10% is a reason for multidisciplinary care, and a 2-year risk above 40% is a signal to plan kidney replacement therapy (1.3, 1.28). At an eGFR of 40 mL/min/1.73 m², a patient with A1 albuminuria and a patient with A3 may have fundamentally different risks, so urgency cannot be judged without UACR; the trend, for example a decline from 75 to 55 mL/min/1.73 m² within a year, is often more important than an isolated value (1.28).

eGFR alone has several weaknesses. Preserved filtration does not exclude damage, because the remaining nephrons maintain total GFR through compensatory hyperfiltration, and CKD can be diagnosed even at an eGFR ≥ 60 mL/min/1.73 m² if UACR ≥ 30 mg/g persists; testing of renal functional reserve has no validated threshold and remains a research tool (1.22). The choice of equation changes the category: among 4,591 Chinese patients with diabetes, CKD-EPI 2021 assigned 37.44% to G1, whereas EKFC shifted a substantial proportion to G2 (33.09% versus 25.90%); the source is a short commentary without complete methods (1.15). In the population-based SHIP Trend-0 cohort (4,211 individuals), higher fat mass was associated with lower eGFR, more markedly in women and older individuals, whereas in younger individuals the direction was reversed, which the author attributes to obesity-related hyperfiltration; the cross-sectional design does not allow causal conclusions (1.25). The common conclusion is consistent with KDIGO: for borderline decisions and at extremes of body

composition, use a combined equation with cystatin C or, if needed, measured GFR (1.3, 1.15, 1.22, 1.25).

The unrecorded diagnosis and the unreferred patient

In the TriNetX database, a CKD code was missing in 68.4% of patients with hypertension without diabetes and in 51.7% of patients with type 2 diabetes; REVEAL-CKD found unrecognized CKD G3 in 95.5% of patients in France, 84.3% in Germany, and 77.0% in Italy (1.27). These data need to be read precisely, however: they reflect administrative nonrecording, not necessarily clinical nonrecognition; the US analysis is a conference abstract, albuminuria was missing, and the study was funded by a pharmaceutical company (1.27). A poster from primary care shows the other side: of 5,270 patients never referred to a nephrologist, 58.6% had intermediate to high cardiovascular risk according to PREVENT, and of 1,383 patients with stages 3 to 5, 4.4% had a 2-year KFRE risk of 3.0% to 9.9% and another 3.8% a risk above 10% (1.6). This is not evidence that consultation reduces mortality, but it shows the size of the group with unaddressed risk (1.6). An educational overview reminds readers that the early stages are painless (1.2).

Prognostic markers beyond eGFR and UACR remain investigational. In a Japanese cohort, a 30% reduction in UPCR or UACR was associated with a lower risk of kidney replacement therapy (HR 0.52 and 0.58), but at an eGFR below 15 mL/min/1.73 m² the association for UPCR was weaker, so UACR is preferable in that setting (1.14). In the All of Us cohort, time in the UACR target range below 30 mg/g predicted MACE better than mean UACR, but this does not change the guidelines (1.7). Low fasting urine osmolality had an aHR of 2.94 for the primary composite in SMART2D, but significance was not replicated in SURDIAGENE, and values such as 350 or 400 mOsm/kg H₂O are not validated thresholds (1.20).

Blood pressure and the four pillars of kidney protection

The strongest evidence in the chapter is unequivocal. A meta-analysis of 46 RCTs with 285,124 participants showed that each 5 mm Hg reduction in systolic blood pressure lowered the risk of major cardiovascular events, with an HR of 0.91 in CKD (95% CI, 0.87 to 0.94) and 0.90 without CKD, with no heterogeneity by stage (including stages 4 to 5), by proteinuria, or by drug class, and the benefit persisted even at a baseline blood pressure below 120/70 mm Hg (1.1). The only signal requiring verification is the combination of CKD and diabetes, with an HR of 0.96 (95% CI, 0.90 to 1.02) and $p = 0.044$ for interaction (1.1).

With the four-pillar regimen, two levels need to be kept apart. The individual classes have outcome RCTs: CREDENCE reduced the relative risk of the primary endpoint by 30%, DAPA-CKD by 39%, EMPA-KIDNEY by 28%, FIDELIO-DKD by 18%, and FLOW by 24%, but the populations differ and the percentages cannot be added together (1.19). An outcome trial of the full combination is lacking, and the model-based estimate of a 5.5-year extension of the period free of kidney events (95% CI, 4.0 to 6.7) assumes additivity and lifelong treatment, so the word “synergy” is overstated (1.19). An expert synthesis checked against KDIGO restates the target populations: an SGLT2 inhibitor at an eGFR ≥ 20 mL/min/1.73 m² and UACR ≥ 200 mg/g, finerenone in type 2 diabetes with an eGFR > 25 mL/min/1.73 m² and normal potassium, and semaglutide when the glycemic target has not been reached (1.24). The data from the Monogram Health program on a 35% reduction in hospitalizations come from the organization itself (1.5).

Finerenone: between RCTs and signals

In a pooled analysis of 14,574 participants, finerenone reduced the risk of kidney failure or CKD progression by 24% regardless of diabetes, and in FIND-CKD in nondiabetic CKD it reduced the composite endpoint by 23% (1.11). In FIND-CKD, the difference in annual eGFR slope was 0.7 mL/min/1.73 m² and the HR was 0.77, with hyperkalemia in 17.0% versus 13.3% (1.19). The prespecified subgroup of 459 patients with hypertensive nephropathy had a slope difference of 0.65 mL/min/1.73 m² per year (95% CI, 0.02 to 1.29) and an HR for the composite of 0.61 (95% CI, 0.38 to 0.99); both intervals approach the null effect, the etiology was determined without histology, only 16.9% of participants were taking an SGLT2 inhibitor, and the indication is not approved in the EU (1.29).

The phase 2 CONFIDENCE trial showed that simultaneous initiation of finerenone and empagliflozin reduced UACR at 180 days by approximately 52%, 29% more than finerenone alone and 32% more than empagliflozin alone, with hyperkalemia in 9.3% versus 11.4% and 3.8% (1.18). An exploratory mediation analysis found that the early change in systolic blood pressure explained less than 10% of the UACR reduction, but blood pressure was measured at a single time point and the primary outcome was a surrogate endpoint, not a clinical event (1.18, 1.24). At the opposite end of the spectrum is a report on 180 Iraqi patients in whom eGFR rose from approximately 32 to 47 mL/min after 1 year of finerenone; without a complete design and with an eGFR increase atypical for this class, it is only a signal (1.10). A practical algorithm for nonsteroidal mineralocorticoid receptor antagonists (nsMRAs) emphasizes a predefined plan for potassium monitoring (1.8).

Gut, inflammation, brain, and functional reserve

The gut-kidney axis remains at the level of associations. TMAO is associated with kidney fibrosis and cardiovascular events, but no intervention targeting it has shown benefit on hard outcomes (1.13). In a cross-sectional cohort of 2,389 patients, higher indoxyl sulfate was associated with cognitive impairment (adjusted OR 1.11), but the design does not allow the claim that lowering it will improve cognition (1.12). The 41% regression of fibrosis after inhibition of TMAO production with fluoromethylcholine is a result from a mouse model, in which other toxins also decreased (1.21). The slower eGFR decline with lubiprostone in a phase 2 trial with 150 patients is a signal that does not justify off-label use (1.4).

Canakinumab in CANTOS reduced MACE (HR 0.85) at the cost of fatal infections, ziltivekimab in ZEUS did not reduce MACE (HR 0.99), and bardoxolone in BEACON increased the risk of heart failure, with an HR of 1.83 (1.26). An elevated hsCRP should therefore prompt a search for the cause and is not an indication for treatment (1.26). A retrospective cohort linked protein intake below 1.0 g/kg/day with a lower risk of the composite outcome (HR 0.77; 95% CI, 0.62 to 0.97) without deterioration in nutritional parameters, but it does not prove causality (1.9). The pooled prevalence of cognitive impairment is 40% in CKD (95% CI, 33% to 46%) and 53% in patients on hemodialysis, but this is not dementia, and no single toxin is a confirmed therapeutic target (1.23). Frailty reaches approximately 70% in some groups with advanced CKD and is dynamic; from the Taiwanese cohort of 120 patients, only excerpts without numerical results are available (1.17).

What is proven, what is probable, and what remains open

Proven at the level of RCTs and meta-analyses are the benefit of blood pressure lowering in CKD regardless of stage and drug class (1.1), the outcome benefit of the individual pillars of kidney protection (1.19), and the effect of finerenone on CKD progression, including in the nondiabetic population (1.11,

1.19). Probable, but based on surrogate endpoints or subgroups, are the additive reduction in albuminuria with finerenone combined with an SGLT2 inhibitor (1.18) and the benefit of finerenone in hypertensive nephropathy (1.29); the value of proteinuria dynamics and the KFRE for referral is also probable, although the evidence for earlier referral is predominantly nonrandomized (1.14, 1.28).

It remains open whether the four-drug combination delivers more than the sum of its parts (1.19, 1.24), whether part of the benefit of finerenone is mediated by blood pressure lowering (1.18, 1.29), whether EKFC or CKD-EPI 2021 better predicts outcomes in patients with diabetes (1.15), whether urine osmolality, time in the UACR target range, or indoxyl sulfate will improve risk reclassification (1.7, 1.12, 1.20), and whether any intervention targeting the gut-kidney axis or inflammation will change hard outcomes (1.4, 1.13, 1.21, 1.26). In the JOINT-KD study, collaboration between the general practitioner and the nephrologist in CKD G5 did not delay dialysis (HR 0.89) but was associated with a lower risk of hospitalization for infection (HR 0.36; 95% CI, 0.15 to 0.87), which is an observational signal (1.16).

Implications for practice

In patients with hypertension, heart failure, atherosclerotic disease, diabetes, or obesity, or those over 60 years of age, measure both eGFR and UACR, confirm chronicity, and record the diagnosis (1.2, 1.27). Refer according to the KFRE, trend, and albuminuria, not only according to an eGFR below 60 mL/min/1.73 m², and add cystatin C for borderline values or extremes of body composition (1.3, 1.15, 1.22, 1.25, 1.28). Lower blood pressure without delay, even in stages 4 to 5 (1.1).

Layer kidney protection according to the indication: titrate RAS blockade, do not combine an ACE inhibitor with an ARB, do not delay an indicated SGLT2 inhibitor, and add finerenone in type 2 diabetes according to albuminuria, eGFR, and potassium, with a check after approximately four weeks (1.8, 1.18, 1.19, 1.24). In hypertensive nephropathy without diabetes, finerenone remains outside the European indication (1.29). An early decline in eGFR after initiation is not automatically a sign of injury, but a decline of more than approximately 30% requires evaluation (1.19, 1.24). After AKI, the discharge summary should include the baseline, peak, and most recent creatinine and a follow-up plan within three months; a past episode of AKI is not a reason to stop RAS blockade permanently (1.24). Prescribe protein restriction below 1.0 g/kg/day on an individualized basis with a nutritional assessment (1.9, 1.17); in patients with advanced CKD, screen specifically for cognitive impairment and frailty and review neurotoxic drugs (1.17, 1.23). A shared plan with the general practitioner, with limits for potassium, blood pressure, and volume, is an inexpensive measure with observational support (1.16).

What to watch next

In the near term, watch for the peer-reviewed publication of ZEUS (1.26), outcome data on combination therapy following CONFIDENCE and the fate of the model-based estimate for the four-drug combination (1.18, 1.19), the evolution of finerenone indications in nondiabetic CKD (1.29), phase 3 trials of lubiprostone and the first clinical studies of TMAO production inhibitors (1.4, 1.21), prospective validation of indoxyl sulfate, time in the UACR target range, and urine osmolality in populations treated with SGLT2 inhibitors (1.7, 1.12, 1.20), a comparison of the EKFC and CKD-EPI 2021 equations

on hard outcomes (1.15), and above all local data on what proportion of Slovak patients with an eGFR of 30 to 60 mL/min/1.73 m² have a recorded diagnosis and measured albuminuria (1.6, 1.27).

Articles in this chapter

- 1.1 Blood pressure lowering in patients with chronic kidney disease: a meta-analysis (May 8, 2026, online, Slovak)
- 1.2 Chronic kidney disease: a silent problem worth detecting early (May 13, 2026, online, Slovak)
- 1.3 KDIGO 2024: from CKD stages to personalized risk estimation (May 15, 2026, online, Slovak)
- 1.4 A constipation drug may slow the decline in kidney function in chronic kidney disease (May 17, 2026, online, Slovak)
- 1.5 From slowing progression to remission: a quiet revolution in nephrology (May 17, 2026, online, Slovak)
- 1.6 Proactive nephrology consultations may reduce the risk of adverse outcomes in patients with CKD (May 19, 2026, online, Slovak)
- 1.7 UACR “time in target range” may help in the management of chronic kidney disease (May 27, 2026, online, Slovak)
- 1.8 When and how to add a nonsteroidal MRA (nsMRA) in a patient with T2D and CKD with persistent albuminuria (May 30, 2026, online, Slovak)
- 1.9 Moderate protein restriction may improve prognosis in chronic kidney disease (June 5, 2026, online, Slovak)
- 1.10 Finerenone: a kidney-protective effect in CKD stages 3 to 4 regardless of diabetes (June 16, 2026, online, Slovak)
- 1.11 Finerenone: potential as foundational therapy in CKD without diabetes and in glomerular diseases (June 16, 2026, online, Slovak)
- 1.12 Indoxyl sulfate and cognitive impairment in chronic kidney disease: what we know about tryptophan derivatives (July 1, 2026, online, Slovak)
- 1.13 Trimethylamine N-oxide (TMAO): a gut metabolite as a uremic toxin in chronic kidney disease (July 3, 2026, online, Slovak)
- 1.14 Urinary protein vs. albumin for assessing kidney failure risk in CKD: does UPCR have the same predictive value as UACR? (July 3, 2026, online, Slovak)
- 1.15 More accurate eGFR estimation in patients with diabetes: when the equation changes the CKD stage (July 5, 2026, online, Slovak)
- 1.16 Collaboration between the general practitioner and the nephrologist in CKD G5: lessons from the JOINT-KD study (July 5, 2026, online, Slovak)
- 1.17 Frailty in chronic kidney disease: why monitoring eGFR alone is not enough (July 5, 2026, online, Slovak)
- 1.18 Finerenone with empagliflozin in CKD and type 2 diabetes: does blood pressure explain the reduction in albuminuria? (July 13, 2026, online, Slovak)
- 1.19 Combination therapy for chronic kidney disease: four pillars, evidence, and open questions (July 13, 2026, online, Slovak)
- 1.20 Fasting urine osmolality and the risk of kidney disease progression in patients with type 2 diabetes (July 13, 2026, online, Slovak)
- 1.21 Targeted inhibition of microbial TMAO production in a CKD model: regression of fibrosis, not clinical evidence (July 15, 2026, online, Slovak)
- 1.22 Renal functional reserve: why a normal eGFR does not exclude significant kidney damage (July 16, 2026, online, Slovak)
- 1.23 Chronic kidney disease also affects the brain: cognitive impairment, vascular damage, and clinical consequences (August 2, 2026, online, Slovak)
- 1.24 Treatment of chronic kidney disease in 2026: layered kidney protection, precise risk stratification, and follow-up after AKI (August 4, 2026, online, Slovak)
- 1.25 Body composition and estimated glomerular filtration rate: what a population-based analysis shows and where the informative value of eGFR ends (August 28, 2026, online, Slovak)
- 1.26 Inflammation in chronic kidney disease: from risk biomarker to therapeutic target (September 6, 2026, online, Slovak)
- 1.27 Chronic kidney disease often remains unrecognized in patients with hypertension and cardiovascular disease (September 6, 2026, online, Slovak)
- 1.28 The nephrologist is not just the doctor for dialysis: when and why early nephrology evaluation matters (September 6, 2026, online, Slovak)
- 1.29 Finerenone in hypertensive nephropathy without diabetes: what does the FIND-CKD analysis show? (September 13, 2026, online, Slovak)

CHAPTER 2

Cardiovascular-kidney-metabolic syndrome and heart failure

The heart, kidneys, and metabolism as one clinical framework. Guidelines, stages of cardiovascular-kidney-metabolic syndrome, and heart failure from a nephrologist's perspective. Number of articles: 20.

The chapter on cardiovascular-kidney-metabolic syndrome and heart failure covers a period in which a working AHA concept became a formal framework with its own guidelines on both sides of the Atlantic. Between April and October 2026, the first multidisciplinary AHA, ACC, ADA, and ASN guideline for CKM syndrome was published (2.2, 2.3, 2.8), as were the first joint ESC and ERA guidelines for cardiovascular disease and CKD (2.18) and new ESC guidelines for heart failure (2.19). KDIGO reviews were added (2.7, 2.16), along with a consensus report from the KDIGO conference on heart failure and CKD (2.13). The chapter is therefore predominantly about guidelines, consensus statements, and reviews. The primary data comprise one post hoc analysis of pooled RCTs (2.5), three observational studies (2.9, 2.15, 2.17), pharmacokinetic data from the registration program with a small pilot study (2.11), and one mechanistic review (2.20).

The main message is twofold. All major professional societies agree that CKD, obesity, diabetes, and cardiovascular disease should be detected and treated together and that detection rests on simultaneous measurement of eGFR and UACR (2.3, 2.7, 2.16, 2.18, 2.19). The level of evidence behind individual recommendations, however, varies considerably: drugs with hard cardiorenal outcomes rest on randomized trials (2.12), whereas the staging classification, combination therapy, and new blood pressure target ranges rely on consensus, surrogate endpoints, or observational associations (2.8, 2.10, 2.17).

One syndrome, two guidelines, and earlier KDIGO recommendations

The US guideline from June 2026 introduces five CKM stages from 0 to 4, the PREVENT equations for estimating 10-year and 30-year risk, and routine screening for CKD with eGFR and UACR (2.3, 2.8). Stage 3 is defined by a coronary artery calcium score above 100, NT-proBNP of at least 125 pg/mL with echocardiographic evidence of dysfunction, CKD at very high risk according to KDIGO or G4 to G5, and a 10-year PREVENT risk above 20% (2.2, 2.8). In type 2 diabetes at stage 2 or 3, the threshold for initiating a GLP-1 receptor agonist, an SGLT2 inhibitor, or their combination is a PREVENT-CVD risk of at least 7.5% (2.2). Very high kidney risk is considered equivalent to high cardiovascular risk, so a patient with CKD belongs in an advanced stage even without a prior event (2.7).

The European ESC and ERA guidelines from August 2026 approach the problem from the opposite direction: in cardiovascular disease, they call for active detection of CKD and for organizing care according to the STAMP framework (2.18). Chronicity should be confirmed by repeated measurement over at least three months, in agreement with the US framework (2.8, 2.18). The differences lie in the details: the systolic blood pressure target of 120 to 129 mm Hg at an eGFR of at least 30 mL/min/1.73 m², with class I and level A (2.18), is not identical to the KDIGO target below 120 mm Hg with stand-

ardized measurement (2.7, 2.16), and the US thresholds for annual eGFR and UACR testing cannot be interchanged with the European ones (2.18).

The reviews by Levin and coauthors point out that the graded KDIGO guidelines, together with conference reports, already form a comprehensive framework (2.7, 2.16). An important caveat, however, is that conference reports are not graded recommendations, that the lipid guideline dates from 2013, and that no guideline for heart failure in CKD existed at the time of publication (2.16). The framework is therefore not uniform in its methodological weight.

What the CKM stage says and what it does not

The most sober reading of the classification comes from the reference reviews themselves: staging is not biologically absolute, progression need not be linear, the stage does not replace an etiologic diagnosis, and the calcium score, NT-proBNP, and echocardiography are not tools for population-wide screening (2.8). Adiposity is a central and often modifiable mechanism, not the only cause, and in a lean patient with a CKM phenotype, risk may be underestimated (2.2, 2.8).

How far the classification can reach is shown by a cross-sectional analysis of 1,283 young adults from the FF-CHAYA cohort with a mean age of 22.9 years, in which 79.2% met the criteria for at least stage 1, mostly because of excess adiposity; only 2.8% were in stage 3 (2.15). Intima-media thickness was greater by only 0.014 mm in stage 1 and by 0.020 mm in stage 2, a population-level signal below the threshold of individual measurability; the abstract does not report the prevalence of albuminuria or reduced eGFR, so it cannot be claimed that four fifths of young people have impaired kidney function (2.15).

At the opposite end of the spectrum is the mechanistic interpretation of CKM as a systemic mitochondrial disorder with amplifying loops of oxidative stress, inflammation, and fibrosis (2.20). The model is biologically plausible but rests on evidence of unequal strength; no validated mitochondrial biomarker exists, and a mitochondrial panel, hsCRP, and IL-6 do not belong in routine practice as standalone decision tests (2.20). An expert commentary on five mistakes in practice names omitting UACR, ignoring stage 0, a siloed approach, assuming that GLP-1 receptor agonists are unavailable, and giving up after a reimbursement denial; this is opinion, not new data (2.4).

Pharmacotherapy with hard outcomes and combinations without them

SGLT2 inhibitors have the most consistent kidney effect: DAPA-CKD achieved an HR of 0.61 with an NNT of 19, EMPA-KIDNEY 0.72, and CREDENCE 0.70, with an effect also in patients without diabetes (2.12). In FLOW, semaglutide in 3,533 patients reduced the primary composite (HR 0.76) and cardiovascular death (0.71) and slowed the annual eGFR decline by 1.16 mL/min/1.73 m², which led to the expansion of its indication to slowing CKD progression (2.1, 2.10, 2.12). Finerenone in FIDELIO-DKD achieved an HR of 0.82, with discontinuation because of hyperkalemia in 2.3% versus 0.9%, and it complements RAS blockade and SGLT2 inhibitors rather than replacing them (2.12).

The post hoc FIDELITY analysis with 12,990 patients adds a CKM perspective: compared with stage II, stage IV had an aHR of 1.87 (95% CI, 1.56 to 2.24) for the cardiovascular composite and an aHR of 1.96 (95% CI, 1.43 to 2.69) for the kidney composite, and finerenone reduced both outcomes regardless of stage (p for interaction 0.86 and 0.65) (2.5). The trials, however, were not designed to assess CKM, not all variables needed for classification were collected, and the result confirms consistency rather than providing definitive evidence (2.5). Combination therapy stands on weaker ground. CONFIDENCE showed that simultaneous initiation of finerenone and empagliflozin reduced UACR by 52% at 180

days in 818 patients, 29% and 32% more than the individual components (2.10, 2.12). This is a surrogate endpoint, hard events were not assessed, and ADA 2026 lists it as a level B recommendation, not a mandatory four-drug regimen (2.10). Triple therapy, the sequence of initiation, and kidney protection with tirzepatide have not been demonstrated (2.12). In type 1 diabetes, FINE-ONE reduced UACR by 34% versus 12% with placebo, with hyperkalemia in 10.1% versus 3.3%, and the European indication for finerenone remains restricted to type 2 diabetes (2.10). The optimistic commentary on a cardio-nephro-metabolic revolution with aldosterone synthase inhibitors (2.1) is therefore an outlook, not the state of the evidence.

Heart failure as the second axis of the chapter

Heart failure affects approximately 10% to 30% of patients with CKD, and CKD affects 30% to 60% of patients with heart failure (2.13). In the CRIC cohort of 3,791 adults, the rate of hospitalization for heart failure was 5.8 per 100 person-years, and after a single hospitalization the HR for CKD progression was 1.93 and for death 2.20; these are observational associations, not evidence of causality (2.13). The Swedish registry with 51,848 patients and the same number of controls adds that the association of CKD is strongest with HFpEF, with an OR of 2.46 versus 1.52 for HFrEF, and strengthens as eGFR declines; however, differential capture of CKD by diagnostic codes inflates the odds ratios, albuminuria data are missing, and the 2005 to 2021 period predates the widespread use of SGLT2 inhibitors (2.9).

The ESC 2021 rule-out thresholds, BNP below 35 pg/mL and NT-proBNP below 125 pg/mL, do not apply universally in CKD, atrial fibrillation, older age, and obesity; a low value rules out better than a high value confirms, and in G4 to G5 and on dialysis, echocardiography is decisive (2.13, 2.14). In patients on hemodialysis, echocardiography should be performed on a nondialysis day whenever possible (2.13), and in CKD, NT-proBNP should be interpreted mainly by its trend in the same patient (2.9).

The biggest conceptual change is the elimination of the HFmrEF category: HFrEF now includes LVEF below 50%, SGLT2 inhibitors and MRAs are recommended regardless of LVEF with class I and level A, and eGFR and UACR are part of the initial evaluation (2.19). For the nephrologist, the key point is the agreement of three documents: CKD in itself is not a reason to withhold prognostic therapy, a hemodynamic eGFR decline of up to 30% after initiation of GDMT should not lead to its discontinuation, and a decline of more than 30% calls for a search for another cause (2.13, 2.16, 2.19). In CREDENCE, only 0.5% of patients had a decline of more than 30% (2.13). KDIGO proposes managing hyperkalemia with an SGLT2 inhibitor, potassium binders, correction of acidosis, and diuretics rather than reflexive discontinuation (2.13). A Chinese cohort that estimated time with blood pressure in the range of 110 to 130 mm Hg found an OR of 0.74 (0.60 to 0.93) for decline in kidney function and an HR of 0.90 (0.83 to 0.98) for death, but blood pressure was measured only three times a year and interpolated; this is an observational signal, not a new treatment target (2.17). Subcutaneous furosemide in an autoinjector, with a relative bioavailability of 107.3% in 19 healthy volunteers and the pilot AT HOME-HF study that did not meet its primary endpoint (win ratio 1.11; 95% CI, 0.48 to 2.50), is a tool for a planned outpatient protocol, not a substitute for urgent intravenous therapy, and FDA approval does not mean marketing authorization or reimbursement in Slovakia (2.11).

What is proven, what is probable, and what remains open

Proven at the level of RCTs is the reduction in hard kidney and cardiovascular events with SGLT2 inhibitors, semaglutide, and finerenone in defined populations (2.5, 2.12, 2.13), as is the fact that pro-

gnostic therapy for heart failure does not harm the kidneys in a way that would justify withholding it (2.13, 2.19). Probable, but based on consensus and observational data, is that joint eGFR and UACR screening detects CKD earlier and that coordinated care improves outcomes; the review of coordinated CKM care is heterogeneous and allows only associative conclusions (2.13). Equally probable, but not proven, is the value of CKM stages for treatment decisions, because FIDELITY showed a consistent, not a differential, effect across stages (2.5).

It remains open whether the combination of finerenone with an SGLT2 inhibitor will reduce hard events and in what order the drugs should be started (2.10, 2.12). The systolic blood pressure target in CKD is open, as the KDIGO target (below 120 mm Hg), the ESC/ERA target (120 to 129 mm Hg), and the observational range of 110 to 130 mm Hg overlap but are not identical (2.7, 2.17, 2.18). Evidence is lacking in G5 and G5D (2.13), as are a validated biomarker for the mechanistic CKM model (2.20) and data on whether assigning four fifths of young adults to stage 1 leads to better long-term outcomes (2.15).

Implications for practice

For a Slovak outpatient clinic, a realistic goal is to turn the agreement among guidelines into a simple internal protocol: in every patient with obesity, diabetes, hypertension, heart failure, or cardiovascular disease, measure eGFR and UACR simultaneously, confirm albuminuria with at least two abnormal results from three samples over three to six months, and use the KFRE in G3 to G5 (2.6, 2.7, 2.10, 2.16). Start drugs according to the dominant risk phenotype and the limits of the marketing authorization: an SGLT2 inhibitor at an eGFR of at least 20 mL/min/1.73 m², finerenone in type 2 diabetes with albuminuria and an eGFR of at least 25 mL/min/1.73 m², with a potassium check after approximately four weeks, and a GLP-1 receptor agonist in atherosclerotic disease or obesity (2.7, 2.10, 2.12, 2.18). The class of recommendation is not the same as marketing authorization, reimbursement, or prescribing restrictions (2.19), and this is precisely where many recommendations founder in the Slovak setting.

On the hospital ward and in the dialysis center, this means not withdrawing GDMT because of an expected hemodynamic eGFR decline, managing hyperkalemia actively, considering HFpEF in advanced CKD with dyspnea even when the ejection fraction is normal, and scheduling echocardiography in patients on hemodialysis on a nondialysis day (2.9, 2.13, 2.19). A written plan for acute illness and prescheduled checks of potassium, eGFR, blood pressure, and volume status are as much a part of treatment as the prescription itself (2.10, 2.12).

What to watch next

In the near term, watch whether trials with hard outcomes emerge for the combination of finerenone and an SGLT2 inhibitor, how the European and US guidelines converge on blood pressure targets and screening thresholds, and whether the new LVEF cutoff of 50% and the ESC/ERA guidelines lead to changes in prescribing restrictions in Slovakia (2.12, 2.18, 2.19). On the evidence side, it would be desirable to see validation of CKM stages on hard outcomes beyond post hoc analyses (2.5, 2.15), prospective data on patients in G5 and G5D with heart failure (2.13), and any validated biomarker of mitochondrial dysfunction that would move the mechanistic model from hypothesis to tool (2.20).

Articles in this chapter

- 2.1** The cardio-nephro-metabolic revolution is running at full speed (April 25, 2026, online, Slovak)
- 2.2** ACC/AHA issue the first guideline on cardiovascular-kidney-metabolic syndrome (June 11, 2026, online, Slovak)

- 2.3** The new multidisciplinary CKM strategy (AHA/ACC/ADA/ASN, 2026): the first guideline for cardiovascular-kidney-metabolic syndrome (June 16, 2026, online, Slovak)
- 2.4** 5 critical mistakes in managing CKM syndrome and practical steps for nephrology (June 27, 2026, online, Slovak)
- 2.5** Finerenone across stages of cardiovascular-kidney-metabolic (CKM) syndrome in patients with T2D and CKD: what the FIDELITY post hoc analysis shows (July 1, 2026, online, Slovak)
- 2.6** CKM syndrome finally as “one framework”: what the new ACC/AHA/ADA/ASN guidelines mean for nephrology practice (July 1, 2026, online, Slovak)
- 2.7** The kidney at the center of cardiovascular-kidney-metabolic syndrome: how KDIGO connects nephrology, diabetology, and cardiology (July 15, 2026, online, Slovak)
- 2.8** CKM syndrome: stages 0 to 4, screening, and treatment according to the 2026 AHA/ACC/ADA/ASN guideline (August 5, 2026, online, Slovak)
- 2.9** Chronic kidney disease and incident heart failure: the strongest association points to HFpEF (August 5, 2026, online, Slovak)
- 2.10** Chronic kidney disease in diabetes: early screening and layered cardiorenal therapy (August 14, 2026, online, Slovak)
- 2.11** Subcutaneous furosemide in an autoinjector: possibilities and limits of home treatment of edema in heart failure and CKD (August 14, 2026, online, Slovak)
- 2.12** Selecting and combining SGLT2 inhibitors, GLP-1 receptor agonists, and dual agonists in type 2 diabetes with cardiorenal risk (August 20, 2026, online, Slovak)
- 2.13** Heart failure and CKD: conclusions of the KDIGO Controversies Conference on bidirectional burden, GDMT, and coordinated care (September 1, 2026, online, Slovak)
- 2.14** Five steps toward earlier detection of heart failure in primary care: a practical diagnostic framework (September 3, 2026, online, Slovak)
- 2.15** Cardiovascular-kidney-metabolic risk begins in youth. What does the finding in 79% of young adults really mean? (September 6, 2026, online, Slovak)
- 2.16** The kidney in the middle: how KDIGO guidelines connect cardiovascular-kidney-metabolic syndrome (September 17, 2026, online, Slovak)
- 2.17** Time in target range for blood pressure in heart failure: association with eGFR decline and mortality (September 24, 2026, online, Slovak)
- 2.18** The 2026 ESC/ERA guidelines: a shared plan of care for the heart and kidneys (September 24, 2026, online, Slovak)
- 2.19** ESC 2026 and heart failure: what changes from the nephrology perspective (September 24, 2026, online, Slovak)
- 2.20** CKM syndrome as a mitochondrial systems disorder: inflammation, energetics, and interorgan connections (September 27, 2026, online, Slovak)

CHAPTER 3

Diabetes, obesity, and incretin-based therapy

Type 1 and type 2 diabetes, obesity, and the new generation of incretin-based drugs. The emphasis is on kidney outcomes, safety, and the limits of evidence for rapidly expanding indications. Number of articles: 56.

The chapter on diabetes, obesity, and incretin-based therapy is the largest chapter of this volume and also the one in which the framework of thinking itself has shifted most visibly over the past six months. Type 2 diabetes is no longer read here as a disorder of glycemia but as a cardiorenal-metabolic disease in which the need to protect organs determines the choice of treatment (3.56, 3.8, 3.1). Obesity appears as a chronic disease with a direct kidney impact (3.49, 3.7), and incretin-based therapy is evolving from a diabetes and obesity tool into a component of layered kidney protection (3.37). Fifty-six articles from April to October 2026 document this shift, ranging from consensus statements and regulatory decisions through randomized trials to conference abstracts, preprints, case reports, and commentaries.

The evidence base of the chapter is uneven. A firm foundation is formed by several randomized trials with hard endpoints, above all FLOW and SELECT (3.37, 3.50, 3.55), SURPASS-CVOT (3.42), and FIND-CKD (3.20). Around them accumulates a large group of observational target trial emulations, post hoc analyses, and conference data that are often read more strongly than their design can bear (3.3, 3.27, 3.51, 3.44). The main message is therefore twofold: early combined cardiorenal protection in type 2 diabetes has a solid randomized foundation, but most of the period's new findings remain at the level of associations, surrogate endpoints, or hypotheses.

A new framework for the treatment of type 2 diabetes

The 2026 ADA and EASD consensus recommends earlier use of SGLT2 inhibitors and GLP-1-based therapy, potentially from diagnosis, and their earlier combination in cardiovascular disease, CKD, and heart failure (3.56). The updated NICE guideline points in a similar direction: extended-release metformin, followed by an SGLT2 inhibitor at an eGFR above 30 mL/min/1.73 m² and, in atherosclerotic disease, subcutaneous semaglutide at doses up to 1 mg weekly, with drugs added sequentially after titration (3.11). Both documents are expert consensus statements supported by randomized data; they are complemented by the rule that HbA1c alone is insufficient to estimate risk and that albuminuria and eGFR should be measured regularly (3.8).

The initial hemodynamic eGFR decline of approximately 10% to 15% after starting an SGLT2 inhibitor is not a reason to discontinue it; a decline of up to 30% during the first three months is acceptable, and a larger or progressive decline should prompt a search for hypovolemia, hypotension, or NSAID use (3.1, 3.56). SGLT2 inhibitors are also not interchangeable: in the EU and in Slovakia, only dapagliflozin and empagliflozin are indicated in heart failure and CKD, canagliflozin has a kidney indication only in the United States, and marketing authorization in the EU is not the same as reimbursement (3.35). The question of whether to start the pillars sequentially or simultaneously in type 2

diabetes with CKD is addressed directly only by the 180-day phase 2 CONFIDENCE trial on albuminuria, in which the combination of finerenone and empagliflozin achieved a UACR ratio of 0.71 versus finerenone and 0.68 versus empagliflozin, at the cost of more frequent hyperkalemia; arguments from VERIFY and TRIPLE-AXEL concern glycemia and are only analogies (3.36).

Studies of glycemic control have a weaker link to the kidneys. A post hoc analysis of ACCORD linked keeping HbA1c and systolic blood pressure on target more than 80% of the time with an HR of 0.71 (95% CI, 0.61 to 0.83) for the kidney outcome, but this is a nonrandomized association (3.44). The CONNECT RCT showed an HbA1c reduction of 1.6% versus 0.7% with CGM in patients not on insulin, but it was funded by the device manufacturer (3.12). The cluster-randomized CREATION trial excluded patients with an eGFR below 60 mL/min/1.73 m² (3.32), the DASH4D feeding trial reduced HbA1c by 0.09 percentage points and systolic blood pressure by 4.6 mm Hg over five weeks (3.54), and CATALYST found nonsuppressed cortisol in 23.8% of patients with difficult-to-control diabetes, which is not the prevalence of confirmed Cushing's syndrome (3.53).

Incretins and the kidney: what is randomized

The strongest kidney evidence in the class remains FLOW: semaglutide 1.0 mg in 3,533 patients with CKD reduced the primary composite by 24% (HR 0.76; 95% CI, 0.66 to 0.88), slowed the eGFR decline by 1.16 mL/min/1.73 m² per year, and reduced all-cause mortality by 20% (3.37, 3.55, 3.27). The prespecified kidney analysis of SELECT showed an HR of 0.78 even without diabetes, with the composite at 1.8% versus 2.2%; because it included macroalbuminuria, it cannot be read as a reduction in the need for dialysis (3.37, 3.50). The FLOW subgroup on an SGLT2 inhibitor had an HR of 1.07 (95% CI, 0.69 to 1.67) without a significant interaction; an incremental effect is therefore not proven, and no harmful interaction was found (3.37).

The kidney signal for tirzepatide is of a different kind. It comes from a post hoc analysis of SURPASS-4 (HR 0.58 for the kidney composite versus glargine) and from an exploratory analysis of SURPASS-CVOT, in which the kidney composite occurred in 6.0% versus 7.6% (HR 0.77), not from a dedicated kidney trial such as FLOW (3.24). The primary outcome of SURPASS-CVOT showed noninferiority to dulaglutide (HR 0.92; 95.3% CI, 0.83 to 1.01), not superiority; the FDA approved the indication for MACE reduction on August 27, 2026, the CHMP did not recommend it in the EU, and the primary publication does not report kidney endpoints (3.42). The retinal substudy is reassuring: retinopathy progression was 21.3% versus 23.3% despite a greater reduction in HbA1c (3.15).

Comparisons within the class remain observational. In the TriNetX database, all-cause mortality was 0.9% with tirzepatide versus 1.6% with semaglutide (RR 0.59) (3.3). A target trial emulation found no difference in the primary kidney composite among four agonists, and the lower secondary composite with semaglutide versus dulaglutide (HR 0.92) and exenatide (HR 0.88) is not evidence of superiority (3.27). A meta-analysis of ten head-to-head comparisons, only three of which were RCTs, showed 4.28 percentage points greater weight loss with tirzepatide and more serious adverse events (5.7% versus 2.9%; RR 1.83), without a breakdown of the composite and without kidney endpoints (3.48). In type 1 diabetes, an emulation showed a 5-year risk of MACE of 4.3% versus 5.0% and of ESKD of 1.6% versus 1.9%, on purely observational grounds (3.9).

In nondiabetic CKD, FIND-CKD attenuated the eGFR decline from approximately 4 to 3.3 mL/min per year and reduced the key secondary composite by approximately 23%, with hyperkalemia in approximately 17% versus 13.3% (3.20). FDA approval of finerenone in type 1 diabetes rests only on a 25% reduction in UACR in the six-month FINE-ONE trial with 242 participants, with hyperkalemia in 10.1% versus 3.3% and no evidence on hard outcomes; in the EU, the indication covers only type 2 diabetes

(3.52). Pentoxifylline has its strongest data from the PREDIAN trial, the remaining studies rely on surrogate endpoints, and VA PTXR_x is expected to provide the answer (3.18). The lower incidence of the kidney endpoint with alogliptin (HR 0.50 to 0.63 versus other DPP-4 inhibitors) rests on administrative codes without eGFR or UACR (3.51).

Obesity as a chronic kidney disease

A narrative review in CJASN summarizes the direct pathway through glomerular hyperfiltration and the indirect pathway through diabetes and hypertension, noting that CKD is linked mainly to visceral and perirenal fat depots rather than to total body weight (3.49). A weight loss of only 5% to 10% can improve blood pressure, glycemia, and steatosis (3.7). BMI alone is therefore insufficient: it does not distinguish fat from muscle mass, in CKD it is distorted by fluid retention as well as sarcopenia, and the obesity paradox in patients on dialysis does not prove a protective effect of fat (3.30, 3.45, 3.49). A Mendelian randomization analysis linking higher BMI to higher risk (OR 1.630) and higher hematocrit to lower risk (OR 0.513) does not make hematocrit a treatment target, particularly when the model includes neither eGFR nor albuminuria (3.46).

The most robust finding of the period concerns the duration of treatment. After semaglutide was stopped in STEP 1, 11.6 of the 17.3 percentage points of weight loss returned; in the randomized SURMOUNT-MAINTAIN trial, continuing tirzepatide 10 to 15 mg maintained a weight change of 21.9%, the 5 mg dose 16.6%, and placebo only 9.9% (3.50). Conference reports from ECO 2026 confirm this and add MRI data showing that approximately 84% of the weight loss consisted of fat mass (3.4, 3.5). Panels at HLTH Europe derive from this a model of chronic treatment with emphasis on protein and muscle mass (3.21), and the US debate on reimbursement shows that unstable access threatens precisely the long-term cardiorenal benefit (3.22).

Weight is not the only measure. A conference abstract from ENDO 2026 recorded a decrease of 560 steps per day after treatment initiation, without a control group (3.40). Intrusive thoughts about food are not a validated measure, and in a patient with CKD, reduced appetite may be a warning sign of wasting (3.47). Lipedema is not a manifestation of volume overload, and diuretics have no place in its treatment (3.39).

New molecules and the gray market

In TRANSCEND-T2D-1, retatrutide lowered HbA_{1c} by 1.94 percentage points (placebo-adjusted, 1.12) and body weight by 15.3% over 40 weeks, and in TRIUMPH-1 it reduced weight by 28.3% after 80 weeks at 12 mg, with a signal of urinary tract infections (7.5% to 8.8% versus 5.3%) (3.13, 3.31). Neither trial had kidney or cardiovascular outcomes (3.31). Similarly, elecoglipron in phase 2 (3.17), HTD1801 in SYMPHONY-2 with diarrhea in 23.8% versus 1.1% (3.43), and liver-directed insulin in OPTI-2 (3.16) deliver metabolic or glycemic results without organ protection.

The gray market is more concerning. Unapproved retatrutide is sold through more than 120 websites, and a preprint based on Reddit posts found at least one symptom in 57.6% of self-reported users (3.26). An EHR preprint showed half the weight loss with gray-market products (7.2% versus 15.5%) and a higher burden of cardiovascular (RR 1.56) and neuropsychiatric symptoms (RR 1.95), with unknown composition and small numbers of patients with CKD (3.41). In Slovakia, § 46 of Act No. 362/2011 Coll. applies, and the availability of approved incretins does not create an entitlement to an unapproved drug (3.38).

Safety signals and hypothesis-generating findings

The GLP-1 class calls for informed vigilance, not reflexive discontinuation. Long-acting agonists are not routinely stopped before surgery; interruption is considered in gastroparesis, vomiting, or a recent dose escalation (3.2, 3.1). An association with NAION has not been shown to be causal (3.14). Wernicke encephalopathy is a pharmacovigilance signal (ROR 10.2 for semaglutide and 11.4 for tirzepatide); the similar value for both drugs supports a nutritional mechanism, and what matters for the nephrologist is that thiamine is removed into the dialysate and that confusion is easily attributed to uremia (3.29). The proportion of prescriptions outside approved indications rose from 4.5% to 17%, so a targeted history of eating disorders makes sense (3.19). The effect on the reward system is strongest for alcohol in small RCTs, but it is not an approved indication (3.25). The common denominator of kidney risk during titration and when switching between agonists is volume depletion with gastrointestinal symptoms (3.55, 3.48, 3.24).

Biomarkers and predictors do not yet have a place at the bedside. A score based on six acylcarnitines improved the prediction of eGFR decline, but external validation was only partial (3.10). Urinary podocalyxin achieved an AUC of 0.96, but the reference standard was UACR itself (3.33). The lower risk of kidney events with higher ketone bodies (aHR 0.46) may reflect SGLT2 inhibitor treatment (3.34). The GLP1R variant rs10305420 explained a small part of the variability in weight loss (3.28), the Apal genotype of the vitamin D receptor requires validation (3.6), and concerns about supplements that increase TMAO rest on mechanistic and observational data (3.23).

What is proven, what is probable, and what remains open

Proven by randomized trials is that semaglutide 1.0 mg reduces hard kidney outcomes in type 2 diabetes with CKD (3.37, 3.55), that finerenone slows eGFR decline in nondiabetic albuminuric CKD (3.20), that tirzepatide is noninferior to dulaglutide for MACE without an increased risk of retinopathy (3.42, 3.15), and that stopping incretin-based obesity treatment leads to weight regain (3.50, 3.4). Probable, but observational or exploratory, are the kidney effect of tirzepatide (3.24), the benefit of GLP-1 receptor agonists in type 1 diabetes (3.9), lower mortality with tirzepatide than with semaglutide (3.3), and synergy of finerenone with empagliflozin on hard outcomes (3.36).

It remains open whether the metabolic results of retatrutide, elecoglipron, and HTD1801 will translate into organ protection (3.13, 3.17, 3.43), whether the incremental effect of GLP-1 receptor agonists on top of an SGLT2 inhibitor is real (3.37), whether NAION and Wernicke encephalopathy are causally related (3.14, 3.29), and whether any of the biomarkers will survive external validation (3.10, 3.33, 3.34). None of the emulated target trials or post hoc analyses in this chapter establishes causality (3.27, 3.51, 3.44).

Implications for practice

For a Slovak outpatient clinic, the framework is clear: in type 2 diabetes, measure eGFR and albuminuria regularly (3.8, 3.56), start an SGLT2 inhibitor with an approved cardiorenal indication early and do not stop it because of the expected eGFR decline (3.35, 3.1), consider a GLP-1 receptor agonist early alongside RAS blockade and an SGLT2 inhibitor in a patient with CKD (3.37), and in persistent albuminuria do not unnecessarily delay the next pillar, checking serum potassium after two to four weeks (3.36). The US labels for tirzepatide and for finerenone in type 1 diabetes do not entitle prescribers in Slovakia to use them; the current summary of product characteristics (SmPC), ŠÚKL (State Institute for Drug Control), and reimbursement are decisive (3.42, 3.52, 3.35).

At every initiation, titration, or switch between agonists, check creatinine, eGFR, electrolytes, blood pressure, and volume status, review diuretics and NSAIDs, give the patient sick day rules, and consider reducing the dose of basal insulin (3.48, 3.55, 3.37). Plan obesity treatment for the long term with monitoring of nutrition, muscle mass, and physical activity (3.50, 3.21, 3.40); in advanced CKD and on dialysis, assess dry weight and nutritional status instead of BMI (3.45, 3.30); and offer patients who have obtained an unapproved peptide a nonjudgmental conversation, approved therapy, and a suspected adverse reaction report to ŠÚKL (3.26, 3.41, 3.38).

What to watch next

In the near term, watch for the EMA decision on the cardiovascular indication of tirzepatide and any kidney publication from SURPASS-CVOT (3.42, 3.24), phase 3 trials with hard outcomes for retatrutide and elecglipton, including TRANSCEND-T2D-2 and TRANSCEND-T2D-3 (3.31, 3.17), a kidney trial of HTD1801 with UACR as the primary endpoint (3.43), the results of VA PTXR_x for pentoxifylline (3.18), any trial of combination therapy with hard endpoints following CONFIDENCE (3.36), long-term data on organ protection with lower maintenance doses (3.50), and the evolution of the pharmacovigilance signals for NAION and Wernicke encephalopathy, which deserve systematic reporting from Slovak centers as well (3.14, 3.29, 3.19).

Articles in this chapter

- 3.1** The 10 most common mistakes in prescribing SGLT2 inhibitors and GLP-1 receptor agonists (May 11, 2026, online, Slovak)
- 3.2** GLP-1 receptor agonists before surgery: stop or continue (May 11, 2026, online, Slovak)
- 3.3** Tirzepatide versus semaglutide: lower mortality and fewer gastrointestinal complications (May 11, 2026, online, Slovak)
- 3.4** Five key advances in obesity treatment from ECO 2026 (May 12, 2026, online, Slovak)
- 3.5** ECO 2026 brought new data on long-term obesity treatment (May 14, 2026, online, Slovak)
- 3.6** Vitamin D receptor genotyping: a new path to type 2 diabetes prevention? (May 17, 2026, online, Slovak)
- 3.7** Obesity in general practice: why it must be treated as a chronic disease (May 18, 2026, online, Slovak)
- 3.8** Type 2 diabetes at the center of cardiometabolic prevention: from lipids to organ health (May 18, 2026, online, Slovak)
- 3.9** Cardiac and kidney benefits of GLP-1 analogues may extend to type 1 diabetes (May 27, 2026, online, Slovak)
- 3.10** Serum acylcarnitines may predict kidney function decline in type 2 diabetes (May 27, 2026, online, Slovak)
- 3.11** Extended-release metformin and an SGLT2 inhibitor as the new first-line treatment of type 2 diabetes (June 5, 2026, online, Slovak)
- 3.12** Continuous glucose monitoring may also help patients with type 2 diabetes not on insulin (June 9, 2026, online, Slovak)
- 3.13** Retatrutide delivers substantial weight loss and metabolic benefits, but the new data must be read with caution (June 9, 2026, online, Slovak)
- 3.14** GLP-1 drugs and ischemic optic neuropathy: a signal to monitor, not a reason to panic (June 10, 2026, online, Slovak)
- 3.15** Tirzepatide did not increase the risk of diabetic retinopathy in high-risk patients (June 10, 2026, online, Slovak)
- 3.16** Liver-directed insulin: a possible path to lower hypoglycemia risk in type 1 diabetes (June 10, 2026, online, Slovak)
- 3.17** Elecglipton, a new oral GLP-1 receptor agonist: phase 2 results with improved HbA1c and weight loss (June 13, 2026, online, Slovak)
- 3.18** Pentoxifylline in diabetic kidney disease: does it still have a place? (AJKD mini review) (June 17, 2026, online, Slovak)
- 3.19** GLP-1 and eating disorders: growing concerns about inappropriate use and how to prevent it (June 20, 2026, online, Slovak)
- 3.20** Beyond hemodynamics: new evidence and perspectives in nondiabetic chronic kidney disease (CKD) (June 20, 2026, online, Slovak)

- 3.21** The GLP-1 era demands a new model of obesity care (and what it means for nephrology) (June 27, 2026, online, Slovak)
- 3.22** Obesity as a “cost” diagnosis in healthcare: what it means for access to GLP-1 therapy and, indirectly, for nephrology (July 2, 2026, online, Slovak)
- 3.23** Choline and L-carnitine supplements in diabetic nephropathy: when TMAO can create a vicious circle (July 3, 2026, online, Slovak)
- 3.24** Tirzepatide and kidney outcomes in the SURPASS program: what they mean for nephrology (July 5, 2026, online, Slovak)
- 3.25** GLP-1 receptor agonists, “food noise,” and compulsive behavior: what it means for nephrology (July 5, 2026, online, Slovak)
- 3.26** Retatrutide outside approval: when an investigational drug outpaces regulation (July 10, 2026, online, Slovak)
- 3.27** Semaglutide and the risk of chronic kidney disease in type 2 diabetes: a real-world comparison of GLP-1 receptor agonists (July 15, 2026, online, Slovak)
- 3.28** Genetic predictors of response to semaglutide and tirzepatide: promise and limits for nephrology practice (July 21, 2026, online, Slovak)
- 3.29** Semaglutide and Wernicke encephalopathy: a rare complication of rapid weight loss (August 5, 2026, online, Slovak)
- 3.30** Adipose tissue is not a passive reservoir: what its biology tells us about obesity and cardiorenal risk (August 5, 2026, online, Slovak)
- 3.31** Retatrutide in type 2 diabetes: marked reduction in HbA1c and body weight, but the cardiorenal benefit is not yet known (August 19, 2026, online, Slovak)
- 3.32** When the physician tries on the patient’s role: can experiential education improve glycemic control in type 2 diabetes? (August 19, 2026, online, Slovak)
- 3.33** Urinary podocalyxin in diabetic nephropathy: remarkable numbers that cannot yet be taken to the bedside (August 24, 2026, online, Slovak)
- 3.34** Serum ketone bodies and the risk of kidney events in type 2 diabetes: what a cohort study shows and what cannot be inferred from it (August 28, 2026, online, Slovak)
- 3.35** Six SGLT2 inhibitors are not six interchangeable drugs: differences in approved cardiac and kidney indications (August 28, 2026, online, Slovak)
- 3.36** Sequentially or all at once? How to start combined cardiorenal therapy in type 2 diabetes and chronic kidney disease (August 28, 2026, online, Slovak)
- 3.37** Are GLP-1 drugs now “kidney drugs”? Kidney benefits in the evidence of recent years (and how to translate them into practice) (September 2, 2026, online, Slovak)
- 3.38** Retatrutide and “pitting the physician against the patient”: barriers in the expanded access process in practice and their relevance for nephrology (September 3, 2026, online, Slovak)
- 3.39** Lipedema: multidisciplinary management and the place of surgery in treatment, a practical clinical review (September 3, 2026, online, Slovak)
- 3.40** “Weight down, steps down”: why physical activity may decline with GLP-1 receptor agonists and how to address it in nephrology and cardiometabolic practice (September 3, 2026, online, Slovak)
- 3.41** The “gray market” for retatrutide: half the weight loss and more cardiovascular symptoms? Clinical interpretation of a real-world signal and implications for the nephrologist (September 3, 2026, online, Slovak)
- 3.42** Mounjaro (tirzepatide): a new FDA indication for cardiovascular risk reduction in type 2 diabetes and what it means for practice (September 3, 2026, online, Slovak)
- 3.43** HTD1801 in type 2 diabetes: results of the SYMPHONY-2 trial and the limits of the evidence (September 5, 2026, online, Slovak)
- 3.44** Time in target range for HbA1c and systolic blood pressure and kidney outcomes: a post hoc analysis of the AC-CORD trial (September 9, 2026, online, Slovak)
- 3.45** Obesity as a multifactorial disease: a new screening tool and its relevance for nephrology practice (September 9, 2026, online, Slovak)
- 3.46** BMI and hematocrit in type 2 diabetes: what they really tell us about the risk of kidney failure (September 11, 2026, online, Slovak)
- 3.47** Intrusive thoughts about food, stigma, and obesity treatment: why tracking body weight is not enough (September 11, 2026, online, Slovak)
- 3.48** Tirzepatide vs. semaglutide for weight reduction: efficacy, safety, and relevance for nephrology practice (September 17, 2026, online, Slovak)
- 3.49** Obesity and cardiometabolic health in kidney disease: from lifestyle to a multicomponent strategy (September 17, 2026, online, Slovak)
- 3.50** Obesity treatment with GLP-1 receptor agonists: discontinuation, maintenance strategy, and a nephrology perspective (September 20, 2026, online, Slovak)

- 3.51** Alogliptin and kidney outcomes: what a comparison of DPP-4 inhibitors shows (September 24, 2026, online, Slovak)
- 3.52** Finerenone in CKD and type 1 diabetes: what FDA approval really means (September 24, 2026, online, Slovak)
- 3.53** Hidden hypercortisolism in difficult-to-control type 2 diabetes: what CATALYST showed (September 24, 2026, online, Slovak)
- 3.54** DASH4D in type 2 diabetes: better glycemia and blood pressure in a controlled trial (September 24, 2026, online, Slovak)
- 3.55** Switching between GLP-1 receptor agonists: timing, titration, and nephrology safeguards (October 2, 2026, online, Slovak)
- 3.56** The 2026 ADA and EASD consensus: type 2 diabetes is no longer a disease of glycemia (October 2, 2026, online, Slovak)

CHAPTER 4

Hypertension, lipids, and vascular risk

Blood pressure measurement and targets, resistant hypertension, dyslipidemia, antithrombotic therapy, and pregnancy with kidney risk. Number of articles: 24.

The chapter on hypertension, lipids, and vascular risk brings together 24 articles from April to October 2026 and is, at its core, a chapter about target values and about what prevents them from being reached in practice. Evidence of intermediate strength predominates: commentaries on new guidelines and critiques of them (4.1, 4.2, 4.8, 4.10, 4.14, 4.17), reviews (4.5, 4.6, 4.11, 4.13, 4.16, 4.19, 4.20, 4.23), observational cohorts (4.9, 4.15, 4.22, 4.24), and post hoc analyses of randomized trials (4.12, 4.18). Only three papers provide primary randomized evidence: the phase 2 trial of baxdrostat in CKD (4.3, 4.4), the phase 3 Bax24 trial (4.21), and the negative TRACK trial of rivaroxaban (4.7).

The main message is twofold. Targets are becoming stricter and more uniform: the systolic blood pressure (BP) target in CKD is moving to below 120 mm Hg (4.22), and the LDL cholesterol target in clinical atherosclerotic disease to below 1.4 mmol/L (4.10, 4.14, 4.17). Real-world data, however, show that only a minority of patients reach these targets (4.22) and that the new drugs so far offer reductions in BP or albuminuria, not evidence on clinical events (4.4, 4.21). The gap between guidelines and practice is larger than the gap between the guidelines themselves.

A low blood pressure target and the reality of reaching it

The observational analysis of records from the Mass General Brigham network is the most sobering. In 2024, only 21.9% of patients with CKD G3 to G4 had a routine mean systolic BP below 120 mm Hg, compared with 18.3% in 2020, and the mean number of antihypertensive drug classes changed only from 1.53 to 1.55 (4.22). At a threshold of 130 mm Hg, the proportion of controlled patients rose from 44.7% to 50.4%, so inadequate control cannot be attributed solely to the strict KDIGO target (4.22). In a cohort of 18,996 patients, a BP of 120 mm Hg or higher was associated with an HR of 1.19 (95% CI, 1.11 to 1.27) for cardiovascular events and an HR of 1.21 (95% CI, 1.08 to 1.35) for kidney failure (4.22). This is an association derived from nonstandardized measurements without data on albuminuria, and it does not prove that lowering BP below 120 mm Hg causally reduces risk (4.22).

Randomized support for a low target in CKD remains indirect: in the SPRINT subgroup with CKD, the composite cardiovascular outcome did not reach significance (HR 0.81; 95% CI, 0.63 to 1.05), mortality decreased (HR 0.72), and kidney protection was not demonstrated (4.22). A commentary on a post hoc analysis of ESPRIT adds an implementation perspective: 62.5% of patients in the intensive arm in community centers reached the target below 120 mm Hg at the cost of approximately 1.3 additional antihypertensive drugs and 1.5 additional visits per year, and the key was titration protocols and centralized monitoring, not new drugs (4.12).

The level of BP is not its only dimension. A pooled analysis of 952 participants in SPRINT MIND and ACCORD MIND showed that higher visit-to-visit systolic BP variability was independently associated with faster progression of white matter lesions; however, variability mediated only 9.15% of the effect

of intensive treatment, and lower confidence limits just above zero make the finding a fragile signal (4.18). A review of diurnal natriuresis links blunted daytime sodium excretion with nocturnal hypertension; in CKD, the differences were on the order of a few mm Hg, and randomized data on chronotherapy are lacking (4.23). Both papers lead to the same conclusion: the target is a stable BP measured in a standardized manner, with the use of ABPM, not variability or salt timing as a separate target (4.18, 4.23).

Resistant hypertension and aldosterone synthase inhibition

A review of resistant hypertension sets out a sequence of steps that the new drugs do not change: exclude apparent resistance by verifying measurement, using out-of-office monitoring, and assessing adherence in a nonjudgmental way, and actively look for primary aldosteronism and BP-raising substances such as nonsteroidal anti-inflammatory drugs, calcineurin inhibitors, or erythropoiesis-stimulating agents (4.20). In PATHWAY-2, spironolactone lowered home systolic BP by 8.70 mm Hg compared with placebo, and it remains the preferred add-on drug at an eGFR of at least 30 mL/min/1.73 m² and a potassium level of no more than 4.5 mmol/L (4.20). This is the reference point for everything new.

Baxdrostat needs to be assessed according to the population studied. In the phase 2 trial in 195 patients with CKD, an eGFR of approximately 44 mL/min/1.73 m², and albuminuria of approximately 714 mg/g, it lowered systolic BP at 26 weeks by 8.1 mm Hg more than placebo and, in an exploratory analysis, albuminuria by 55%; hyperkalemia occurred in 41% versus 5% (4.3, 4.4). In the phase 3 Bax24 trial in 217 patients with resistant hypertension and preserved kidney function, 24-hour systolic BP fell by 14.0 mm Hg more than with placebo, and 3% of treated patients had potassium above 6.0 mmol/L (4.21). The difference between 41% and 3% is the most important safety finding of the chapter, and it concerns precisely nephrology patients (4.3, 4.21). Shared weaknesses: follow-up of 12 to 26 weeks, no comparison with spironolactone, no data on events, manufacturer funding, and, in Bax24, a primary outcome available for only 184 of 217 patients (4.4, 4.21). The FDA approved baxdrostat on May 15, 2026; data are lacking for an eGFR below 45 mL/min/1.73 m², and US approval does not mean availability in Slovakia (4.20). Renal denervation belongs in the same framework: a reduction of approximately 5 to 10 mm Hg, about a third of patients do not respond, few patients with an eGFR below approximately 40 mL/min/1.73 m² were included in the trials, and the procedure complements pharmacotherapy rather than replacing it (4.6).

Lipids: the return of targets and a critique of the process

The 2026 ACC/AHA guideline replaces the Pooled Cohort Equations with the PREVENT calculator, which yields a risk estimate roughly half as high, reinstates LDL cholesterol targets, below 1.4 mmol/L in secondary prevention, and retains CKD stages 3 and 4 as an independent indication for a statin at ages 40 to 75 years (4.10). Non-HDL cholesterol, apoB, and lipoprotein(a) gain greater weight (4.2). For CKD G3 or higher, a nephrology commentary proposes a moderate-intensity statin or a combination with ezetimibe at an LDL-C of 1.8 to 4.9 mmol/L, targets of LDL-C below 1.4 mmol/L and non-HDL-C below 2.2 mmol/L in clinical ASCVD, and does not support routine new initiation of a statin in patients on hemodialysis (4.14).

The critique of the process is substantive: 12 of the 33 authors had financial conflicts of interest, an independent systematic review and the participation of the AAFP and ACP are missing, and the panel had a single patient representative (4.1). For Slovak practice, what matters is that PREVENT was validated in the US population, it does not replace SCORE2, and European guidelines remain decisive

(4.14). The October 2025 ESC and EAS update retains the 2019 targets, below 1.4 mmol/L for very high risk and below 1.8 mmol/L for high risk, with a reduction of at least 50%, and recalls that lowering LDL cholesterol by approximately 1 mmol/L reduces major vascular events by about one fifth in relative terms (4.17). On dialysis, both sides agree: neither 4D, AURORA, nor the dialysis subgroup of SHARP showed a benefit of new statin initiation, and established treatment can be continued (4.14, 4.17).

Lipoprotein(a) should be measured at least once in a lifetime; a level of 125 nmol/L or higher corresponds to an approximately 1.4-fold risk of ASCVD and a level of 250 nmol/L or higher to an approximately twofold risk; values in mg/dL and nmol/L must not be converted with a fixed factor; and in CKD, acquired elevation in nephrotic syndrome and peritoneal dialysis should be kept in mind (4.14, 4.19). Targeted lowering of Lp(a) does not yet have outcome evidence (4.17, 4.19).

Antithrombotic therapy and vascular risk in CKD

TRACK is the only large negative randomized result in the chapter and the most important one for dialysis centers. In 1,458 patients with an eGFR of no more than 29 mL/min/1.73 m² or on dialysis, rivaroxaban 2.5 mg twice daily did not reduce major cardiovascular events after a median of 1.7 years (22.6% versus 20.7%; HR 1.09; 95% CI, 0.87 to 1.36) and increased major bleeding (8.8% versus 6.0%; HR 1.51; 95% CI, 1.02 to 2.22) (4.7). The benefit seen in COMPASS and VOYAGER PAD cannot be extrapolated, and the result does not apply to clear indications for anticoagulation (4.7). Early termination and treatment discontinuation in almost a third of patients weaken the conclusion of no efficacy, not the conclusion about bleeding (4.7).

Hope is shifting to factor XI inhibitors: OCEANIC-STROKE with asundexian showed a marked reduction in recurrences without excessive bleeding, although the review is based on an expert meeting rather than on data in CKD (4.11). Peripheral artery disease completes the picture: in the CRIC cohort of 3,627 patients with CKD, a conventionally normal ABI of 0.9 to 1.0 carried an HR of 2.76 for PAD, and at an ABI of 1.4 or higher the HR was 4.85; however, there is no randomized evidence that screening is beneficial (4.24).

Pregnancy as a window into kidney risk

A review explains the kidney manifestations of preeclampsia with a two-stage model of placental hypoperfusion and endothelial dysfunction (4.5). A Danish cohort of 286,078 women showed that in preeclampsia with moderate or severe proteinuria, the 10-year risk of hypertension was 16.0% versus 3.6% and that of CKD 5.1% versus 0.4%; an association with cardiovascular disease was not confirmed (4.9). A Stockholm cohort of 171,693 primiparous women found an HR of 2.43 for the composite of albuminuria above 300 mg/g and eGFR below 60 mL/min/1.73 m²; the absolute risk was only 0.6%, but in the first year after delivery, albuminuria had been measured in only 10% of women, and at a threshold of 30 mg/g the number of events rose from 113 to 428 (4.15).

CHAP provides randomized evidence: treating mild chronic hypertension in 2,408 women to a target below 140/90 mm Hg reduced the composite adverse outcome from 37.0% to 30.2% (adjusted relative risk 0.82; 95% CI, 0.74 to 0.92) (4.16). BP peaks on the third to sixth day after delivery (4.16). CKD occurs in approximately 3% to 4% of pregnancies, eGFR equations are not reliable in pregnancy, and RAAS blockers, statins, and SGLT2 inhibitors should be reviewed before conception (4.13, 4.16).

What is proven, what is probable, and what remains open

Little has been proven in randomized trials, and most of it is negative or based on surrogate outcomes: low-dose rivaroxaban in advanced CKD has no benefit and increases bleeding (4.7), baxdrostat lowers BP and albuminuria over a period of weeks (4.3, 4.21), treatment of mild hypertension in pregnancy improves maternal outcomes (4.16), and a target below 120 mm Hg is achievable in practice (4.12). It is probable, but based on observational data, that a BP above 120 mm Hg in CKD carries a higher risk (4.22), that preeclampsia with proteinuria increases the risk of CKD (4.9, 4.15), that a normal ABI in CKD does not exclude PAD (4.24), and that BP variability and the rhythm of natriuresis are markers rather than targets (4.18, 4.23).

It remains open whether aldosterone synthase inhibition will reduce clinical events and whether it is safe at an eGFR below 45 mL/min/1.73 m² (4.20, 4.21), whether kidney protection exists with a low BP target (4.22), whether lowering Lp(a) will improve outcomes (4.17, 4.19), whether factor XI inhibitors will bring safer anticoagulation in CKD (4.11), and whether ABI screening or UACR screening after preeclampsia will improve prognosis, not merely detection (4.15, 4.24).

Implications for practice

The greatest potential for improvement lies not in new drugs but in measurement and organization. Standardized measurement, home or 24-hour monitoring before every decision on escalation, and verification of adherence and of BP-raising drugs are the common denominator of all the guidelines (4.8, 4.20, 4.22). Protocol-driven titration with monitoring for hypoperfusion, kidney function, and electrolytes makes it possible to aim for a lower target without the J-curve becoming an excuse (4.12). In resistant hypertension, the aldosterone axis and sleep apnea should be investigated; spironolactone remains the first choice, and aldosterone synthase inhibitors are not available in Slovakia and, in CKD, carry a risk of hyperkalemia in approximately two of five patients (4.3, 4.20, 4.21).

Lipid-lowering therapy should be guided by European targets and should distinguish nondialysis CKD, dialysis, and transplantation: a statin or a combination with ezetimibe before dialysis, continuation rather than new initiation after dialysis has started, and a one-time Lp(a) measurement, rechecked after remission of nephrotic syndrome (4.14, 4.17, 4.19). Prophylactic rivaroxaban has no place at an eGFR below 30 mL/min/1.73 m² or on dialysis (4.7). Women after preeclampsia should be recorded as being at permanently increased risk, their BP should be checked between the third and sixth day after delivery, and they should be followed with UACR from a threshold of 30 mg/g, not with creatinine alone (4.9, 4.15, 4.16). In CKD, the ABI should be read as a continuous variable, and at a value of 1.4 or higher, the toe-brachial index should be added (4.24).

What to watch next

In the near term, the decisive data will come from phase 3 trials of baxdrostat with hard endpoints, including the combination with dapagliflozin, and safety data at an eGFR below 45 mL/min/1.73 m² (4.4, 4.20), the results of Lp(a)HORIZON and OCEAN(a)-Outcomes (4.17, 4.19), LIBREXIA-STROKE and LIBREXIA-AF with milvexian (4.11), and the inclisiran outcome trials ORION-4 and VICTORION-2P (4.17). It will be worth watching whether KDIGO updates its 2013 lipid guideline (4.17), whether UACR-based follow-up after preeclampsia becomes established (4.15), and whether the proportion of pa-

tients with CKD reaching the target BP begins to change faster than by one percentage point per year (4.22).

Articles in this chapter

- 4.1** The new dyslipidemia guidelines: why they need to be read critically (May 17, 2026, online, Slovak)
- 4.2** The 2026 dyslipidemia management guideline: what it brings to clinical practice (May 17, 2026, online, Slovak)
- 4.3** A new pill lowers difficult-to-control blood pressure and may protect the kidneys (May 19, 2026, online, Slovak)
- 4.4** New treatment for resistant hypertension in CKD: baxdrostat lowers both blood pressure and albuminuria (May 19, 2026, online, Slovak)
- 4.5** Preeclampsia and the kidney: why damage occurs and what it means for practice (May 24, 2026, online, Slovak)
- 4.6** Renal denervation in resistant hypertension: a promising method, but not a shortcut in treatment (June 5, 2026, online, Slovak)
- 4.7** Low-dose rivaroxaban in advanced CKD provided no cardiovascular benefit (June 10, 2026, online, Slovak)
- 4.8** New hypertension guidelines: less improvisation, more accurate measurement, and practical decisions (June 10, 2026, online, Slovak)
- 4.9** Proteinuria in preeclampsia may signal a higher long-term risk of hypertension and CKD (June 10, 2026, online, Slovak)
- 4.10** New dyslipidemia guidelines: PREVENT replaces the old calculator, and LDL targets are lower (June 10, 2026, online, Slovak)
- 4.11** New approaches in antithrombotic therapy: from “good” and “bad” clots to safer prevention (June 16, 2026, online, Slovak)
- 4.12** Implementation strategies for intensive blood pressure control: lessons for everyday clinical practice (with a focus on nephrology) (June 20, 2026, online, Slovak)
- 4.13** Kidney disease in pregnancy: early risk assessment and multidisciplinary care (July 5, 2026, online, Slovak)
- 4.14** Dyslipidemia in chronic kidney disease: what in the 2026 ACC/AHA guideline changes nephrology practice (July 15, 2026, online, Slovak)
- 4.15** Kidney risk after preeclampsia: the hidden burden and missed opportunities for early detection (August 5, 2026, online, Slovak)
- 4.16** Hypertension in pregnancy and after delivery: why the postpartum visit must not be the last (August 5, 2026, online, Slovak)
- 4.17** Dyslipidemia in 2026: cardiovascular risk assessment, LDL cholesterol targets, and treatment in chronic kidney disease (August 15, 2026, online, Slovak)
- 4.18** Visit-to-visit systolic blood pressure variability and faster progression of white matter lesions: implications for nephrology practice (August 19, 2026, online, Slovak)
- 4.19** Lipoprotein(a): a cardiovascular risk factor we often miss in primary care (August 22, 2026, online, Slovak)
- 4.20** Uncontrolled and resistant hypertension: diagnosis, the aldosterone axis, and new treatment options (August 28, 2026, online, Slovak)
- 4.21** Baxdrostat markedly lowers 24-hour blood pressure in resistant hypertension: results of the Bax24 trial (September 6, 2026, online, Slovak)
- 4.22** A systolic blood pressure target below 120 mm Hg in chronic kidney disease has seen only limited uptake in practice (September 11, 2026, online, Slovak)
- 4.23** Diurnal sodium excretion and nocturnal hypertension: salt timing as a clinical problem (September 14, 2026, online, Slovak)
- 4.24** Peripheral artery disease in CKD: why an ABI threshold below 0.9 is not enough (October 2, 2026, online, Slovak)

CHAPTER 5

Glomerular and immune-mediated kidney diseases

IgA nephropathy, membranous nephropathy, lupus nephritis, ANCA-associated vasculitis, and complement-mediated diseases. New targeted therapies and their place according to current guidelines. Number of articles: 31.

The chapter on glomerular and immune-mediated kidney diseases brings together 31 articles from April to October 2026, and it is the area in which knowledge changed fastest. IgA nephropathy dominates, alongside FSGS, C3 glomerulopathy, membranous nephropathy, lupus nephritis, and ANCA-associated vasculitis. The mix of evidence is uneven: one final analysis of a phase 3 RCT with a kidney function endpoint (5.22), several interim analyses of RCTs with the surrogate endpoint of proteinuria (5.14, 5.17, 5.28), one network meta-analysis (5.2), regulatory decisions by the FDA and EMA (5.7, 5.15, 5.17), guidelines and commentaries from KDIGO, KDOQI, ACR, and EULAR (5.8, 5.9, 5.28), consensus conference reports (5.16, 5.27), retrospective cohorts (5.18, 5.19, 5.23, 5.26), and review or opinion pieces without original data (5.1, 5.3, 5.5, 5.10, 5.11).

In IgA nephropathy, the first evidence has emerged that targeted therapy slows the loss of glomerular filtration rate, not merely proteinuria (5.22), and proteinuria has been confirmed as a prognostically strong but still surrogate endpoint (5.21, 5.28). The period also brought a warning against taking the numbers on trust: the revocation of the European authorization of avacopan because of data integrity problems (5.15, 5.31), company announcements that have not been peer reviewed (5.29), uncontrolled cohorts (5.26), and a reminder that even baseline eGFR in observational studies is an estimate with its own uncertainty (5.20).

IgA nephropathy: from proteinuria to kidney function

The pathogenetic framework is stable: a four-hit model with production of Gd-IgA1, autoantibodies, immune complexes, and mesangial deposition with inflammation, which underpins the targeting of the APRIL and BAFF axis, complement, mucosal IgA production, and the endothelin pathway (5.3, 5.5, 5.28). Opinion pieces speak of a golden age (5.1), but optimism needs to be separated from what has been proven.

The strongest finding of the chapter is the final 24-month analysis of APPLAUSE-IgAN: in 477 patients, iptacopan slowed the annualized eGFR decline to 3.10 versus 6.12 mL/min/1.73 m² per year, a difference of 3.02 mL/min/1.73 m² per year (95% CI, 2.02 to 4.01), and reduced the composite kidney endpoint from 33.5% to 21.4% (HR 0.57), which corresponds to an NNT of approximately 8 to 9 over two years (5.22). This is the first evidence of an effect of complement inhibition on loss of kidney function (5.25). The price is clear: serious infections in 6.7% versus 2.1% of patients, and mandatory vaccination against encapsulated bacteria, which does not eliminate the risk completely (5.22, 5.25). Limitations include low use of SGLT2 inhibitors at baseline and the absence of a direct comparison with other targeted drugs (5.22).

A post hoc analysis of PROTECT gives proteinuria prognostic weight: patients with complete remission below 0.3 g/day by week 110 had a chronic eGFR slope of 0.4 versus 4.2 mL/min/1.73 m² per year and the composite endpoint in 1% versus 14% (5.21). This is an association burdened by residual confounding, reverse causality, and immortal time bias, and in 35% of patients proteinuria later rose above 0.75 g/day (5.21). The KDIGO 2025 goals, proteinuria below 0.5 g/day and ideally below 0.3 g/day with an eGFR decline of less than approximately 1 mL/min per year, are thus supported by prognostic data, not by evidence that reaching the goal with a drug guarantees a kidney benefit (5.4, 5.9, 5.21).

New molecules: big numbers, surrogate endpoints

Drugs targeting the BAFF and APRIL axis have produced the largest reductions in proteinuria. On July 7, 2026, atacicept received FDA accelerated approval on the basis of an ORIGIN 3 analysis in 203 patients in which UPCR fell by 46% versus 7% with placebo, a relative reduction of 42% (95% CI, 29 to 52%) expressed as a ratio of geometric means (5.17). The final analysis announced by the sponsor reports an annualized eGFR slope through week 104 of 0.6 versus 5.6 mL/min/1.73 m² and an HR of 0.24 for the composite endpoint; however, these are company data that have not been peer reviewed, at week 104 only 50 and 37 of 214 patients per arm had an eGFR measurement, and dialysis, transplantation, or death together accounted for 0 versus 8 events (5.29). In the interim analysis of TELIGAN in 318 patients, telitacicept reduced UPCR by 58.9% versus 8.8%, with a higher incidence of adverse events of any type (89.3% versus 78.6%) and a lower incidence of serious adverse events (2.5% versus 8.2%) (5.14). In VISIONARY, sibeprenlimab reduced UPCR by 51.2% and Gd-IgA1 by 67.1% (5.28). The KDIGO conference report confirms that this axis is more promising in IgAN than rituximab (5.16, 5.27).

The KDIGO commentary is restrained about these figures: the trials differ in time point, population, and expression of effect, ranking drugs by percentages is not acceptable, and eGFR slope data are still awaited for three of the four molecules (5.28). The network meta-analysis of 17 RCTs says the same: Nefecon had the most favorable point estimate for eGFR slope, but the confidence intervals included the null, and the network was sparse with short follow-up (5.2). For Nefecon, which KDIGO 2025 prefers, KDOQI points to a possible loss of response after the course ends (5.9).

The endothelin pathway rests on firmer but more modest data. In PROTECT, proteinuria with sparsentan after 110 weeks was approximately 40% lower than with irbesartan, and the chronic eGFR slope was 2.7 versus 3.8 mL/min/1.73 m² per year, but the composite kidney endpoint did not differ significantly (5.26). In ALIGN, atrasentan reduced UPCR by 38.1% versus 3.1%, with fluid retention in 11.2% versus 8.2% (5.28). The combination of sparsentan with an SGLT2 inhibitor has only short-term surrogate data: SPARTACUS was open-label and uncontrolled, and the PROTECT OLE substudy of 63 patients showed a UPCR ratio of 0.75 (95% CI, 0.58 to 0.98) at week 12, without data on hyperkalemia or hypotension (5.24). The Spanish cohort of 41 patients with a 44.1% reduction in proteinuria at 3 months is uncontrolled (5.26).

Podocytopathies and complementopathies: selection by mechanism

In FSGS, the FDA approved sparsentan on April 13, 2026, as the first drug specifically for FSGS without nephrotic syndrome; in DUPLEX, however, the difference in eGFR slope was not significant, and immunosuppression achieves remission in fewer than 25% of patients (5.7). More interesting is the shift toward selection by mechanism: up to 15% of adults carry a causal mutation, approximately 10% of patients have anti-nephrin autoantibodies, and in a phase 2 RCT in 62 patients, the TRPC6 in-

hibitor BI 764198 achieved a response in 35% versus 7% with placebo, with 4 of 4 treated patients with a pathogenic TRPC6 variant responding (5.7, 5.13). The sample is small and follow-up lasted 12 weeks; phase 3 trials will be decisive (5.13).

In C3 glomerulopathy, where KDIGO 2021 relied on consensus, phase 3 RCTs have brought a consistent reduction in proteinuria: iptacopan in APPEAR-C3G by 35.1% compared with placebo, and pegcetacoplan in VALIANT by 67.3%, with an eGFR change of 1.6 versus 7.9 mL/min at a nominal $p = 0.0322$ (5.12). The subgroup with a clear benefit from terminal blockade has not been defined, and long-term kidney outcomes remain uncertain (5.12). The Core Curriculum points out that normal serum C3 and C4 do not exclude local activation and that the failure of narsoplimab in ARTEMIS-IGAN does not refute the lectin hypothesis (5.25).

Two retrospective studies are hypothesis-generating only. In a cohort of 236 patients with podocytopathies, persistent microscopic hematuria within three months after biopsy was associated with kidney failure, with an HR of 3.55 (95% CI, 1.78 to 7.05) based on 38 events; the direction of the association is convincing, its magnitude uncertain, and it does not create a therapeutic target (5.18). The anti-PLA2R titer in severe hypoalbuminemia was associated with thrombotic risk (adjusted hazard ratio, 1.51 per 10 RU/mL), but the model rests on at most ten events and does not account for immunosuppression, so thromboprophylaxis decisions are still based on serum albumin and bleeding risk (5.19).

Systemic diseases and the right depth of intervention

In lupus nephritis, ACR 2025, EULAR 2025, and KDIGO 2024 agree on hydroxychloroquine, rapid glucocorticoid tapering, and a proteinuria goal below 0.5 g/day, but they differ on whether triple therapy should be the standard for all; rituximab is not first-line in any of the documents, and class V remains without robust data (5.8). Timing is clearer: treatment should not be delayed for a biopsy, but the biopsy should be performed as soon as possible, because conventional markers cannot replace histology; the benefit of early response comes from observational cohorts (5.30). For connective tissue diseases, two reviews without original data emphasize that no single test establishes the diagnosis, antibodies serve as signposts, and the intensity of treatment is determined by the threat to organs (5.10, 5.11).

In ANCA-associated vasculitis, the EMA identified serious data integrity problems in the ADVOCATE trial and recommended not starting avacopan in new patients and switching treated patients to an alternative with liver monitoring (5.15); NEJM retracted the trial publication, and the European Commission revoked the authorization on August 4, 2026 (5.31). The remaining recommendations rest on unchallenged RCTs: the reduced-dose glucocorticoid regimen from PEXIVAS is noninferior, with fewer serious infections, plasma exchange provided no benefit (HR 0.86; 95% CI, 0.65 to 1.13), and maintenance with rituximab in RITAZAREM has an HR for relapse of 0.41 compared with azathioprine (5.31).

The KDIGO Controversies Conference adds an interpretive key: anti-CD20 therapy is first-line in membranous nephropathy, the benefit of rituximab in steroid-dependent nephrotic syndrome is transient, and in lupus nephritis, obinutuzumab and CAR T cells open up the concept of longer remission at the cost of serious adverse events (5.16, 5.27). CD20 is not expressed on plasma cells, what matters is the right depth of depletion, not maximal intensity of intervention, and validated biomarkers as well as criteria for stopping treatment are lacking (5.27).

Quality of evidence and pitfalls of interpretation

A methodological analysis of 2,874 patients showed that the bias in baseline eGFR grows with the width of the time window around biopsy and silently distorts associations and progression thresholds (5.20). The retrospective study of transverse coaxial biopsy with the highest number of glomeruli contains four internal inconsistencies, the technique was chosen by the operator, and the only RCT did not confirm a difference, so it is a promising technique, not a new standard (5.23). A cross-sectional analysis of 186 videos showed that engagement correlated with visual quality, not with reliability according to mDISCERN and JAMA (5.6).

What is proven, what is probable, and what remains open

Proven at the RCT level are the slowing of eGFR decline with iptacopan in IgAN, with a demonstrated infection risk (5.22, 5.25), the antiproteinuric effect of sparsentan, atrasentan, atacicept, telitacicept, and sibeprenlimab (5.14, 5.17, 5.26, 5.28), the reduction in proteinuria with alternative pathway inhibition in C3G (5.12), and the noninferiority of reduced-dose glucocorticoids without a benefit of plasma exchange in AAV (5.31). Probable, but based on surrogate or non-peer-reviewed data, are a disease-modifying effect of atacicept (5.29), a benefit of combining sparsentan with an SGLT2 inhibitor (5.24), the efficacy of TRPC6 inhibition in genetically selected patients (5.13), and the persistence of the effect of Nefecon after the course ends (5.9). Open questions include the sequencing and combination of targeted drugs (5.2, 5.28), the duration of B-cell-targeted therapy (5.16, 5.27), the value of hematuria and anti-PLA2R as decision-making parameters (5.18, 5.19), the initial regimen in lupus nephritis (5.8, 5.30), and the selection of responders in C3G (5.12).

Implications for practice

For a Slovak outpatient clinic and hospital ward, a stepwise framework emerges. Diagnosis rests on biopsy with MEST-C scoring and risk stratification; prediction tools serve shared decision-making, not drug selection (5.4, 5.9). Kidney protection remains the foundation: RAS blockade, an SGLT2 inhibitor, sodium below 2 g/day, and BP of no more than 120/70 mm Hg (5.4, 5.9). Proteinuria should be measured repeatedly, and a value sustained over time should be documented before treatment is changed (5.21); a positive dipstick should be confirmed by microscopy of the urinary sediment, and hematuria should not be regarded as a threshold for escalating immunosuppression (5.18). Of the targeted drugs for IgAN, iptacopan has the strongest evidence, with vaccination against meningococcus, pneumococcus, and Hib ideally at least 2 weeks before treatment (5.22, 5.25); atacicept is not authorized in the EU or in Slovakia (5.17, 5.29). With sparsentan, BP, potassium, and liver tests should be monitored (5.26).

In AAV, a local protocol is needed for switching patients from avacopan with liver monitoring, together with an explanation that this reflects doubts about the evidence, not a fluctuation in efficacy (5.15, 5.31). In lupus nephritis, the rule is to treat now and perform the biopsy as soon as possible, with hydroxychloroquine for every patient without a contraindication (5.30). Before B-cell-targeted therapy, immunoglobulins, blood count, and vaccination status should be assessed, but treatment should not be delayed when an organ is threatened (5.27). Patients also need guidance on social media videos, where popularity is no guarantee of quality (5.6).

What to watch next

In the near term, the decisive developments will be the peer-reviewed publication of the final ORIGIN 3 analysis and a confirmatory eGFR assessment (5.17, 5.29), completion of TELIGAN with data on dur-

ability of effect (5.14), the eGFR slope data for sibeprenlimab and atrasentan that KDIGO is awaiting (5.28), phase 3 trials of TRPC6 inhibition (5.13), long-term kidney outcomes in C3G (5.12), the consequences of the revocation of the avacopan authorization for the KDIGO and EULAR guidelines (5.15, 5.31), and whether head-to-head trials with hard kidney endpoints, whose absence the network meta-analysis highlighted, will appear (5.2). It will also be worth watching whether observational studies begin to report the uncertainty of baseline eGFR, without which prognostic signals will remain difficult to compare (5.20).

Articles in this chapter

- 5.1** A golden age for IgA nephropathy and rare glomerulopathies (April 25, 2026, online, Slovak)
- 5.2** Comparative effectiveness of treatments for IgA nephropathy: a network meta-analysis (May 8, 2026, online, Slovak)
- 5.3** IgA nephropathy: the role of APRIL in the four-hit model of pathogenesis (May 11, 2026, online, Slovak)
- 5.4** IgA nephropathy in the era of new therapeutic options: from supportive care to targeted therapy (May 11, 2026, online, Slovak)
- 5.5** IgA nephropathy: new insights bring a wave of targeted therapies (May 13, 2026, online, Slovak)
- 5.6** TikTok and IgA nephropathy: short videos can help, but the quality of information is uneven (May 19, 2026, online, Slovak)
- 5.7** Overview of the latest research: FSGS and diabetic kidney disease (2025 to 2026) (June 6, 2026, online, Slovak)
- 5.8** Comparison of the three major lupus nephritis guidelines: ACR 2025, EULAR 2025, and KDIGO 2024 (June 16, 2026, online, Slovak)
- 5.9** IgA nephropathy according to KDIGO 2025 and the KDOQI US Commentary: from biopsy to more targeted treatment (June 19, 2026, online, Slovak)
- 5.10** Connective tissue diseases in practice: diagnostic and therapeutic challenges in systemic autoimmune diseases (June 20, 2026, online, Slovak)
- 5.11** Connective tissue diseases: a nephrologist's practical clinical view of diagnosis, organ risk, and treatment decisions (June 20, 2026, online, Slovak)
- 5.12** Complement 3 glomerulopathy (C3G): standard of care, nonspecific therapy, and targeted complement inhibition (June 20, 2026, online, Slovak)
- 5.13** TRPC6 inhibition in focal segmental glomerulosclerosis (FSGS): what a randomized phase 2 trial shows and why it is a significant step toward precision nephrology (June 20, 2026, online, Slovak)
- 5.14** Telitacicept in IgA nephropathy: a phase 3 interim analysis (TELIGAN) shows a marked reduction in proteinuria with a higher incidence of adverse events (June 22, 2026, online, Slovak)
- 5.15** New data cast doubt on efficacy: EMA has recommended revoking the authorization of Tavneos (avacopan) in the EU, and what it means for ANCA-associated vasculitis and nephrology (June 26, 2026, online, Slovak)
- 5.16** B-cell-targeted therapies in immune-mediated kidney diseases: what the KDIGO Controversies Conference shows (June 26, 2026, online, Slovak)
- 5.17** Atacicept in primary IgA nephropathy: what FDA accelerated approval means (July 15, 2026, online, Slovak)
- 5.18** Persistent microscopic hematuria in podocytopathies: a prognostic signal, not a therapeutic target (August 4, 2026, online, Slovak)
- 5.19** Anti-PLA2R and thrombosis risk in membranous nephropathy: the antibody as a possible complement to albumin (August 24, 2026, online, Slovak)
- 5.20** Baseline eGFR at kidney biopsy: how its selection and imputation can bias the results of observational studies (August 28, 2026, online, Slovak)
- 5.21** Complete remission of proteinuria in IgA nephropathy is associated with slower decline in glomerular filtration rate (September 6, 2026, online, Slovak)
- 5.22** Iptacopan slows kidney function decline in IgA nephropathy: final 24-month results of the APPLAUSE-IgAN trial (September 6, 2026, online, Slovak)
- 5.23** Transverse coaxial biopsy of the native kidney: higher specimen yield, but insufficient evidence to change practice (September 11, 2026, online, Slovak)
- 5.24** Sparsentan with an SGLT2 inhibitor in IgA nephropathy: SPARTACUS and PROTECT OLE (September 14, 2026, online, Slovak)
- 5.25** Complement in nephrology: diagnosis, infection risk, and targeted inhibition in IgA nephropathy (September 14, 2026, online, Slovak)

- 5.26** Sparsentan and rapid, sustained reduction of proteinuria in IgA nephropathy in real-world practice (September 17, 2026, online, Slovak)
- 5.27** Targeting B cells in immune-mediated kidney diseases: conclusions of the KDIGO conference and why “the stronger, the better” does not apply (September 17, 2026, online, Slovak)
- 5.28** Complement, B cells, and the endothelin system in IgA nephropathy: where new therapies fit according to the KDIGO commentary (September 17, 2026, online, Slovak)
- 5.29** Trutakna in IgA nephropathy: the final ORIGIN 3 analysis and its implications for practice (September 20, 2026, online, Slovak)
- 5.30** Lupus nephritis: treat now, biopsy as soon as possible (September 24, 2026, online, Slovak)
- 5.31** ANCA-associated vasculitis with kidney involvement: which treatment standards still apply in 2026 (September 29, 2026, online, Slovak)

CHAPTER 6

Acute kidney injury, critical care, and toxicology

Acute kidney injury, perioperative and intensive care, nephrotoxicity of drugs and environmental agents, and clinical toxicology. Number of articles: 17.

The chapter on acute kidney injury, critical care, and toxicology brings together seventeen texts, ranging from randomized trials of kidney replacement therapy (KRT) through observational cohorts from intensive care units to reviews of environmental metals and mushroom toxins. The level of evidence is unusually varied: randomized trials or reviews of them (6.1, 6.4, 6.6, 6.10, 6.11), large retrospective cohorts (6.12, 6.15), a population-based study conveyed through an interview with an expert (6.2), a case-control study (6.13), a mechanistic pilot study (6.5), narrative reviews and clinical texts (6.7, 6.8, 6.9, 6.14, 6.16, 6.17), and a case report (6.3). For every claim, the reader should therefore distinguish whether it reflects an experimental result, an association, or the author's opinion.

Despite this diversity, the main message of the period from April to October 2026 is consistent. A reflexive intervention in AKI, whether immediate dialysis, a fixed dialysis schedule, an early consultation alert, or a preventive drug, has repeatedly failed to outperform decision-making based on a clearly defined indication. At the same time, randomized data show that an intervention with sound pathophysiologic logic can harm the patient and that surrogate endpoints are not enough to justify a change in practice (6.6, 6.11). The toxicology texts add a second lesson: the cause of AKI is uncovered by a targeted history and a systematic approach, not by blanket screening (6.7, 6.9, 6.14, 6.17).

Timing and intensity of kidney replacement therapy

The most solid part of the chapter concerns when and how intensively to dialyze. A review of the ELAIN, AKIKI, IDEAL-ICU, STARRT-AKI, and AKIKI-2 trials gives a consistent picture: immediate initiation of KRT without an urgent indication generally does not improve important outcomes (6.4). Only the single-center ELAIN trial showed lower 90-day mortality with KRT started within 8 hours, with the limitation of a low fragility index, whereas AKIKI, IDEAL-ICU, and STARRT-AKI showed no difference in mortality, and in STARRT-AKI the accelerated arm had more hypotension and hypophosphatemia (6.4). The opposite extreme is not safe either: in AKIKI-2, prespecified analyses suggested higher 60-day mortality with a markedly delayed strategy (6.4). The result is a narrow window that neither creatinine nor urine output reliably defines when indications are relative (6.4).

The same logic applies to what to do after KRT has started. A multicenter RCT in JAMA with 221 patients compared dialysis only when metabolic or clinical criteria were met with scheduled dialysis three times weekly (6.1). Recovery of kidney function at discharge was achieved in 64% versus 50% of patients, an absolute difference of 13.8 percentage points, but significance was not confirmed in the prespecified adjusted analysis (6.1). The conservative group received a median of 1.8 versus 3.1 dialysis sessions per week and had 69 versus 97 episodes of dialysis-associated hypotension (6.1). This is a signal in favor of an indication-guided approach in hemodynamically stable patients, not proof (6.1).

Independently of each other, both texts lead to the same conclusion: standardize the triggers for starting KRT and, in case of uncertainty, consider a time-limited trial with agreed goals and a date for review (6.4). In pilot data, the furosemide stress test identified patients at higher risk of needing KRT, but biomarkers are not suitable as the sole basis for a decision (6.4).

Prediction, prevention, and their clinical cost

The second line of evidence concerns the difference between recognizing risk and modifying it. Machine learning models predict AKI with an area under the ROC curve of 0.85 to 0.86 approximately 27 to 39 hours before creatinine doubles (6.10). The single-center ESTOP-AKI RCT with 180 patients, however, showed that the time gained does not automatically translate into an outcome: the maximum change in creatinine over seven days did not differ ($p = 0.30$), AKI developed in 38.9% of the cohort, and with early consultation, the nephrologist's recommendations were implemented in only 41% of cases versus 68% in the control group (6.10). The author's interpretation is convincing: the model predicts risk, not a treatable mechanism, because the same score arises from hypovolemia, congestion, sepsis, drug toxicity, or obstruction (6.10).

Pharmacologic prevention has produced a positive but difficult-to-interpret result. The MERCURI-2 RCT with 784 patients administered four doses of dapagliflozin 10 mg around elective cardiac surgery and reduced AKI within 7 days from 52% to 28% (RR 0.54) (6.11). However, the incidence of AKI with placebo was almost 2.4 times the planned 22%, according to the author's calculation the urine output criterion explains approximately four fifths of the difference, and with assessment by creatinine alone, the figures would be 22% versus 27% (6.11). Mortality, length of hospital stay, and major cardio-renal events within 30 days did not change (6.11). Preserved urine output is not proof of kidney protection (6.11).

The most serious warning of the chapter is VICTORY, a double-blind phase 3 RCT in 238 patients with deep burns that administered vitamin C 50 mg/kg IV every 6 hours for 96 hours (6.6). The primary composite outcome was worse (40.8% versus 29.7%; adjusted RR 1.28; 95% CI, 0.99 to 1.65), mortality at 28 days was higher (15.0% versus 7.6%; RR 1.96; $p = 0.001$), and the trial was stopped early (6.6). Both AKI (15.0% versus 11.0%) and KRT initiation (10.8% versus 5.9%) were more frequent (6.6). The positive result of MERCURI-2 and the negative result of VICTORY lead to the same conclusion: an intervention should be judged by clinical outcomes, not by its pathophysiologic appeal (6.6, 6.11).

Prognostic context from observational data

The third line of evidence consists of cohorts that do not say what to do but what to expect. A population-based study from Ontario included 531,090 adults admitted to the ICU, a quarter of whom had CKD (6.12). The adjusted odds ratio for 90-day mortality rose from 1.14 in G3a to 2.32 in nondialysis G5, and among survivors after KRT initiation, 7.2% of patients without CKD, 50.3% in G4, and 83.8% in nondialysis G5 remained dialysis dependent (6.12). The authors point out that dialysis dependence is an outcome conditional on survival and that the lower mortality of patients on dialysis compared with nondialysis G5 reflects selection, not a protective effect of dialysis; this is a prognostic tool, not causal evidence (6.12).

A Dutch cohort of 1,408 patients with *Staphylococcus aureus* bacteremia found AKI in 35.2% of patients on flucloxacillin and in 20.2% on cefazolin, an adjusted odds ratio of 2.37 (95% CI, 1.30 to 4.55) (6.15). The observational result points in the same direction as the randomized trials SNAP (13.9% versus 19.6%) and CloCeBa (1% versus 12%), which lends the association plausibility (6.15). The time course is surprising: the median onset was 1.3 days, and 80.3% of episodes developed within seven

days, which suggests a convergence of several mechanisms, including sepsis-associated AKI; the small cefazolin group of 84 patients and confounding by indication prevent attributing the entire difference to the drug (6.15). The study of more than 3 million patients after COVID-19, with a higher risk of AKI and CKD progression than after influenza, particularly in the first six months, should be read in a similar way; these are observational data supplemented by expert opinion (6.2).

Toxicology and environmental influences

The fourth line of evidence is the most heterogeneous. A narrative review in the American Journal of Kidney Diseases classifies nephrotoxins by strength of evidence: strong evidence exists for lead, cadmium, and mercury, whereas for PFAS, bisphenol A, microplastics, and pesticides, the available data are mostly experimental or cross-sectional and do not confirm causality (6.9). A review of uranium develops this framework: with chronic exposure, tubular markers change before eGFR does, and elevated urinary uranium proves exposure, not injury; the association with CKD below the Slovak limit of 30 µg/L comes only from observational studies (6.7).

A Taiwanese case-control study is an example of a strong association without clinical application: the adjusted odds ratio for CKD was 12.45 (95% CI, 6.04 to 25.66) for MMP-2 and 3.77 for MMP-9, but the cross-sectional design does not allow the direction of the relationship to be determined, and polymorphisms of the MMP2 and MMP9 genes were not associated with CKD, which supports the notion of an epiphenomenon (6.13). A study of 174 plasma exchange procedures in 114 patients is proof of mechanism, not of benefit: in patients with a higher burden, microplastic levels decreased, whereas at the lowest values an increase was also observed, probably because of contamination from plastic components (6.5).

Clinical toxicology is more pragmatic. For fly agaric products, a series of 34 cases recorded not a single case of AKI, and the only documented report of kidney failure is a conference abstract (6.17). More important, psilocin was detected in seven of the eight analyzed products, and two products declaring fly agaric extract contained it instead of the declared substances (6.17). In the differential diagnosis, latency is decisive: white *Amanita* species cause AKI after 10 to 12 hours, and orellanine after 1 to 2 weeks (6.17). The review of ketamine-associated uropathy is based on cohorts and case reports; in a cohort of 59 selected patients, 51% had hydronephrosis, but this is not a population prevalence, and randomized trials have not confirmed the efficacy of interventions (6.16). The case report of DRESS three weeks after allopurinol, with granulomatous interstitial nephritis, is a reminder that creatinine can continue to rise even after the drug is withdrawn and that biopsy speeds up treatment (6.3). The review of ketoacidosis adds a laboratory pitfall: the urine nitroprusside test does not detect β-hydroxybutyrate, and in euglycemic DKA, blood glucose is usually below 11.1 mmol/L, particularly with SGLT2 inhibitors (6.8).

What is proven, what is probable, and what remains open

RCTs have proven that immediate initiation of KRT without an urgent indication does not improve mortality and leads to more adverse events (6.4), that high-dose vitamin C in burns increases mortality and the need for KRT (6.6), that perioperative dapagliflozin reduces AKI defined mainly by urine output without an effect on hard outcomes (6.11), and that an algorithm-triggered consultation did not change a short-term creatinine endpoint (6.10). It is probable but not proven that a more conservative dialysis strategy supports recovery of kidney function (6.1), that flucloxacillin is more nephrotoxic than cefazolin (6.15), and that COVID-19 increases kidney risk more than influenza does (6.2). Prognostic gradients by CKD stage in the ICU are robust but descriptive (6.12).

It remains open which trigger or biomarker will define the optimal window for KRT (6.4), whether the difference in MERCURI-2 persists when assessed by creatinine alone (6.11), how to turn prediction into an implemented intervention (6.10), and whether uraemia below the current limit, microplastics, or MMPs have causal significance (6.5, 6.7, 6.13). For ketamine-associated uropathy and DRESS, the evidence comes from case reports and defines vigilance, not a standard (6.3, 6.16).

Implications for practice

A Slovak hospital ward or ICU should standardize the triggers for starting KRT in writing and, when the prognosis is uncertain, set up a time-limited trial with a review date (6.4); in stable patients on KRT, it is reasonable to assess the indication for each dialysis session instead of following an automatic schedule, bearing in mind that this is a signal (6.1). Dapagliflozin should not be started to prevent perioperative AKI outside a trial, and withholding SGLT2 inhibitors before surgery, usually 3 to 4 days in advance, remains appropriate in line with the summary of product characteristics and the local protocol (6.8, 6.11). High-dose vitamin C in burns has no place outside a clinical trial (6.6). With flucloxacillin, creatinine should be checked from the first days, and cefazolin should be considered in suitable patients with MSSA, with caution in CNS infection (6.15). CKD stage should not be a reason to refuse ICU admission, but when dialysis is started in a patient with G5, honesty requires discussing the low probability of returning to treatment without dialysis (6.12).

In the outpatient clinic, a simple framework applies: confirm a rise in creatinine with a repeat sample, consider pseudo-AKI with trimethoprim or cobicistat, systematically search for prerenal, drug-related, obstructive, parenchymal, and vascular causes, and exclude obstruction by ultrasound within 24 hours (6.14). In older adults, a combination of volume depletion, diuretics, RAS blockade, and NSAIDs is common; interruption of treatment should be temporary, and fluids should not be given merely because of an elevated creatinine level (6.14). After COVID-19, monitoring of UACR and kidney function is advisable (6.2). An environmental history belongs in the evaluation of unexplained tubulopathy (6.9), and testing for uraemia and metals is justified in a Fanconi phenotype or in a person with CKD who uses well water in an at-risk area, not across the board (6.7, 6.13). In AKI after mushroom ingestion, another species or another substance should be sought, and the National Toxicological Information Center should be consulted (6.17); in an adolescent with voiding symptoms and hydronephrosis, ask about ketamine without judgment (6.16); with fever, rash, eosinophilia, and AKI 2 to 6 weeks after a new drug, think of DRESS (6.3); and confirm ketoacidosis with blood β -hydroxybutyrate (6.8).

What to watch next

In the near term, it will be worth watching whether a larger multicenter trial confirms the signal for indication-guided dialysis from 221 patients (6.1) and whether a trigger for starting KRT that is more accurate than creatinine and urine output can be validated (6.4). For MERCURI-2, an analysis based solely on creatinine criteria and verification of the data on stage 2 or 3 AKI will be decisive (6.11). After ESTOP-AKI, any system that links prediction to a prioritized prompt and to a check on its implementation deserves attention (6.10). In toxicology, it will be worth following the STOMP initiative for reproducible measurement of microplastics (6.5), a possible revision of the limits for uraemia in

drinking water (6.7), randomized comparisons of flucloxacillin and cefazolin in other populations (6.15), and long-term data on the kidney consequences of COVID-19 with current variants (6.2).

Articles in this chapter

- 6.1** A more conservative dialysis strategy in AKI may support recovery of kidney function (May 11, 2026, online, Slovak)
- 6.2** COVID-19 and kidney health: what impact the infection may have and whom to monitor (May 24, 2026, online, Slovak)
- 6.3** DRESS syndrome after allopurinol: the diagnostic trap of granulomatous interstitial nephritis with pancreatitis (June 16, 2026, online, Slovak)
- 6.4** When to start kidney replacement therapy (KRT) in acute kidney injury (AKI) (June 17, 2026, online, Slovak)
- 6.5** Can we remove microplastics from the body? What the first clinical study of therapeutic plasma exchange suggests (June 19, 2026, online, Slovak)
- 6.6** VICTORY: high-dose intravenous vitamin C does not improve outcomes in severe burns and may be harmful, with signals important for nephrology (June 22, 2026, online, Slovak)
- 6.7** Uranium and the kidneys: chemical nephrotoxicity, radiation injury, and the risks of environmental metals (July 11, 2026, online, Slovak)
- 6.8** Ketoacidosis in nephrology practice: from starvation to euglycemic diabetic ketoacidosis (July 24, 2026, online, Slovak)
- 6.9** Environmental toxins and kidney injury: what the nephrologist should specifically ask about (July 25, 2026, online, Slovak)
- 6.10** ESTOP-AKI: the algorithm recognized the risk, but early nephrology consultation did not improve outcomes (August 5, 2026, online, Slovak)
- 6.11** Dapagliflozin before cardiac surgery reduced the incidence of acute kidney injury. Changing perioperative practice is still premature (August 19, 2026, online, Slovak)
- 6.12** Severity of chronic kidney disease and prognosis of patients admitted to the intensive care unit (August 19, 2026, online, Slovak)
- 6.13** Matrix metalloproteinases, toxic metals, and chronic kidney disease: a strong association without evidence of causality (September 6, 2026, online, Slovak)
- 6.14** Sudden rise in creatinine in an older patient with hypertension: causes, diagnosis, and clinical approach (September 8, 2026, online, Slovak)
- 6.15** Flucloxacillin in *Staphylococcus aureus* bacteremia: more acute kidney injury than with cefazolin (September 9, 2026, online, Slovak)
- 6.16** Ketamine-associated uropathy in children and adolescents: diagnosis and management (September 24, 2026, online, Slovak)
- 6.17** Fly agaric in "gummies": toxicological risks and their true nephrological dimension (September 25, 2026, online, Slovak)

CHAPTER 7

Hemodialysis and extracorporeal therapies

Dose and technique of hemodialysis and hemodiafiltration, vascular access, volume management, complications, quality of life, and the organization of dialysis care. Number of articles: 37.

The chapter on hemodialysis and extracorporeal therapies summarizes 37 articles published from May to the end of September 2026, and its composition faithfully mirrors the current dialysis literature. Firm randomized evidence with a hard endpoint concerns virtually a single question, the convection dose in online hemodiafiltration (7.16, 7.27). The remainder is based on observational cohorts, small randomized trials with surrogate endpoints, reviews, technical descriptions, and qualitative studies. Registry analyses and cohorts from provider networks (7.13, 7.29, 7.31) bring large numbers, but also all the limitations of observational data, including conflicts of interest.

The main message is twofold. High-volume postdilution OL-HDF has become a treatment with a GRADE-based recommendation, and its benefit is tied to the convective volume actually delivered (7.27). At the same time, what really troubles patients, namely volume instability, cramps, itch, frailty, and vascular access complications, remains the domain of observational data and clinical judgment, and the articles repeatedly show that we do not systematically detect these problems (7.20, 7.26, 7.28, 7.35). Here, algorithms and sensors so far offer a complement to clinical assessment, not a replacement for it (7.9, 7.25, 7.32).

Convective volume as the treatment dose

The starting point is CONVINCENCE: with high-dose postdilution OL-HDF, 17.3% of patients died versus 21.9% with high-flux hemodialysis (HR 0.77), an absolute difference of approximately 4.6 percentage points over a median of 30 months (7.16). The recommendations of the Italian Society of Nephrology combine ESHOL, CONVINCENCE, and an individual participant data meta-analysis of five trials with 4,153 patients: an HR for all-cause mortality of 0.84 (95% CI, 0.74 to 0.95; moderate certainty), mortality of 23.3% versus 27.0%, and a graded relationship between convective volume and the risk of death (7.27). The caveats are stated honestly: CONVINCENCE was open-label and enrolled only patients able to achieve at least 23 L, so the absence of heterogeneity does not prove that no unsuitable subgroup exists; a benefit of low-volume HDF or of the predilution mode, and an effect on quality of life, have not been demonstrated (7.16, 7.27). The correct interpretation is therefore not about the modality but about the dose, which is limited by the filtration fraction, blood flow, treatment time, and above all the vascular access (7.27).

The EuCliD registry cohort of 18,515 incident patients extends this logic to new patients, for whom randomized evidence is lacking: after IPTW weighting, the HR for all-cause mortality was 0.80 (95% CI, 0.75 to 0.86) at a mean convective volume of 24.9 L (7.31). In an analysis approximating the intention-to-treat principle, however, the effect shrank to an HR of 0.88, the groups differed before weighting in the proportion of fistulas (40% versus 28%), and the data come from a single provider's

network (7.31). The stricter the method, the smaller the effect, which is the signature of residual confounding, not proof of causality.

MCO dialyzers improve the laboratory clearance of larger middle molecules, but they have not shown a reduction in mortality or hospitalizations and may slightly lower albumin (7.16). At the opposite end of the dosing scale stands incremental hemodialysis: observational cohorts suggested slower loss of residual function, but the Fernández Lucas RCT of 150 patients did not confirm this (difference in slope, $-0.09 \text{ mL/min/1.73 m}^2$ per month; $p = 0.230$), and meta-analyses showed no difference in mortality (7.36). The Obi cohort observed a higher risk of death at a KRU of $3 \text{ mL/min/1.73 m}^2$ or less (HR 1.61; 95% CI, 1.07 to 2.44), a warning that a twice-weekly regimen requires periodic measurement of residual function (7.36). The same caution applies to reducing Qd for environmental reasons, which in the pilot study was not accompanied by Kt/V data (7.12).

Dry weight and intradialytic stability

Volume management has no gold standard. A review describes dry weight as a dynamic target in which the clinical picture and blood pressure trends form the foundation, complemented by bioimpedance, the relative blood volume curve, and lung ultrasound (7.8). A Korean retrospective study of 1,672 patients shows the limits of algorithms: machine learning models had an accuracy below 40% in the entire cohort and failed precisely in extracellular overhydration and malnutrition, that is, in the sickest patients (7.9). A French observational study of 56 high-risk patients adds that relative blood volume alone predicts hypotension poorly in new patients (AUC 0.62), and that the combined model, with an AUC of 0.85 to 0.86, has only internal validation (7.32).

Interventional data are limited. A crossover RCT of low-sodium dialysate in 23 analyzed patients did not demonstrate a significant overall decrease in tissue sodium; a higher ultrafiltration rate was associated with lower sodium in both skin and muscle (7.3). A twelve-week RCT of a traffic-light diary in 80 patients increased adherence to recommendations from 75% to 95%, but for IDWG, blood pressure, and hypotension the abstract does not report effect sizes, and the intervention was unblinded (7.15).

According to a meta-analysis of 94 observational studies, 55% of patients have cramps and 33% have intradialytic cramps, but the wide prediction intervals (20% to 86%) mean that the prevalence in one's own unit cannot be inferred from the meta-analysis (7.20). First-line management is adjustment of dry weight, ultrafiltration rate, and electrolytes, including magnesium, not drugs (7.20). A prospective study of 48 patients starting hemodialysis surprisingly found dialysis disequilibrium syndrome in 22.9%, and the only independent risk factor was intradialytic hypertension (OR 4.43; 95% CI, 1.38 to 18.15), not predialysis urea (7.11). Cardiac arrest in the dialysis unit requires a written protocol: blood from the circuit is returned with a fluid bolus without delaying compressions, and the lock solution is aspirated before a tunneled catheter is used (7.24).

Vascular access, catheters, and safety

A Korean nationwide cohort of 24,713 patients showed that an AVF created less than one month before dialysis carried an OR of 14.03 for catheter use and an HR of 1.67 for mortality, that an interval of 1 to 3 months was also unfavorable (OR 1.85), whereas for AVG the risk was concentrated in the last month (7.14). A late procedure may be a marker of late referral, but the conclusion supports a lead time of at least approximately three months (7.14). A Portuguese cohort of 40 patients for whom the team did not recommend arteriovenous access covers the other end of the spectrum: 90% started hemodialysis unplanned, mortality reached 42.5%, and only one of 11 patients chose PD (7.34). A neg-

ative recommendation is a prognostic marker that should trigger assessment of PD, conservative management, and palliative care (7.34).

For catheters, the strongest evidence came from a pragmatic cluster-randomized trial of 1,468 ICU patients: a 4% tetrasodium EDTA lock reduced the combined incidence of infection and occlusion from 19.9 to 13.1 events per 1,000 catheter-days (incidence rate ratio, 0.68; 95% CI, 0.47 to 0.96), but the effect was driven by fewer occlusions, not fewer infections, and the population was critically ill (7.4). The SWAM technique is a technical description without data on long-term patency (7.10). An Irish cohort with 522 bloodstream infection episodes shows a fall in the proportion of staphylococci from 60% to 34.8% without an absolute increase in gram-negative pathogens; the catheter remains the main modifiable risk, and the local antibiogram guides empirical treatment (7.23).

Symptoms, frailty, and weak signals

Patients do not report symptoms, and staff do not ask. A Turkish study of 294 patients found moderate to severe itch in 16.3%, and of 212 treated patients, most often with antihistamines, 47.2% reported no effect (7.28). A review of neuroimmune mechanisms explains the failure of antihistamines and points out that, in the pooled KALM analysis, difelikefalin reduced itch by at least 3 points in 51.1% of patients versus 35.2% with placebo; the four-node model is a hypothesis-generating framework (7.18). Olfactory dysfunction was present in 91% of patients on hemodialysis versus 44% of controls and was associated with cognition and albumin, not with Kt/V (7.35).

Frailty is ceasing to be tied to age. In Tbilisi, among 50 patients aged 64 years or younger, 20% were frail and 44% prefrail, but with only ten frail patients the regression findings are unstable (7.26). A French cohort of 792 patients validated the nurse-assessed CFS: each point carried a subdistribution hazard ratio for death of 1.33 (95% CI, 1.19 to 1.49), and at 24 months 45.87% of frail patients had died versus 9.78% of fit patients; agreement between the nurse raters was only moderate (kappa 0.60) (7.33). CT measures of myosteatosis in 114 patients do not replace the dynamometer (7.22). A review of palliative care points out that in patients aged 80 years and older, a survival difference between dialysis and conservative management was not confirmed (2.1 versus 1.4 years; $p = 0.08$), and it offers low-threshold tools, including deprescribing and a plan for dialysis withdrawal (7.21).

Other signals do not change treatment. Blood group A was associated with lower mortality (HR 0.780), but the C-statistic rose only from 0.749 to 0.751 (7.17). The Fresenius cohort of 2,468 pairs showed lower ESA use with GLP-1 receptor agonists without a difference in hemoglobin, in a per-protocol analysis without an active comparator; this is not a reason to start such treatment for anemia (7.29). A post hoc analysis of tenapanor found a borderline significant association with TRACP-5b, but the drug is not available in the EU (7.30). In the pharmacotherapy of obesity, approximately 50% of topiramate is removed by dialysis, so it is given after dialysis, and data on tirzepatide in patients on dialysis are limited (7.6). Intravenous L-carnitine attenuates but does not abolish TMAO formation, and a benefit of lowering TMAO for hard outcomes has not been demonstrated (7.7). Among wearable sensors, only a patch hemoglobin sensor has validation data, with limits of agreement of approximately ± 2 g/dL, which makes it suitable for trends but not for adjusting the ESA dose (7.25).

The system, access to care, and shared decision-making

The ERA Registry describes an increase in KRT incidence in Central and Eastern Europe of 1.5% per year but an increase in prevalence of 5.1% per year; in Slovakia, the incidence of dialysis initiation was stable, hemodialysis prevalence rose by 2.0% per year, and PD prevalence fell by 3.3% per year, while

the overall totals are not comparable because transplant recipients are missing (7.13). A comparison of the ERA Registry and USRDS shows a KRT incidence of 388.7 versus 146.2 per million population, with a larger difference in women, and an opposite relationship between sex and mortality in Europe and the US (7.5). A US analysis links Medicaid expansion to an approximately 3.9% higher proportion of optimal starts of treatment, mainly owing to home dialysis, again as an association without causality (7.2).

Access to care is not only the distance to the dialysis center but also access to education, vascular access, and home modalities, and preparation begins months to years before the first dialysis (7.1). A Japanese study of 474 patients showed that 24.2% were unwilling to talk about their values, and these patients rated their involvement 11.39 points lower on CollaboRATE; trust in the nephrologist was protective, whereas depressive symptoms increased unwillingness (7.19). A qualitative study of older patients and care partners adds that education should be spread over time and that clinicians should also ask about the burden on the patient's loved one (7.37).

What is proven, what is probable, and what remains open

What is proven is the survival benefit of high-volume postdilution OL-HDF in prevalent patients able to achieve approximately 23 L per session, with moderate certainty (7.27), and the benefit of an EDTA lock for catheter patency in intensive care, but not for infections (7.4). What is probable, although based on observational data, is that the benefit of HDF also applies to incident patients (7.31), that late AVF creation worsens outcomes (7.14), that frailty measured with a simple scale predicts mortality (7.33), and that cramps and itch are considerably more common than the documentation shows (7.20, 7.28).

It remains open whether incremental hemodialysis preserves residual function (7.36), whether algorithms for dry weight and hypotension will improve clinical outcomes after external validation (7.9, 7.32), whether low-sodium dialysate is clinically relevant (7.3), whether mechanism-based treatment of pruritus works (7.18), and whether the associations involving blood group, GLP-1 receptor agonists, and TMAO reflect causal effects (7.7, 7.17, 7.29). Modeled scenarios for reducing the carbon footprint (7.12) and wearable dialysis (7.25) remain hypotheses.

Implications for practice

With OL-HDF, the delivered, not the prescribed, convective volume should be documented at every session together with blood flow and treatment time, and vascular access should be managed as part of the dose; an MCO dialyzer is an alternative when a catheter is used or blood flows are low (7.16, 7.27, 7.31). An AVF should be created at least approximately three months in advance, and a negative surgical recommendation should prompt assessment of PD and a palliative plan, not waiting for an urgent catheter (7.14, 7.34). Dry weight should be reassessed multimodally and in small steps, with bioimpedance and relative blood volume used as adjuncts, especially in malnourished patients (7.8, 7.9, 7.32).

The fixed structure of the monthly review should include questions about cramps, itch, and appetite; frailty screening without an age cutoff at entry, once a year, and after hospitalization; and, in frail patients, a goals-of-care conversation with deprescribing (7.20, 7.21, 7.26, 7.28, 7.33). For infections, the local antibiogram and minimizing catheter use are decisive (7.23); cardiac arrest calls for a written protocol with rehearsed catheter handling (7.24). Before decisions about modality, it is reasonable to screen for depressive symptoms, to spread the conversations over several visits, and, with the patient's consent, to involve the care partner (7.1, 7.19, 7.37).

What to watch next

Key items to watch are whether a randomized trial of HDF in incident patients or an independent replication of the EuCliD data outside a single provider's network will appear (7.31), the results of the VA IncHVets trial of incremental hemodialysis (7.36), external validation of relative blood volume and machine learning models against clinical endpoints (7.9, 7.32), the transfer of the EDTA lock data from intensive care to maintenance hemodialysis (7.4), external validation of the CFS threshold of 6 (7.33), and comparative studies of the SWAM technique with patency follow-up of up to 12 months (7.10). In Slovakia, the top priority is a complete KRT registry, including preemptive transplantation and conservative care, without which the declining PD prevalence can be neither interpreted nor influenced (7.13).

Articles in this chapter

- 7.1** Trends in dialysis care: access, home modalities, and the patient at the center (May 13, 2026, online, Slovak)
- 7.2** Medicaid expansion and “optimal starts” in incident kidney failure: results of an observational cohort analysis (2008 to 2019) (May 26, 2026, online, Slovak)
- 7.3** Low-sodium dialysate and tissue sodium: what a randomized crossover study showed (May 26, 2026, online, Slovak)
- 7.4** EDTA lock solutions: can they improve central venous catheter patency? (June 10, 2026, online, Slovak)
- 7.5** Sex differences in dialysis and transplantation: Europe vs. the US according to the ERA Registry and USRDS (June 19, 2026, online, Slovak)
- 7.6** Pharmacological treatment of obesity in patients with advanced CKD and on dialysis: practical dosing, dialysis-specific considerations, and safety (June 20, 2026, online, Slovak)
- 7.7** Does injectable L-carnitine prevent TMAO formation? What the data really say (July 4, 2026, online, Slovak)
- 7.8** Determining dry weight (EDW) in hemodialysis: clinical assessment, BCM, BVM, and POCUS (July 6, 2026, online, Slovak)
- 7.9** Artificial intelligence in determining dry weight in hemodialysis patients (July 6, 2026, online, Slovak)
- 7.10** The SWAM technique for hemodialysis access thrombosis: a new option for mechanical thrombectomy (July 9, 2026, online, Slovak)
- 7.11** Dialysis disequilibrium syndrome: an underdiagnosed complication at the start of hemodialysis (July 9, 2026, online, Slovak)
- 7.12** The carbon footprint of hemodialysis: measurement, practical measures, and the limits of emission reduction (July 13, 2026, online, Slovak)
- 7.13** Kidney replacement therapy in Central and Eastern Europe: incidence and prevalence trends in the ERA Registry (July 13, 2026, online, Slovak)
- 7.14** Timing of vascular access before hemodialysis: an AVF needs more lead time than an AVG (July 15, 2026, online, Slovak)
- 7.15** A “traffic light” diary in hemodialysis: better short-term fluid management in a small RCT (July 15, 2026, online, Slovak)
- 7.16** Beyond diffusion: online hemodiafiltration and MCO dialyzers in the treatment of kidney failure (July 25, 2026, online, Slovak)
- 7.17** Blood group A and lower mortality on hemodialysis: an intriguing association with no implications for current treatment (August 2, 2026, online, Slovak)
- 7.18** The neuroimmune architecture of uremic pruritus: a four-node model and its therapeutic implications (August 5, 2026, online, Slovak)
- 7.19** When the patient does not want to talk about their values: a hidden barrier to shared decision-making about dialysis (August 5, 2026, online, Slovak)
- 7.20** Skeletal muscle cramps on dialysis: a global prevalence of 55% and what to do about it in practice (August 5, 2026, online, Slovak)
- 7.21** Palliative care in routine nephrology: low-threshold tools for frail and older patients (August 5, 2026, online, Slovak)
- 7.22** Myosteatosis in hemodialysis: CT measures of muscle quality are promising but do not yet replace functional assessment (August 15, 2026, online, Slovak)
- 7.23** Bloodstream infections in hemodialysis: incidence is falling, but the microbiological spectrum may be shifting (August 15, 2026, online, Slovak)

- 7.24** Cardiac arrest during hemodialysis: the extracorporeal circuit, the catheter lock, and the silent burden on dialysis nurses (August 23, 2026, online, Slovak)
- 7.25** Wearable sensors in nephrology and dialysis: what already has validation data and what is still in development (August 23, 2026, online, Slovak)
- 7.26** Frailty in dialysis patients younger than 65 years: what a cross-sectional study from Tbilisi adds (August 23, 2026, online, Slovak)
- 7.27** Online hemodiafiltration as a dose-dependent therapy: recommendations of the Italian Society of Nephrology (August 23, 2026, online, Slovak)
- 7.28** Itch (pruritus) on dialysis: why reported prevalence varies twofold and what patients will not tell you unprompted (August 24, 2026, online, Slovak)
- 7.29** GLP-1 receptor agonists, inflammation, and anemia in patients on hemodialysis: what real-world evidence shows and how to translate it into practice (September 3, 2026, online, Slovak)
- 7.30** Which dialysis patients need a higher dose of tenapanor? The role of bone resorption and intestinal transit (September 6, 2026, online, Slovak)
- 7.31** High-volume hemodiafiltration and mortality in incident dialysis patients: a convincing association, not proof of causality (September 6, 2026, online, Slovak)
- 7.32** Continuous relative blood volume monitoring may predict intradialytic hypotension (September 6, 2026, online, Slovak)
- 7.33** Nurse-assessed frailty predicts mortality on dialysis: a multicenter validation of the Clinical Frailty Scale (September 9, 2026, online, Slovak)
- 7.34** When arteriovenous vascular access is not possible: a negative surgical recommendation as a prognostic marker (September 9, 2026, online, Slovak)
- 7.35** Olfactory dysfunction in hemodialysis patients: a common but overlooked marker of overall morbidity (September 9, 2026, online, Slovak)
- 7.36** Incremental hemodialysis and residual kidney function: what the evidence shows (September 24, 2026, online, Slovak)
- 7.37** Living with hemodialysis: advice from older patients and their care partners (September 27, 2026, online, Slovak)

CHAPTER 8

Peritoneal and home dialysis

Peritoneal dialysis and home therapies: patient selection, prevention and treatment of infections, solute transport, and program sustainability. Number of articles: 13.

The chapter on peritoneal and home dialysis summarizes thirteen articles from June to September 2026, and its evidence base is typical of the field: retrospective cohorts and registry analyses predominate (8.5, 8.7, 8.12), together with expert statements and consensus documents (8.1, 8.3, 8.9), methodological commentaries (8.8, 8.10), one mechanistic modeling study (8.6), a questionnaire survey (8.4), an institutional report (8.2), a case report (8.13), and a critical analysis of a media story (8.11). Randomized data appear only indirectly, in the context of home hemodialysis (8.9).

Nevertheless, the main message is consistent across these heterogeneous sources: the success of home treatment depends more on a functioning system around the patient than on the choice of modality or solution itself. The Barcelona program is growing thanks to shared infrastructure and nursing continuity (8.2), the KDIGO consensus builds the development of home HD on funding, regulation, and safety backup (8.9), the Polish and Albertan studies locate the main loss of candidates between the recognition of suitability and the start of treatment (8.7, 8.8), and the Catalan registry shows that the risky moment is not the initial choice but an unplanned transfer between modalities (8.12).

The system matters more than the modality

Hospital Clínic Barcelona announced 100 patients on home dialysis, 80 on PD and 20 on home HD; the number of patients on PD has doubled from 40 to 80 since December 2023, and 61 people started home treatment in 2025 (8.2). The authors attribute the growth to shared facilities for training and monitoring of both modalities and to a nursing team that provides training both in the center and at home, so that support forms a single continuum (8.2). However, this is a program description without clinical outcomes, that is, inspiration for organizing work, not evidence of efficacy (8.2).

The KDIGO consensus report from Kawagoe is an implementation framework for Hong Kong, Japan, South Korea, and Taiwan, not a clinical guideline (8.9). The three proposed phases, namely reimbursement and regulatory conditions, a pilot program with a multidisciplinary team, and expansion after evaluation of adverse events, retention, and costs, are transferable to Europe as well; for smaller countries, the author proposes regional reference centers with centralized training, round-the-clock support, and an outcomes registry (8.9).

The flip side is the story of a Chinese patient who built his own hemodialysis setup and, according to a media report from January 2013, survived on it for thirteen years (8.11). Approximately 120 liters of dialysate pass the membrane per session, and the ISO 23500 standards require less than 100 CFU/mL and less than 0.25 EU/mL for dialysis water, which an improvised setup without air and blood leak detectors, conductivity measurement, and volume-controlled ultrafiltration cannot ensure; the story also lacks any data on microbiology and infections (8.11). The message is the same as in the KDIGO

consensus, only from the opposite side: home treatment is safe only within a supervised program (8.9, 8.11).

Intensive home hemodialysis and the strength of evidence

Article 8.9 is the only one that works systematically with randomized data. FHN Daily, with 245 patients receiving in-center dialysis six times a week, reduced the risk of death or increase in left ventricular mass (HR 0.61; 95% CI, 0.46 to 0.82) but increased the number of vascular access interventions (HR 1.71) (8.9). FHN Nocturnal, with 87 patients on nocturnal home HD, did not improve the primary outcomes, and in long-term follow-up with a median of 3.7 years there were 14 versus 5 deaths (HR 3.88; 95% CI, 1.27 to 11.79), an uncertain but serious signal given the small sample (8.9). ACTIVE, with 200 patients, did not change quality of life as measured by EQ-5D (difference, 0.04; $p = 0.29$), although it improved serum phosphate, serum potassium, and hemoglobin (8.9).

A registry comparison of 1,873 patients on daily home HD with 9,365 matched patients showed an HR for death of 0.87, but these are observational data burdened by selection bias, and KDIGO 2023 considers outcomes across modalities to be largely similar (8.9). The contradiction between favorable registries and neutral or cautionary randomized trials is the chapter's main evidentiary tension; a causal effect of home HD on survival has not been proven, and its value lies in autonomy and laboratory parameters, which are goals other than mortality (8.9).

Candidate selection and missed opportunities

A retrospective study from Warsaw evaluated 118 preselected patients with CKD G4 to G5 after an individual 90- to 120-minute meeting; 62.7% chose PD, and 51.7% of the entire sample started it (8.7). In univariable models, older age (OR 0.967 per year), diabetes (OR 0.353), higher BMI (OR 0.884), and the need for assistance (OR 0.264) were associated with lower odds of choosing PD, but these are associations without adjustment for confounders (8.7). The most telling finding is that of 19 patients requiring assistance, only six chose PD, because help could come only from family (8.7).

A commentary on the Albertan model of Yang et al., based on 598 patients, shows the other side of the same problem (8.8). The model achieved a sensitivity of 91.3%, an accuracy of 0.68, and an area under the ROC curve of 0.69 to 0.71, which represents only moderate discrimination with necessarily low specificity; external validation, calibration, and decision curve analysis are all lacking, and the model predicts the physician's decision, not the patient's outcome, so it may formalize existing biases (8.8). Crucially, of 391 patients deemed suitable, only 22.3% started PD within six months, so the main loss occurs between the decision and treatment, because of waiting for a catheter, limited training capacity, or lack of support (8.8). Both studies concur in recommending that unpreparedness not be confused with unsuitability, that assisted PD be considered, and that concordance between the preferred and the initiated modality be monitored instead of a mandatory PD-first approach (8.7, 8.8).

The Catalan registry adds data on the oldest patients to this axis. A cohort of 4,205 patients aged 80 years and older found no difference in five-year survival by initial modality after adjustment (PD without transfer, HR 1.19; 95% CI, 0.95 to 1.48), but did find one for transfer from PD to HD (HR 1.46; 95% CI, 1.02 to 2.08) (8.12). The apparently worse survival with PD in the matched analysis ($p = 0.008$) disappeared once transfer was modeled as a time-dependent variable (8.12). The reasons for transfer were loss of membrane function (30.4%), intercurrent illness (30.4%), peritoneal infection (23.2%), and patient preference (16.1%), so transfer is more likely a marker of deterioration than its cause; the limitations are that only 63 patients transferred and that data on frailty and nutrition are missing (8.12).

Solutions, acid-base balance, and muscle assessment

A Taiwanese cohort of 1,800 patients on PD with heart failure associated icodextrin with lower all-cause mortality (3.2 versus 14.0 events per 100 patient-years; adjusted HR 0.16; 95% CI, 0.13 to 0.20) and fewer MACE (HR 0.68), whereas hospitalizations for heart failure (HR 1.06) and myocardial infarction (HR 0.94) did not differ (8.5). Such an effect size is implausible for a dialysis solution, and icodextrin was also associated with a lower risk of transfer to HD (HR 0.52) and of transplantation (HR 0.22), which points to patient selection and unmeasured factors; volume status was not measured (8.5). This is an observational association, not proof of causality (8.5).

A mechanistic study combined the three-pore transport model with a whole-body acid-base model and calibrated it on data from six patients during 4-hour dwells with Physioneal and Dianeal PD4 solutions (8.6). The model reproduced the data with an RMSE of 6.6% and 6.4% and estimated a transfer of 65.7 ± 10.8 mmol of bicarbonate into the dialysate with PD4 versus 23.1 ± 6.8 mmol with B/L, without accounting for reabsorption, and ventilation was derived, not measured (8.6). The study demonstrates neither superiority nor equivalence of either solution and, without external validation, remains a physiologically interesting hypothesis (8.6).

A review of the Hong Kong studies on the modified creatinine index shows a typical pattern: at a single time point, the index correlated with muscle mass ($r = 0.684$ with creatinine kinetics, $r = 0.641$ with bioimpedance) and predicted 12-month mortality, but in serial follow-up of 351 patients, the limits of agreement exceeded 60 percentage points, and the change in the index was not associated with survival, hospitalizations, or peritonitis (8.10). The index is a screening and short-term prognostic marker, not a diagnostic test for sarcopenia or a tool for monitoring an intervention (8.10).

Infections as a team and diagnostic problem

The expert overview from the ASN podcast presents no new data but formulates a useful framework: gaps arise from inconsistent practices in education, technique checks, and infection documentation, and every episode treated without root cause analysis increases the risk of repeating the same error; quantified effects of individual measures are lacking (8.1). In relapsing peritonitis, defined by the ISPD as recurrence within 4 weeks with the same organism, taurolidine as a lock solution may be an adjunct within a narrow window for catheter salvage when biofilm is suspected, but the evidence comes only from case reports and small series, and it must never delay catheter removal in refractory or fungal peritonitis, sepsis, or tunnel involvement (8.3).

A Slovak case report from Košice and Martin draws attention to *Mycobacterium fortuitum* in refractory culture-negative peritonitis: the median diagnostic delay is six to thirty days, and complete cure was achieved in only 14.8% of episodes in an observational series (8.13). When cultures remain negative after three to five days, the ISPD recommends adding fungal and mycobacterial cultures with seven-day incubation and treating NTM peritonitis with antibiotics and simultaneous catheter removal (grade 2D), without macrolide monotherapy; the conclusions rest on the ISPD recommendations, not on the details of the case (8.13). A green nephrology questionnaire study of 90 patients adds that 21.3% also recycled solution bags, which are medical waste, and that 84.4% preferred in-person visits to telemedicine (8.4).

What is proven, what is probable, and what remains open

Little is proven at the RCT level: more frequent in-center dialysis improved the composite outcome in FHN Daily at the cost of more access interventions, whereas nocturnal home HD and extended

dialysis in ACTIVE did not meet their primary endpoints (8.9). What is probable, although based on observational data, is that the initial modality does not determine survival in patients older than 80 years and that an unplanned transfer marks a period of risk (8.12), that systematic education increases the proportion of patients choosing PD (8.7), that the modified creatinine index has prognostic value at a single time point (8.10), and that organizational pillars determine program growth (8.2, 8.9).

It remains open whether icodextrin reduces mortality in heart failure, because an HR of 0.16 is evidently a product of selection (8.5), whether taurolidine has a place beyond the level of case reports (8.3), whether the integrated acid-base model will withstand external validation (8.6), whether the predictive suitability model offers incremental value over clinical judgment (8.8), and whether the mortality signal from FHN Nocturnal reflects a true risk or the small sample (8.9).

Implications for practice

Modality education should begin as patients approach CKD G4, should offer transplantation, PD, in-center and home HD, and conservative care, and should track how many patients actually start their preferred modality (8.7, 8.8). Assisted PD is a legitimate model, and its absence is probably the largest single source of lost candidates (8.7). Age is not a reason to exclude PD, but when starting it, a plan for possible technique failure, including vascular access, should be made in advance (8.12). Home HD should be developed only within a supervised program, which in a smaller country should first take the form of a regional reference center (8.9, 8.11).

Icodextrin should be considered individually in cases of insufficient ultrafiltration during the long dwell or in volume overload, with a glucose-specific glucometer, because maltose can mask hypoglycemia for up to two weeks after discontinuation (8.5). Every peritonitis episode should be audited (8.1); in relapse, the microbiology should be compared and a decision about the catheter made promptly (8.3); and with repeatedly negative cultures, microbiological testing should be extended with prolonged incubation (8.13). When sarcopenia is suspected, strength should be measured directly with handgrip or a chair stand test (8.10), the patient's financial and transport burden should be actively explored (8.11), and education on waste sorting should become part of routine care (8.4).

What to watch next

In the near term, it is worth watching whether an external validation of the Albertan model with calibration appears and whether anyone measures what the model currently does not, namely technique failure, peritonitis, and quality of life (8.8). For icodextrin, the observational signal should lead to a target trial emulation with measured volume status or to a randomized comparison; otherwise, the HR of 0.16 will remain a curiosity (8.5). For home HD, it will be important to see how implementation programs following the KDIGO framework evaluate adverse events, retention, and costs, and whether further cohorts confirm or refute the signal from FHN Nocturnal (8.9). In the Slovak setting, it makes sense to introduce a local peritonitis audit with extended culture (8.1, 8.13) and to record concordance between the preferred and the initiated modality, so that one's own loss of candidates can be quantified (8.7, 8.8).

Articles in this chapter

- 8.1** Core interventions in peritoneal dialysis: infection prevention as a shared team responsibility (June 9, 2026, online, Slovak)

- 8.2** One hundred patients on home dialysis: a model of training and home treatment monitoring (June 16, 2026, online, Slovak)
- 8.3** Adjuvant tauroldine for relapsing peritonitis in peritoneal dialysis: catheter salvage or a dead end? (July 5, 2026, online, Slovak)
- 8.4** A sustainable future for peritoneal dialysis: patients, recycling, and green nephrology (July 5, 2026, online, Slovak)
- 8.5** Icodextrin in peritoneal dialysis and heart failure: how to read the new observational data (July 13, 2026, online, Slovak)
- 8.6** Buffer transport and acidosis correction in peritoneal dialysis: the integrated 3PM/ABE model (July 13, 2026, online, Slovak)
- 8.7** Predialysis education and the choice of peritoneal dialysis: what a Polish cohort showed (July 15, 2026, online, Slovak)
- 8.8** Predicting suitability for peritoneal dialysis: a model that needs validation and clear incremental value (August 5, 2026, online, Slovak)
- 8.9** Home hemodialysis: greater patient autonomy is not enough without a functioning system (August 6, 2026, online, Slovak)
- 8.10** Sarcopenia in peritoneal dialysis: why the modified creatinine index is not enough (August 15, 2026, online, Slovak)
- 8.11** Improvised home hemodialysis: why one person's survival is not evidence of safety (August 23, 2026, online, Slovak)
- 8.12** Dialysis modality and survival in patients aged 80 years and older: the course of treatment matters more than the initial choice (September 9, 2026, online, Slovak)
- 8.13** Refractory culture-negative peritonitis in peritoneal dialysis: think of *Mycobacterium fortuitum* (September 9, 2026, online, Slovak)

CHAPTER 9

Kidney transplantation

Living donation, immunology, long-term outcomes, and new directions in kidney transplantation, from biomarkers of immunosuppression to xenotransplantation. Number of articles: 14.

The chapter on kidney transplantation summarizes fourteen articles from June to September 2026, and its common denominator is an effort to separate what genuinely advances transplant medicine from what so far only holds promise. Thematically, it ranges from the timing of transplantation through long-term care of the recipient and protection of the living donor to experimental approaches such as cell therapy, xenotransplantation, and desensitization with bispecific antibodies. Only biomarker-guided tacrolimus dosing (9.1), allopurinol in donors (9.11), and selected organ preservation methods (9.14) have randomized support. The rest consists of observational registries and cohorts, a meta-analysis of observational studies, qualitative research, curriculum reviews, expert commentaries, and single case reports.

The main message is twofold. The greatest clinical benefit comes not from technological novelties but from organizational discipline: timely education and referral of the candidate, prevention of sensitization, and systematic lifelong follow-up of the recipient (9.12, 9.13, 9.5). At the same time, the chapter shows how wide the gap is between feasibility and proven benefit: a functioning xenograft, a decline in anti-HLA antibodies in a single patient, or a rise in circulating regulatory T cells is not evidence of clinical benefit (9.8, 9.10, 9.9). Most of today's decisions can rest only on observational data, which is precisely why their interpretation should be worded cautiously.

Timing of transplantation and preparation of the candidate

The strongest clinical thread of the chapter is the importance of time spent on dialysis. A review based on the work of Prof. Reischig states that the duration of dialysis before transplantation is a risk factor for mortality and graft failure with both living and deceased donors, and that in the French registry of more than 20,000 patients, the difference was significant even for dialysis shorter than 6 months; 10-year graft survival after preemptive transplantation reached 80% versus 61% (9.12). KDIGO recommends preemptive transplantation with grade 1A at an eGFR below 10 mL/min/1.73 m² (9.12). However, the article honestly points out that randomization is not ethically possible, that after correction for lead-time bias, an Australian analysis found no significant difference compared with very short dialysis, and that patients receiving preemptive transplants are younger and healthier (9.12). The grade 1A recommendation thus rests on registries and two meta-analyses, and part of the benefit is probably explained by selection.

Nevertheless, the practical implication is unequivocal. Transplantation at an eGFR above 10 to 20 mL/min/1.73 m² brings no additional benefit, education should begin at an eGFR below 30, and the choice of kidney replacement therapy modality should be made at an eGFR below 20 mL/min/1.73 m² (9.12). The proportion of preemptive procedures is low: in the US, approximately 9% of deceased donor and 33% of living donor transplants, and in Europe, 19% of all transplants, whereas University Hospital Plzeň (FN Plzeň) achieves more than 25% through education (9.12). The Plzeň example is the most relevant for Slovak centers because it shows that the organization of preparation, not organ availability,

is decisive; in a small randomized trial, enhanced nurse-led education increased the number of pre-emptive transplants (9.12).

The same logic applies to retransplantation. A Canadian qualitative study of 23 patients, 23 caregivers, and 11 transplant nephrologists identified four main barriers: sensitization, comorbidities, a history of nonadherence, and delayed referral with poor coordination (9.13). A purposive sampling design does not establish the prevalence of the barriers or their impact (9.13), but it names a familiar problem: a failing graft is often addressed only after the patient returns to dialysis. Nor should obesity be an automatic barrier, because with a BMI up to 40 kg/m², transplantation reduces mortality by more than 60% compared with remaining on the waiting list, and KDIGO 2020 does not recommend excluding patients on the basis of obesity alone; at a BMI of 40 kg/m² or higher, the risk of delayed graft function, rejection, and wound complications rises, but the survival benefit persists (9.4). The review of standards and innovations adds that age should not be an automatic contraindication either (9.14).

Individualization of immunosuppression

The randomized phase 2 TTVguideIT trial from 13 European centers tested tacrolimus dosing guided by Torque teno virus (TTV) load in 260 stable low-risk recipients; the primary composite endpoint occurred in 35% versus 38%, meeting noninferiority, and rejection on protocol biopsies at 12 months was comparable (9.1). This is the methodologically cleanest result of the chapter, but although patients in the TTV group had lower tacrolimus levels and doses, a statistically significant reduction in infections was not demonstrated (9.1). The trial establishes the safety of lower dosing in the first year in low-risk patients, not clinical benefit (9.1).

The observational counterpart is the Irish national cohort of 428 transplants performed from 1970 to 1983. Thirty-three grafts (7.7%) survived forty years or more, and a living donor was associated with better ultra-long-term graft survival (OR 3.51; 95% CI, 1.17 to 11.1; $p = 0.027$) (9.5). The wide confidence interval and the historical context with different immunosuppression limit transferability to current patients (9.5), but the cohort supports the principle that TTVguideIT attempts to formalize: graft function was excellent in patients on low, individualized immunosuppression with regular risk reassessment (9.5).

Regarding the choice of immunosuppressant, belatacept in the seven-year BENEFIT follow-up yielded a higher eGFR and a lower risk of death or graft loss, but it was compared with cyclosporine and is indicated only for EBV-seropositive recipients; tegoprobart remains experimental, because the phase 2 BESTOW trial in 127 recipients ended in September 2025 without published results (9.14). The vision of tolerance through regulatory T cells remains at the level of small phase 1 and 1/2 trials, mostly nonrandomized and with short follow-up; they have not demonstrated complete withdrawal of immunosuppression, and the authors of the review have commercial ties to Treg therapies (9.9). A realistic goal is a safe dose reduction with stable graft function and histology, which must be confirmed by randomized trials (9.9).

The price of immunosuppression: tumors, infections, and bone

The Core Curriculum 2026 states that cancer risk is approximately 2 to 4 times higher than in the general population and that tumors are the second most common cause of death; skin tumors are the most common malignancy, and PTLN is a risk mainly in EBV-seronegative recipients (9.3). The Irish cohort gives this risk a concrete form: nonmelanoma skin cancers affected 22 of 33 ultra-long-term survivors, approximately 67% (9.5). The review recommends photoprotection (SPF 30+), an annual

full-body dermatologic examination, and, in cases of recurrence, reduction of immunosuppression or conversion to an mTOR inhibitor; in PTLN, the first step is minimization of immunosuppression, and the benefit of mTOR inhibitors has not been clinically documented (9.3).

Infection prevention rests on vaccination, and here the sources diverge. For the posttransplant period, the vaccine review lists influenza vaccination from 4 weeks and pneumococcal and RSV vaccines from 6 months, always without live vaccines (9.2), whereas the Core Curriculum recommends waiting at least 3 months (9.3). Both sources agree that vaccination should ideally take place before transplantation (9.2, 9.3). Differences in age thresholds for herpes zoster vaccination (ACIP from 50, RKI from 60, NHS from 65 years) show that an outpatient clinic needs its own locally adapted protocol, not a mechanically adopted table (9.2).

A meta-analysis of 23 observational studies with 26,266 recipients links persistent posttransplant hyperparathyroidism to all-cause mortality (HR 1.74) and graft failure (HR 2.16), with a gradient by serum calcium: HR 1.66 for the normocalcemic form and 2.67 for the hypercalcemic form; pretransplant hyperparathyroidism was not associated with mortality (HR 0.80) (9.6). This is a consistent association, not proof of causality; inconsistent PTH thresholds and the lack of evidence that treatment improves graft survival mean that the meta-analysis justifies monitoring, not a specific PTH target or routine parathyroidectomy (9.6).

The living donor: more questions than answers

The randomized, double-blind phase 2b AL-DON trial randomly assigned 71 healthy donors to allopurinol 300 mg daily or placebo for 9 months; uric acid decreased significantly ($p < 0.001$), but left ventricular mass did not change ($p = 0.25$), nor did blood pressure ($p = 0.98$) or insulin sensitivity ($p = 0.62$) (9.11). All three serious adverse events occurred in the active arm (9.11). The result is consistent with the negative CKD-FIX and PERL trials; urate-lowering therapy is indicated in donors only for gout or urate nephrolithiasis (9.11).

APOL1 genotyping is an example of an association that weakens after adjustment. In the LETO cohort of 653 donors with a median follow-up of 18.5 years, an eGFR below 45 mL/min/1.73 m² occurred in 14.7% of donors with the high-risk genotype versus 6.4% (RR 2.31; 95% CI, 1.16 to 4.61), but after adjustment for predonation eGFR, the RR fell to 1.91 (95% CI, 0.90 to 4.03; $p = 0.09$) (9.7). A single measurement, the absence of a nondonor control group, and only three cases of kidney failure make it impossible to quantify the risk of kidney failure; KDIGO 2017 states that the test may be offered, and the KDIGO 2025 consensus does not recommend routine screening (9.7). In the Slovak setting, the principle matters more than the test: the genotype is interpreted together with age, eGFR, UACR, blood pressure, and family risk (9.7).

The frontier of experimentation

In organ preservation, hypothermic machine perfusion has randomized support: in a 2009 paired trial with 336 donors and 672 recipients, DGF occurred in 70 versus 89 recipients (OR 0.57; $p = 0.01$) (9.14). One hour of normothermic perfusion in 277 kidneys from donors after circulatory death did not reduce DGF (OR 1.13; 95% CI, 0.69 to 1.84), and a pilot study of a C1-esterase inhibitor in 20 kidneys yielded only an exploratory signal, with the authors holding related patents (9.14).

Xenotransplantation shows that a functioning graft does not mean immunologic quiescence. In the first living recipient of a kidney from a pig with 69 genomic edits, Banff 2A rejection developed on day 8 despite intensive induction, a biopsy on day 34 showed fibrosis and tubular atrophy in approxi-

ately 30% of the cortex, and the patient died on day 52 of a probable arrhythmia (9.8). A single patient without a control group does not allow estimation of the risk of rejection or of zoonotic risk (9.8).

The letter on teclistamab should be read with similar caution. A patient with two previous transplants who had waited more than 12 years achieved, after 31 weeks of treatment, a decline in anti-HLA antibodies and transplantation with a functioning graft at short-term assessment (9.10). This is an off-label proof of biological concept with substantial safety risks; allocation priority, an acceptable mismatch program, paired exchange, and standard desensitization should come before any experiment (9.10), which overlaps with the recommendation to avoid unnecessary transfusions and to update the antibody profile when the graft is failing (9.13).

What is proven, what is probable, and what remains open

Randomized evidence supports the reduction of DGF with hypothermic machine perfusion (9.14), the noninferiority of TTV-guided tacrolimus dosing in low-risk patients in the first year (9.1), and the lack of effect of allopurinol on cardiovascular surrogate endpoints in healthy donors (9.11). What is probable, although based on observational data, is that preemptive transplantation improves both patient and graft survival (9.12), that living donation benefits ultra-long-term graft survival (9.5), that persistent hyperparathyroidism identifies patients at increased risk (9.6), and that transplantation also benefits obese patients up to a BMI of 40 kg/m² (9.4).

It remains open whether treatment of posttransplant hyperparathyroidism will improve hard outcomes (9.6), whether biomarker-guided immunosuppression will yield clinical benefit beyond the first year (9.1), what the true benefit of APOL1 genotyping is (9.7), whether regulatory T cells will allow safe reduction of immunosuppression (9.9), and whether desensitization with bispecific antibodies will move from a case report to controlled trials (9.10). Xenotransplantation and tegoprubart remain experimental (9.8, 9.14).

Implications for practice

For a Slovak outpatient clinic, the organizational dimension matters most: begin transplant education at an eGFR below 30 mL/min/1.73 m², refer the candidate ideally before dialysis, and do not regard age or obesity up to a BMI of 40 kg/m² as an automatic contraindication (9.12, 9.4, 9.14). When graft function declines progressively, HLA antibody status should be updated promptly, living donors identified, vascular access prepared, psychological support offered, and a responsible coordinator designated; continuation of immunosuppression after graft failure is decided individually, because both withdrawal and transplant nephrectomy are associated with sensitization (9.13).

The long-term recipient needs monitoring of eGFR and proteinuria, blood pressure, immunosuppressant toxicity, infections, tumors, metabolic and bone complications, drug interactions, and frailty (9.5), an annual dermatologic examination with photoprotection (9.3), review of the vaccination history with catch-up of missing vaccines according to the required intervals (9.2), and regular assessment of PTH, calcium, phosphate, and 25-OH vitamin D in the context of graft function (9.6). The living donor should have blood pressure, BMI, eGFR, and albuminuria checked at least once a year (9.7). Experimental approaches belong exclusively in clinical trials with predefined endpoints and stopping rules (9.9, 9.10).

What to watch next

In the near term, it is worth watching the follow-up French trial of TTV-guided dosing from the second year after transplantation (9.1), publication of the results of the BESTOW trial of tegoprubart (9.14), any randomized trials of regulatory T cells assessed by protocol biopsies, DSA, and donor-derived cell-free DNA (9.9), further cases and any controlled trial of teclistamab in extreme sensitization (9.10), and further xenotransplants with longer follow-up of humoral injury and zoonotic risk (9.8). For routine practice, interventional data with hard outcomes on the treatment of persistent hyperparathyroidism (9.6) and the experience of European centers in increasing the proportion of preemptive transplants through education on the Plzeň model (9.12) will be useful.

Articles in this chapter

- 9.1** Biomarker-guided reduction of immunosuppression after kidney transplantation: TTV as a measure of immune activity (June 16, 2026, online, Slovak)
- 9.2** Vaccination in chronic kidney disease (CKD) and after kidney transplantation: a practical overview of vaccines and timing (June 17, 2026, online, Slovak)
- 9.3** Malignancy and kidney transplantation: what the nephrologist should know (Core Curriculum 2026) (June 17, 2026, online, Slovak)
- 9.4** Obesity in nephrology: screening, management, and impact on dialysis and transplantation (a practical perspective based on the AJKD Core Curriculum) (June 20, 2026, online, Slovak)
- 9.5** Forty years with a functioning kidney transplant: what ultra-long-term survivors show (July 5, 2026, online, Slovak)
- 9.6** Persistent hyperparathyroidism after kidney transplantation as a risk marker for mortality and graft failure (July 9, 2026, online, Slovak)
- 9.7** APOL1 genotyping in living kidney donation: more precise stratification, not automatic exclusion (July 15, 2026, online, Slovak)
- 9.8** Kidney xenotransplantation from a gene-edited pig: a functioning graft does not yet mean immunologic quiescence (July 24, 2026, online, Slovak)
- 9.9** Regulatory T cells after kidney transplantation: a path to immunologic tolerance without lifelong immunosuppression? (July 25, 2026, online, Slovak)
- 9.10** Teclistamab before kidney transplantation: a first clinical signal in extreme HLA sensitization (August 4, 2026, online, Slovak)
- 9.11** Allopurinol in living kidney donors lowered uric acid but did not improve cardiac structure or blood pressure (August 19, 2026, online, Slovak)
- 9.12** Preemptive transplantation: the optimal form of kidney replacement (September 3, 2026, online, Slovak)
- 9.13** After kidney graft failure: barriers to retransplantation through the eyes of patients, caregivers, and physicians (September 9, 2026, online, Slovak)
- 9.14** Kidney transplantation: what is standard, what is innovation, and what is still experimental (September 24, 2026, online, Slovak)

CHAPTER 10

Anemia, mineral metabolism, electrolytes, and acid-base balance

Anemia of chronic kidney disease, iron metabolism, potassium, vitamin D, and other disorders of the internal milieu. Number of articles: 13.

The chapter on anemia, mineral metabolism, electrolytes, and acid-base balance brings together thirteen articles from April to October 2026, and it centers on two areas: anemia of chronic kidney disease (CKD) after publication of the KDIGO 2026 guideline, and hyperkalemia as a barrier to effective cardiorenal therapy. Hyponatremia and vitamin D receive less space. The predominant type of evidence is not new randomized trials but interpretation of guidelines (10.2, 10.8, 10.13), practical reviews without original data (10.6, 10.7, 10.10), and retrospective or observational cohorts (10.4, 10.9, 10.11, 10.12).

The main message is twofold. In anemia, knowledge has consolidated into a clear sequence of steps in which iron deficiency takes precedence over stimulation of erythropoiesis and ESAs take precedence over HIF-PHIs, although the iron recommendations are graded only 2D (10.2, 10.8, 10.13). In hyperkalemia, the question has shifted from how to lower potassium to how to avoid interrupting RAAS blockade and MRAs; observational data show that this is precisely where practice falls short (10.5, 10.11). The common denominator is the tension between a formal recommendation and real-world data that call it into question but do not refute it.

Anemia of CKD after KDIGO 2026

The main thresholds of the KDIGO 2026 guideline are recorded consistently in three texts in this chapter. Anemia is defined as Hb below 130 g/L in men and below 120 g/L in women, and screening includes hemoglobin, ferritin, TSAT, and a complete blood count, plus B12, folate, inflammation, and blood loss as the situation requires (10.2). The KDOQI commentary adds reticulocytes and calls for a search for bleeding when ferritin is below 45 ng/mL, MCV below 80 fL, or TSAT below 20% (10.8). The logic is the same in all documents: first eliminate correctable causes, and only then consider an ESA (10.2, 10.8, 10.13).

The targets are conservative. In patients on dialysis, an ESA is considered at an Hb of approximately 90 to 100 g/L; outside dialysis, the decision is individual and usually not automatic at an Hb of 100 g/L or higher. Maintenance Hb during ESA or HIF-PHI therapy should usually be below 115 g/L, and routine targeting of an Hb of approximately 130 g/L or higher should be avoided because of cardiovascular and thromboembolic risk (10.2, 10.8). The goal is symptom improvement, not normalization of Hb at any cost (10.2).

The position of HIF-PHIs illustrates the gap between the tone of the commentary and the tone of the evidence. A brief commentary without figures or studies describes roxadustat and daprodustat as an established oral alternative that benefits patient comfort and staff logistics (10.1). Both KDIGO 2026 and KDOQI, however, place ESAs as first choice ahead of HIF-PHIs (grade 2D); a trial of HIF-PHI therapy in hyporesponsiveness is possible, although the evidence is weak, and combining the two is not

recommended. For roxadustat, the EMA lists hypertension, hyperkalemia, edema, and vascular access thrombosis (10.8, 10.9). The optimistic commentary is therefore not false, but it is premature relative to the strength of the evidence.

Iron: thresholds, proactive repletion, and safety

All four recommendations in the iron chapter of KDIGO 2026 are graded 2D (10.13). Initiation thresholds differ by dialysis status: in hemodialysis, ferritin up to 500 ng/mL and TSAT up to 30%, with a preference for the IV route; outside hemodialysis, including PD, ferritin below 100 ng/mL and TSAT below 40%, or ferritin 100 to 299 ng/mL and TSAT below 25% (10.2, 10.8, 10.13). After 1 to 3 months of ineffective oral therapy, treatment switches to the IV route (10.8, 10.13).

Randomized support is strongest in hemodialysis. In PIVOTAL (2,141 patients), proactive iron sucrose 400 mg monthly reduced the primary composite endpoint (29.3% versus 32.3%; HR 0.85; 95% CI, 0.73 to 1.00) and the ESA dose without an increase in infections; ERBP points out that the trial enrolled incident patients treated with ESAs (10.13). In FIND-CKD (626 patients not on dialysis), ferric carboxymaltose reduced the primary endpoint compared with oral iron (23.5% versus 31.8%; HR 0.65) (10.13). The safety limits of ferritin above 700 ng/mL or TSAT of 40% or higher are taken from the proactive arm of PIVOTAL, so both KDOQI and ERBP regard them as justified mainly in hemodialysis, whereas data for PD and CKD not on dialysis are limited (10.8, 10.13).

Two safety signals concern IV iron. Ferric carboxymaltose has the highest incidence and longest duration of FGF23-mediated hypophosphatemia, phosphate should be measured before repeated doses, and ERBP does not recommend the product in recent transplant recipients (10.8, 10.13). A shortened 60-minute iron dextran protocol reduced the total administration time from 189.2 to 80.1 minutes ($p < 0.01$), but under the new regimen six events occurred in 123 administrations (4.9%), including a grade 4 anaphylactic shock during the test dose, compared with one grade 3 hypertensive event (0.8%) under the extended regimen. The difference was not significant, but with a single serious case the estimate is very imprecise, and the study demonstrates feasibility, not safety equivalence (10.12).

The greatest tension concerns iron during infection. Both KDIGO 2026 and KDOQI recommend withholding iron during systemic infection (10.8, 10.13). A retrospective TriNetX cohort (2000 to June 2025) of hospitalized adults with iron deficiency anemia and acute infection, however, found no worsening of infection, and IV iron was associated with better hemoglobin recovery and fewer transfusion days, without prolonged hospitalization (10.4). The available materials do not report specific figures, and these are associations without a causal interpretation that do not change the recommendation (10.4).

ESA hyporesponsiveness

Throughout the chapter, ESA hyporesponsiveness in hemodialysis is consistently framed as a signal of correctable causes, not as a standalone diagnosis (10.9). A secondary analysis of a retrospective cohort compared roxadustat (78 patients) with continued ESA therapy (30 patients) in 108 adults without randomization. Roxadustat was associated with a higher adjusted probability of achieving an Hb of at least 10 g/dL in three of six months (OR 4.44; 95% CI, 1.24 to 15.84; $p = 0.022$) (10.9).

Interpretation calls for restraint. Hb variability with roxadustat was only numerically lower, without statistical significance, and after adjustment for baseline Hb, treatment group was not independently associated with final Hb; an OR of 4.44 therefore does not represent a fourfold causal effect, and six months of follow-up does not allow safety to be assessed (10.9). The conclusion agrees with KDIGO

2026: first verify dosing, adherence, iron, blood loss, inflammation, and dialysis adequacy, and only then choose an ESA, HIF-PHI, or transfusion on an individual basis (10.9).

Hyperkalemia as a problem of undertreatment

Three articles agree that the main harm is not potassium itself but discontinuation of heart- and kidney-protective therapy. In the CARE-HK registry, RAASi or MRAs were not adequately used in about one third of patients, mainly because of concerns about hyperkalemia, and potassium binders were underused (10.5). A Spanish prospective observational study of 429 patients with serum potassium above 5.5 mmol/L confirms this: recurrence occurred in 43.8% of patients, more often at an eGFR below 30 mL/min (49.1% versus 38.8%; $p = 0.03$), and mineralocorticoid receptor antagonists were discontinued in 21.82% of users versus 6.80% for ACE inhibitors (10.11).

The observational nature of these data should be taken seriously. The Spanish study reports only unadjusted proportions without confidence intervals, has no definition of recurrence, and the comparison of binders was not randomized; the authors themselves describe causality as unclear (10.11). Maintaining RAAS blockade with a binder is therefore supported not by this study but by the randomized AMBER and DIAMOND trials (10.5, 10.11). KDIGO 2024, as interpreted in an educational activity, states that hyperkalemia alone should not lead to discontinuation of RAAS inhibition; the recommendation is to maintain RAASi and MRAs and add a potassium binder (10.5). Patiromer exchanges potassium for calcium and SZC for sodium; in REALIZE-K, SZC with spironolactone was accompanied by more serious heart failure events, whereas in a DIAMOND subanalysis NT-proBNP decreased with patiromer. The signal comes from smaller RCTs and pooled analyses (10.5).

The practical framework is dual-track: titrate RAASi or MRAs while actively managing potassium, and check K⁺ within the first days to weeks after every dose change (10.6). The sequence runs from excluding pseudohyperkalemia through removal of causes, diet, and a loop diuretic to a binder, with dose reduction only as a last step and with a plan to restore therapy (10.6, 10.11). Dietary education should be individualized, without universal lists of forbidden foods: boiling diced potatoes with a change of water reduces their potassium content by approximately 50% to 75%, a cup of melon or green grapes contains about 180 mg of potassium, and treatment of constipation is part of potassium control, because the gut takes on a compensatory share of potassium excretion as renal excretion declines (10.7).

Hyponatremia and vitamin D

How fast chronic hyponatremia should be corrected remains controversial. The proponent of faster correction relies on a Canadian observational analysis of more than 22,000 hospitalizations in which osmotic demyelination syndrome occurred in only approximately 0.05% of cases (10.3). The opponent notes that the limit of 8 mmol/L per 24 hours applies to the chronic, high-risk form, and that in preliminary PRONATREOUS data at serum sodium up to 105 mmol/L, 4.5% of patients had confirmed ODS and another 14.7% had suspected ODS, all with correction faster than 8 mmol/L per day (10.3). Both sides rest on observational analyses subject to confounding by indication (10.3).

For vitamin D, the picture is clear in a negative sense. In VITAL-DKD, 2,000 IU daily for five years did not significantly preserve eGFR in 1,312 adults with type 2 diabetes, so vitamin D is not a proven form of kidney protection (10.10). In CKD G3a to G5D, 25(OH)D can be considered as part of the assessment of calcium, phosphate, PTH, and alkaline phosphatase, but calcitriol and its analogs should not be used routinely in CKD not on dialysis and are reserved for G4 to G5 with severe, progressive secondary hyperparathyroidism (10.10).

What is proven, what is probable, and what remains open

Randomized evidence supports proactive IV iron in hemodialysis in incident patients on ESAs (PIVOTAL, HR 0.85), ferric carboxymaltose versus oral iron in CKD not on dialysis (FIND-CKD, HR 0.65), and maintenance of RAAS blockade with a potassium binder (AMBER, DIAMOND) (10.5, 10.11, 10.13). Randomized evidence refutes kidney protection with nutritional vitamin D (10.10). Probable, but supported only by consensus or observational data, are Hb targets below 115 g/L, the precedence of ESAs over HIF-PHIs, and the ferritin and TSAT safety limits outside hemodialysis, all graded 2D (10.2, 10.8, 10.13).

It remains open whether systematically withholding IV iron during infection does more harm than good (10.4), whether roxadustat offers a causal benefit in hyporesponsiveness and what its long-term safety is (10.9), whether shortened infusion protocols are equivalent in safety (10.12), whether SZC worsens fluid retention to a clinically relevant degree in heart failure (10.5), and what the true incidence of ODS is with faster correction of profound hyponatremia (10.3).

Implications for practice

For a Slovak outpatient clinic and dialysis center, the most practical takeaway is the sequence of steps in anemia: complete blood count, reticulocytes, ferritin, and TSAT; treatment of iron deficiency first, according to thresholds that depend on dialysis status; and only then an ESA, within shared decision-making, with maintenance Hb below 115 g/L (10.2, 10.8, 10.13). With ferric carboxymaltose, phosphate should be monitored before repeated doses (10.8). The shortened US iron dextran protocol cannot be transferred without modification, and any in-house protocol must include observation for at least 30 minutes (10.12). Withholding iron during infection remains the recommendation, but rigid postponement in confirmed deficiency can be reconsidered on an individual basis (10.4).

In hyperkalemia, the key change is in the way of thinking: hyperkalemia should not be a reason to stop RAASi and MRAs. The sequence runs from excluding pseudohyperkalemia through correction of acidosis, volume, and constipation, targeted education, and diuretics to a potassium binder, and the dose is reduced only as a last step (10.5, 10.6, 10.7, 10.11). The choice of binder in a patient with heart failure warrants collaboration between the cardiologist and the nephrologist (10.5). In chronic profound hyponatremia in high-risk patients, correction of no more than 8 mmol/L per 24 hours with frequent sodium monitoring remains the rule (10.3), and decisions about vitamin D in CKD are based on CKD-MBD trends, not on an isolated 25(OH)D value (10.10).

What to watch next

In the near term, watch for the primary publication of the TriNetX cohort with specific figures (10.4), safety data on HIF-PHIs in hyporesponsiveness beyond six-month retrospective cohorts (10.9), prospective PRONATREOUS results (10.3), regulatory developments regarding SZC and heart failure and randomized comparisons of potassium binders (10.5, 10.11), safety data for ferritin and TSAT limits in PD and CKD not on dialysis (10.8, 10.13), and prospective validation of shortened iron infusion protocols with systematic recording of reactions (10.12).

Articles in this chapter

- 10.1** Innovations in dialysis and anemia management (April 25, 2026, online, Slovak)
- 10.2** Anemia in chronic kidney disease according to KDIGO: a brief summary of the recommendations (May 14, 2026, online, Slovak)

- 10.3** Fast or slow correction of chronic hyponatremia? A controversy returning to clinical practice (May 17, 2026, online, Slovak)
- 10.4** IV iron during acute infection: real-world data versus the fear of "worsening the infection" (May 22, 2026, online, Slovak)
- 10.5** Case-based: how to manage hyperkalemia in CKD and heart failure without reducing effective therapy (May 27, 2026, online, Slovak)
- 10.6** Optimizing RAASi/MRA therapy in patients with heart failure, CKD, and hyperkalemia (a practical approach) (June 2, 2026, online, Slovak)
- 10.7** Potassium control in kidney disease: educate, don't scare (June 10, 2026, online, Slovak)
- 10.8** Anemia of CKD according to KDIGO 2026: what the KDOQI US Commentary brings to practice (June 19, 2026, online, Slovak)
- 10.9** Roxadustat in ESA hyporesponsiveness: more frequent repeated achievement of target Hb, not lower variability (July 15, 2026, online, Slovak)
- 10.10** Vitamin D in clinical practice: whom to test, whom to treat, and when treatment can cause harm (August 13, 2026, online, Slovak)
- 10.11** Hyperkalemia in CKD in real-world practice: recurrence is the rule, and mineralocorticoid receptor antagonists pay the highest price (August 24, 2026, online, Slovak)
- 10.12** 60-minute iron dextran infusion at a smaller center: benefits, risks, and limits of the evidence (September 24, 2026, online, Slovak)
- 10.13** Iron in anemia of CKD according to KDIGO 2026: initiation thresholds, proactive IV repletion, and the European ERBP perspective (October 1, 2026, online, Slovak)

CHAPTER 11

Differential diagnosis, genetics, onconeurology, and rare diseases

Diagnostic reasoning, hematuria, genetics of kidney disease, onconeurology, monoclonal gammopathies, and rare conditions that should be considered early. Number of articles: 18.

The chapter on differential diagnosis, genetics, onconeurology, and rare diseases is the most heterogeneous part of the book. It brings together eighteen articles ranging from communication tips for history-taking and a qualitative study of diagnostic reasoning, through a survey on genetic testing and reviews of rare syndromes, to oncology texts written for the nephrologist and updated guidelines on hematuria and nephrolithiasis. The common denominator is how the nephrologist arrives at a diagnosis in a patient who does not fit the usual categories, and how incomplete evidence is handled along the way.

The level of evidence is mostly low to moderate. Randomized trials appear for only three interventions: caplacizumab in immune-mediated TTP (11.13), adjuvant pembrolizumab in kidney cancer (11.15), and diazoxide choline in Prader-Willi syndrome (11.9). The rest consists of professional society guidelines, consensus statements, observational cohorts, retrospective series, pilot studies, qualitative research, and educational materials. The main message of the chapter is therefore not “we have a new drug,” but “we have a more precise idea of where we go wrong in diagnosis and what to do about it.”

The discipline of diagnostic reasoning

A short practical text without original data notes that, according to the cited studies, physicians interrupt the patient after 11 to 18 seconds, and it recommends letting the patient speak for at least a minute, specifically looking for red flags, and summarizing what was heard at the end (11.1). This is expert opinion, not a validated intervention, but it costs nothing.

A Canadian qualitative study of nine nephrologists shows that diagnostic reasoning is learned implicitly and that experienced physicians switch between Occam's razor and Hickam's dictum depending on the complexity of the patient (11.10). The most important finding is that time pressure pushes toward the simplest defensible diagnosis regardless of case complexity, so the choice of framework comes to depend on operating conditions (11.10). The limitations are obvious: nine participants, self-reported rather than observed practice, and an unverified effect of the proposed teaching on clinical outcomes. The recommendations to define in advance a threshold for revising the diagnosis and to distinguish between a working and a definitive diagnosis in the medical record are nevertheless reasonable (11.10).

The German initiative Klug entscheiden, with its lean laboratory panel, belongs to the same line: creatinine with eGFR and urinalysis in at-risk patients, quantification of albuminuria at an eGFR below 60 mL/min/1.73 m², measurement of Ca, P, iPTH, and 25-OH vitamin D3 at an eGFR below 45, and careful consideration of contrast at an eGFR below 30, so as to avoid systematic underdiagnosis (11.5).

These are professional society recommendations, and their strength lies in tying each test to a specific clinical question.

Genetics: the indication exists, but the test is not done

The genetics line is internally consistent, and its message is uncomfortable. In a self-reported survey of 1,168 patients from the National Kidney Foundation, 61.4% of respondents had at least one indication for testing, but a physician had discussed it with them in only 11% to 29% of cases, depending on the number of indications, and only 8.6% had been tested (11.4). The survey comes from a US patient organization, but the barriers it identifies, namely physicians' uncertainty in communicating about genetics, concerns about cost, and low patient awareness, are universal. The author does not recommend routine testing but rather active consideration of testing in CKD of unclear etiology, early onset, a positive family history, extrarenal manifestations, and before related-donor transplantation (11.4).

A negative result, moreover, is not the end of the road. A retrospective single-center study from Cleveland Clinic showed that exome sequencing after a negative or inconclusive panel yielded a new or reclassified pathogenic finding in 7 of 54 patients (13%); the sample was small and selected, and the diagnostic yield of panels and exome sequencing across the whole cohort cannot be compared, because the groups were not randomized (11.14). The list of reasons why a panel fails is valuable: narrow scope, blind spots for CNVs and deep intronic variants, regions that are difficult to analyze, such as PKD1 or the VNTR region of MUC1, and outdated variant interpretation. This leads to a requirement for laboratory reports that state the genes, technology, limitations, and year of testing, and to the rule that a VUS should not be used for treatment decisions or for excluding a donor (11.14).

Rare diagnoses show how genetics changes management. In Bartter syndrome, the classification into types 1 to 5 requires a panel that also covers deletions and duplications, particularly in CLCNKB; a Korean cohort of 54 patients, 11% of whom had CKD G3 to G5 after a median of eight years, is small but sufficient to conclude that the disease is not benign, and treatment rests on expert consensus (11.12). In Prader-Willi syndrome, the randomized evidence for diazoxide choline comes only from a 16-week withdrawal phase with 77 pretreated participants, which demonstrates maintenance of effect, not efficacy when treatment is initiated; in the EU, the marketing authorization application was withdrawn in April 2026, and the drug has not been studied in patients with impaired kidney function (11.9).

ADPKD: from observation to active management

A review based on an educational program describes the shift from passive observation to risk-oriented management of ADPKD: identification of rapid progressors, treatment of hypertension with ACE inhibitors or ARBs, and, in selected patients, tolvaptan under the supervision of an experienced nephrologist with monitoring of liver tests (11.2). The article does not report trial figures; its contribution is a framework for long-term care, including individually considered screening for intracranial aneurysms (11.2).

Translational research at Mayo Clinic addresses the main obstacle to tolvaptan, namely polyuria and nocturia. The vasopressin-independent urate, AMPK, and AQP2 pathway it describes is a mechanistic finding; in preclinical models, probenecid attenuated the increase in urine volume after tolvaptan, and in a small phase 2 study it reduced urine volume by approximately 30% (11.6). The data come from material intended for the general public, probenecid has drug interactions, and the author himself

does not recommend adding it off-label outside protocols (11.6). The concept of separating the anti-cystic effect from aquaresis and making nocturnal diuresis a measurable target is of interest.

Onconeurology: nephritis after ICIs and common causes of AKI

The oncology line is the most consistent. Three articles agree that the typical kidney complication of immune checkpoint inhibitors is acute tubulointerstitial nephritis and that a rise in creatinine should first be distinguished from hypovolemia, sepsis, obstruction, contrast media, and nephrotoxic drugs, including proton pump inhibitors and NSAIDs (11.7, 11.8, 11.15). The firmest data come from the review on kidney cancer: in an observational cohort of 429 patients with ICI-associated AKI, the median time to onset was 16 weeks, nephritis was found in 82.7% of biopsies, and AKI recurred after rechallenge in 16.5% (11.15). In the five-year analysis of the randomized KEYNOTE-564 trial, overall survival with adjuvant pembrolizumab was 87.7% versus 82.3% (HR 0.66), at the cost of grade 3 or 4 adverse events in 18.6% versus 1.2%; the trial excluded patients on dialysis and transplant recipients (11.15). In LITES-PARK-005, belzutifan increased the objective response rate but did not significantly improve overall survival (HR 0.88; 95% CI, 0.73 to 1.07) (11.15).

The melanoma review is based on an educational quiz, and its nephrology value lies in two points: creatinine, eGFR, albuminuria, and urine sediment should be known before immunotherapy, and in kidney transplant recipients ICIs may increase the risk of rejection (11.8). The post-congress commentary on EHA and ASCO 2026 explicitly acknowledges that it does not cite specific studies because the source video could not be verified; its overview of risks beyond ICIs is useful, namely hypertension and proteinuria with anti-VEGF therapy, multifactorial AKI with CAR T-cell therapy, and overestimation of creatinine-based eGFR in sarcopenia (11.7).

The hematology section rests on two pillars of differing strength. In iTTP, the evidence is strong: in the randomized HERCULES trial, caplacizumab reduced the composite endpoint of death, recurrence, or thromboembolism from 49% to 12% (11.13). The plasma exchange-free regimen from the single-arm MAYARI study, with remission in 93.5% of 46 evaluable patients, has no control group and belongs only in specialized centers; severe kidney injury is less typical in iTTP than in HUS, but it is not an absolute distinguishing feature (11.13). In contrast, the concept of monoclonal gammopathies of clinical significance beyond MGRS rests on small case series, and the full text of the review was not available when the article was prepared (11.11). The message is nevertheless practical: a paraprotein should be assessed by its pathogenicity, not by clone size, and electrophoresis alone, without immunofixation and free light chains, is not sufficient (11.11).

Hematuria, stones, and what is not a diagnosis

The review of hematuria emphasizes that a positive dipstick, red urine, and confirmed hematuria are not synonymous, that microhematuria requires at least 3 RBC/HPF, and that further evaluation is guided by risk stratification; neither anticoagulants nor physical exertion should be accepted as a definitive explanation, and a negative ultrasound in a high-risk patient does not replace cystoscopy (11.17). Nutcracker syndrome is a textbook example of the difference between a finding and a diagnosis: among 324 asymptomatic potential donors, 30.5% had an aortomesenteric angle below 41°, but only 0.7% met a diameter ratio of 4.9 or higher; no validated criteria exist, and treatment data come from nonrandomized series (11.18).

The AUA 2026 guideline on nephrolithiasis rejects automatic approaches: instead of a universal ban on oxalate-containing foods, it rests on stone analysis, 24-hour urine collection, a urine volume of at least 2.5 liters per day, a normal calcium intake of 1,000 to 1,200 mg, and phenotype-directed treat-

ment, while the urine output target should be individualized in heart failure, advanced CKD, or hyponatremia (11.16). A urine fibrosis test with a TG2-activated reporter achieved a sensitivity of 84% and a specificity of 94% in a cohort of 35 participants; this is a pilot study without confirmation in large cohorts (11.3).

What is proven, what is probable, and what remains open

Little is proven at the level of randomized trials, but what is proven is clear: caplacizumab in iTTP (11.13), adjuvant pembrolizumab in a defined population with kidney cancer at the cost of greater toxicity (11.15), and maintenance of the effect of diazoxide choline in the withdrawal phase (11.9). On the basis of observational data and consensus, it is probable that nephritis after ICIs is common and recurs in one sixth of cases after rechallenge (11.15), that Bartter syndrome leads to advanced CKD in some patients (11.12), that a negative panel does not close the diagnostic workup when suspicion persists (11.14), and that risk stratification of hematuria is better than an automatic approach (11.17). None of these observational conclusions is evidence of causation.

It remains open whether explicit teaching of diagnostic reasoning improves outcomes (11.10), whether the anticystic effect of tolvaptan can be separated from aquaresis (11.6), whether the urine fibrosis test holds up in large cohorts (11.3), what the criteria for nutcracker syndrome are (11.18), how to treat monoclonal gammopathies of clinical significance beyond case reports (11.11), and whether a plasma exchange-free regimen in iTTP holds up in a controlled comparison (11.13).

Implications for practice

For a Slovak outpatient clinic, several inexpensive steps follow: let the patient speak, distinguish between a working and a definitive diagnosis, and decide in advance what would change the diagnosis (11.1, 11.10); in CKD of unclear etiology, early onset, a positive family history, and before related-donor transplantation, actively raise the question of genetic testing and read the result in the context of the scope of the test, because a generally negative panel may not be sufficient for a living donor (11.4, 11.14); and in CKD, adhere to a lean but complete laboratory panel, including albuminuria and mineral and bone parameters (11.5).

For hospital wards and oncology centers, creatinine, eGFR, albuminuria, and urine sediment should be known before immunotherapy, a nephrologist should be involved early in CKD G3b to G5, significant albuminuria, or a history of AKI, and every rise in creatinine after ICIs should first be distinguished from common causes (11.7, 11.8, 11.15). With a paraprotein, immunofixation and free light chains should be added (11.11), and with thrombocytopenia and microangiopathic hemolysis, TTP should be considered urgently and plasma exchange should not be delayed (11.13). The evaluation of hematuria or stones should not be closed without risk stratification (11.16, 11.17), radiologic compression of the renal vein without clinical and hemodynamic concordance does not need treatment (11.18), and in rare tubulopathies a plan for acute illness should be prepared in advance (11.12).

What to watch next

In the near term, watch for controlled data on mitigating aquaresis with tolvaptan (11.6), validation studies of the urine fibrosis test (11.3), the outcome of the phase 3 pitolisant trial in Prader-Willi syndrome and any change in the regulatory status of diazoxide choline in the EU (11.9), whether the Delphi consensus on nutcracker syndrome evolves into validated criteria (11.18), whether plasma exchange-free iTTP regimens reach a controlled comparison (11.13), how ICI rechallenge after nephritis

will be handled in practice (11.15), and whether genetic testing can be moved from indication to implementation outside the setting in which the NKF survey was conducted (11.4).

Articles in this chapter

- 11.1** Five practical tips for better history-taking and a more accurate diagnosis (May 13, 2026, online, Slovak)
- 11.2** Optimizing treatment outcomes in polycystic kidney disease (May 19, 2026, online, Slovak)
- 11.3** Molecular imaging from urine: a new biomarker test that may detect early kidney fibrosis (May 24, 2026, online, Slovak)
- 11.4** Genetic testing in kidney disease is still offered too rarely (June 5, 2026, online, Slovak)
- 11.5** Klug entscheiden in nephrology: which laboratory tests are really necessary? (June 20, 2026, online, Slovak)
- 11.6** A new vasopressin-independent pathway of water regulation in the kidney and its significance for the treatment of ADPKD (July 2, 2026, online, Slovak)
- 11.7** Post-congress notes from EHA and ASCO 2026: what oncology updates have in common with nephrology (July 9, 2026, online, Slovak)
- 11.8** Modern treatment of melanoma: what internists and nephrologists should know (July 9, 2026, online, Slovak)
- 11.9** Prader-Willi syndrome across the lifespan: genetics, hyperphagia, and coordinated care (August 4, 2026, online, Slovak)
- 11.10** Occam or Hickam? How nephrologists switch between diagnostic frameworks (August 5, 2026, online, Slovak)
- 11.11** Monoclonal gammopathy of clinical significance beyond MGRS: what nephrologists should know (August 5, 2026, online, Slovak)
- 11.12** Bartter syndrome: diagnosis, genetic forms, and long-term treatment of renal salt wasting (August 13, 2026, online, Slovak)
- 11.13** Immune-mediated thrombotic thrombocytopenic purpura: an urgent diagnosis in nephrology practice too (September 12, 2026, online, Slovak)
- 11.14** A negative genetic test does not rule out a genetic cause of kidney disease (September 13, 2026, online, Slovak)
- 11.15** New trends in the diagnosis and treatment of kidney cancer: a nephrology-oriented review (September 27, 2026, online, Slovak)
- 11.16** Kidney stones according to AUA 2026: metabolic evaluation, prevention, and treatment (September 27, 2026, online, Slovak)
- 11.17** Evaluation of hematuria in nephrology practice: confirmation, emergencies, and identification of the source (September 27, 2026, online, Slovak)
- 11.18** Nutcracker syndrome: when compression of the left renal vein really explains hematuria (October 2, 2026, online, Slovak)

CHAPTER 12

Clinical pharmacology and drug safety

Drug safety and dosing in reduced kidney function, polypharmacy, analgesics, psychotropic drugs, and dietary supplements with a nephrological dimension. Number of articles: 27.

The chapter on clinical pharmacology and drug safety is the broadest in scope in this volume. Its 27 articles from May to October 2026 cover analgesics, antibiotics, proton pump inhibitors, beta-blockers, gabapentinoids, lithium, dietary supplements, and cannabis. They are united by a single question: how does a drug's benefit-risk balance change as glomerular filtration rate declines and the patient takes five or more drugs at the same time? Chronic kidney disease (CKD) appears here not as a coexisting comorbidity but as an independent driver of polypharmacy that also alters the volume of distribution, protein binding, and renal elimination of drugs (12.13).

The type of evidence is uneven, and this is precisely the main message of the chapter. Where randomized trials exist, they are often negative or neutral: allopurinol did not slow the decline in kidney function in either CKD-FIX or PERL (12.16), and pantoprazole did not increase all-cause mortality in the COMPASS trial (12.24). Most safety signals, by contrast, come from retrospective cohorts, administrative databases, conference abstracts, and educational activities burdened by confounding by indication and reverse causation (12.1, 12.12, 12.23). The editorial conclusion is therefore not a list of forbidden drugs but a discipline: for every drug, verify the indication, the eGFR-based dose, and interactions, and do not mistake an observational association for causation.

Randomized evidence versus observational signals

Allopurinol offers the clearest contrast. In CKD-FIX, among 369 patients with CKD G3 to G4, the annual eGFR decline was 3.33 with allopurinol versus 3.23 mL/min/1.73 m² with placebo, and the confidence interval of the difference excludes a clinically meaningful benefit; in PERL, among 530 patients with type 1 diabetes, iohexol-measured GFR did not differ after 3 years despite a fall in urate from 6.1 to 3.9 mg/dL (12.16). A retrospective cohort of 80 patients in whom mean eGFR rose from approximately 35 to 47 mL/min/1.73 m² is exactly the type of evidence produced by confounding by indication, regression to the mean, and lack of adjustment (12.16). KDIGO 2024, in recommendation 3.14.2 graded 2D, therefore suggests not using urate-lowering drugs to slow CKD progression in asymptomatic hyperuricemia; gout, urate nephrolithiasis, and prevention of tumor lysis syndrome remain fully valid indications (12.16).

Proton pump inhibitors teach a similar lesson. In the COMPASS trial, pantoprazole 40 mg had an HR for all-cause mortality of 1.03 (95% CI, 0.92 to 1.15) after approximately 3 years, and the only confirmed signal was an increase in enteric infections from 1.0% to 1.4% (12.24). Observational studies had reported an OR of 1.68 and a veterans cohort an HR of 1.17; the author rightly describes the convergence of these figures as a pharmacoepidemiologic hypothesis, not evidence (12.24). A South Korean cohort of patients with CKD taking NSAIDs found an adjusted HR of 1.38 (95% CI, 1.10 to 1.74) for CKD progression with PPIs compared with H₂-receptor antagonists, but the association was strongest with treatment lasting only 1 to 15 days, which is of limited biological plausibility and suggests AKI,

acute illness, or reverse causation instead (12.23, 12.24). Equally observational, and without data on the reason for starting a PPI, is the TriNetX network analysis presented at the SGIM 2026 meeting, according to which patients combining PPIs with GLP-1 receptor agonists had an RR of 2.37 (95% CI, 2.30 to 2.44) for upper gastrointestinal adverse effects; GLP-1 receptor agonists themselves cause re-flux-like symptoms (12.1).

The Korean cohort on beta-blockers in CKD without known cardiovascular disease belongs to the same category. After propensity score matching, users had an HR for all-cause mortality of 2.09 (95% CI, 1.96 to 2.24) and an HR for 4-point MACE of 1.33 (95% CI, 1.26 to 1.41) (12.12). The effect size is surprising and, for that very reason, suspect: these are administrative data with possible residual confounding by indication and undiagnosed subclinical disease (12.12). The study does not change the role of beta-blockers when there is a clear cardiac indication; in uncomplicated hypertension in CKD, however, it confirms the preference for RAAS inhibitors, calcium channel blockers, and diuretics (12.12).

Pain, gabapentinoids, and opioids in CKD

Analgesic therapy in CKD is a series of compromises. Thresholds for NSAIDs are not supported by randomized data, and the main argument for caution is a retrospective cohort in which a single peri-operative dose of ketorolac in patients with CKD was associated with AKI in 12.7% versus 2.0% ($n = 114$) (12.26). The review on low back pain allows short-term NSAIDs at the lowest effective dose, but they should be avoided in active AKI, volume depletion, hyperkalemia, and decompensated heart failure, and the combination with a diuretic and RAAS blockade is particularly risky (12.15, 12.26). Acetaminophen has a more favorable kidney profile but a modest effect; morphine and codeine accumulate renally excreted active metabolites, and duloxetine is not recommended at a creatinine clearance below 30 mL/min, because metabolite exposure on hemodialysis is 7- to 9-fold higher (12.15, 12.26).

Gabapentinoids receive the most detailed safety assessment in the chapter. According to the USRDS registry, 19% of patients on hemodialysis were taking gabapentin, although it is excreted unchanged by the kidneys and has a half-life of up to 132 hours on dialysis (12.25). The cohort of Ishida et al., with 140,899 patients on hemodialysis, linked doses above 300 mg daily to a 50% higher risk of altered mental status, a 55% higher risk of falls, and a 38% higher risk of fracture, and almost one third of users had no documented indication (12.25). The cohort of Muanda et al. showed that a higher starting dose was associated with a 30-day incidence of the composite endpoint of 1.9% versus 1.5%, that is, an RR of 1.27 and an NNH of approximately 250 (12.25). The data are observational but consistent with the 2019 FDA warning on respiratory depression when gabapentinoids are combined with opioids (12.25, 12.26). In contrast, a preprint reporting an HR of 7.39 for dementia in CKD is not peer reviewed and is probably a biased signal that should not change practice (12.25). The Slovak SmPC provides concrete guidance: gabapentin 150 to 600 mg daily at a creatinine clearance of 15 to 29 mL/min, 150 to 300 mg daily below 15 mL/min, and 200 to 300 mg after each hemodialysis session in an anuric patient; pregabalin 25 mg daily below 15 mL/min, with a maximum of 75 mg (12.22). Gabapentinoids are not used routinely for sciatica (12.15), and ineffective treatment should be tapered over at least one week (12.22, 12.25).

Antimicrobial therapy and specific toxicities

For infections, the review in Nephrology Dialysis Transplantation shifts the emphasis from a stronger drug to a more precise one: do not treat asymptomatic bacteriuria without an indication, obtain a culture before treating a complicated infection, dose according to eGFR without subtherapeutic

doses, de-escalate according to susceptibility, and address the urologic factor (12.9). A retrospective French study of 228 older men with a median age of 85 years found adverse events in 48.5% of patients on co-trimoxazole versus 29.7% on fluoroquinolones (adjusted OR 2.10), most often AKI and metabolic disturbances; the observational design and the small number of frail patients limit generalizability, but the signal is clinically understandable (12.3). Outpatient parenteral antimicrobial therapy should be a formally governed process, and several of its recommendations rest on low-quality evidence; what matters for nephrology is vein preservation before PICC placement in CKD G3b to G5, dose adjustment according to the trend in kidney function, and laboratory checks at least once weekly (12.14).

Gentamicin can cause permanent bilateral vestibular hypofunction without hearing loss: in a series of 35 patients, all 35 had imbalance but only three had hearing impairment, toxicity has also been described at recommended concentrations, and there is no antidote (12.17). In heparin-induced thrombocytopenia, the greatest strength of the 4Ts score lies in the negative predictive value of a low score of 0 to 3; with an intermediate or high score, all sources of heparin, including catheter flushes, should be stopped (12.11). Lithium deserves a special note: the claim of low renal toxicity is contradicted by a concentrating defect in at least 54% of users, an HR of 1.93 (95% CI, 1.76 to 2.12) for stage 3 CKD, and ESRD in approximately 1.5% (12.19).

Supplements, herbal products, and nonprescription substances

Substances that patients do not regard as drugs form a separate line. Controlled human studies of wellness peptides are lacking, gray-market products have shown arsenic and lead contamination, and kidney dysfunction has also been reported (12.2). Pielotax rests on a single report in 42 patients without placebo, in which the control group also improved and neither group reached the target serum uric acid; notification of a supplement to the Public Health Authority of the Slovak Republic is an administrative registration, not an assessment of efficacy (12.20). Confusing ursolic acid with ursodeoxycholic acid illustrates the risk of similar names: UDCA is the first-line treatment for primary biliary cholangitis, whereas a meta-analysis of six studies of ursolic acid showed no change in any metabolic parameter (12.21). St. John's wort, through induction of CYP3A4 and P-glycoprotein, lowers tacrolimus and cyclosporine concentrations, with reported cases of acute rejection (12.8).

For cannabis, the randomized crossover MARY-JANE trial in 108 young users found 9% lower daily ecotopy on inhalation days (RR 0.91; 95% CI, 0.85 to 0.98), but this is an acute surrogate endpoint in a cohort without heart failure or arrhythmia and cannot be transferred to CKD (12.18). More practical for the outpatient clinic is a meta-analysis of 80 publications according to which driving-related abilities recover within approximately 5 hours after inhalation of 20 mg of THC, and blood THC levels do not correlate with impairment (12.27). The following should be read as hypotheses: mouse data on the immune response to mRNA vaccines through muscle cells (12.5), a proof-of-concept study of prucalopride in 49 adults that improved cognitive test performance without evidence of functional benefit (12.10), and an analysis in rheumatoid arthritis in which eGFR_{cys} was 20.9 mL/min/1.73 m² lower than eGFR_{cr} and, without directly measured GFR, a change in filtration cannot be distinguished from an inflammatory artifact (12.6).

What is proven, what is probable, and what remains open

Randomized trials have proven that allopurinol does not slow CKD progression in asymptomatic hyperuricemia (12.16) and that pantoprazole does not increase all-cause mortality over a three-year horizon, although COMPASS was not designed to capture rare or late events (12.24). Probable, with con-

sistent observational and pharmacokinetic support, are the risk of falls, confusion, and respiratory depression with gabapentinoids above kidney-adjusted doses (12.22, 12.25), the renal toxicity of lithium (12.19), the poorer tolerability of co-trimoxazole in frail older patients (12.3), and the interaction of St. John's wort with calcineurin inhibitors (12.8). It remains open whether PPIs contribute to CKD progression other than through acute tubulointerstitial nephritis (12.23, 12.24), whether the association of beta-blockers with mortality has a causal component (12.12), what the safety threshold for NSAIDs in CKD is (12.26), and how to interpret cystatin-based estimates in the presence of inflammation (12.6). Wellness peptides and Pielotax remain without human evidence (12.2, 12.20). The analysis of postmarketing failures is a reminder that safety issues emerged for almost one third of drugs approved by the FDA from 2001 to 2010, at a median of 4.2 years after approval, so even an approved drug has not been proven safe over the long term (12.7).

Implications for practice

For a Slovak outpatient clinic and dialysis center, the chapter points above all to an organizational habit: the medication list is a dynamic document, and at every change in eGFR, hospitalization, episode of AKI, or transition to dialysis, the indication and dose of each drug, duplications, and effects on potassium, acidosis, and blood pressure should be verified (12.13). Specifically, this means tying PPIs to a clear indication and reconsidering the NSAID first (12.23, 12.24); with a beta-blocker lacking a cardiac indication, checking the ejection fraction and arrhythmias and tapering gradually (12.12); starting gabapentinoids low according to creatinine clearance and considering their toxicity when a patient on dialysis develops unexplained confusion (12.22, 12.25); and giving allopurinol only for gout, urate nephrolithiasis, or risk of tumor lysis syndrome (12.16).

The second habit concerns history-taking. Ask actively about supplements, herbal products, peptides, and cannabis, record the exact product and dose, and in transplant recipients proceed in cooperation with the transplant center (12.2, 12.8, 12.20, 12.27). In patients on dialysis, ask about itching at regular intervals and do not rely on antihistamines, which do not target the pathophysiology of CKD-aP (12.4); with new gait instability after an aminoglycoside, consider vestibulotoxicity (12.17); with a fall in platelet count, use the 4Ts score before ordering antibody tests (12.11); with lithium, monitor eGFR, lithium levels, TSH, and calcium (12.19); and with OPAT in a patient with CKD G3b to G5, do not allow PICC placement without a nephrology consultation (12.14). Suspected adverse reactions should be reported, because case reports were the basis for withdrawal in 71% of cases (12.7).

What to watch next

In the near term, watch for pragmatic randomized data on PPIs and beta-blockers in CKD that confirm or refute the observational signals (12.12, 12.23, 12.24), whether the preprint on gabapentin and dementia passes peer review and whether an optimal gabapentinoid dose in dialysis is established (12.25), and whether studies with directly measured GFR in inflammatory diseases appear (12.6). For supplements, it will be important whether regulators respond to contaminated peptide products (12.2); for antimicrobial resistance, whether prediction models and bacteriophage therapy gain stand-

ardization and clinical data (12.9); and for the Slovak setting, whether gabapentinoids are brought under controls similar to those in the UK (12.25).

Articles in this chapter

- 12.1** Concurrent use of proton pump inhibitors (PPIs) and GLP-1 receptor agonists may increase gastrointestinal adverse effects (May 24, 2026, online, Slovak)
- 12.2** "Wellness" peptides: reliable verification of safety and efficacy is lacking (May 24, 2026, online, Slovak)
- 12.3** Co-trimoxazole for UTI in older patients causes adverse events more often than fluoroquinolones (May 27, 2026, online, Slovak)
- 12.4** Stuck on antihistamines for CKD-aP? Time to reconsider (May 27, 2026, online, Slovak)
- 12.5** mRNA vaccines apparently work more broadly than we thought (June 5, 2026, online, Slovak)
- 12.6** Cystatin C and creatinine-based eGFR estimates vs. biomarkers of rheumatoid arthritis activity: insights from a secondary analysis of a randomized trial (June 20, 2026, online, Slovak)
- 12.7** Postmarketing drug safety failures: an in-depth analysis of 10 drugs with fatal consequences in the FDA and EMA regulatory environment (June 22, 2026, online, Slovak)
- 12.8** St. John's wort for depression: when to consider it and when nephrology is better off without it (July 5, 2026, online, Slovak)
- 12.9** Antimicrobial resistance in urinary tract infections: global burden and a nephrology perspective (July 5, 2026, online, Slovak)
- 12.10** Prucalopride and "brain fog" after depression: can 5-HT₄ agonism improve cognition? (July 5, 2026, online, Slovak)
- 12.11** The 4Ts score in suspected heparin-induced thrombocytopenia: a simple tool of great clinical importance (July 9, 2026, online, Slovak)
- 12.12** Beta-blockers in chronic kidney disease without known cardiovascular disease: caution is warranted (July 9, 2026, online, Slovak)
- 12.13** Chronic kidney disease as an independent driver of polypharmacy (July 10, 2026, online, Slovak)
- 12.14** Outpatient parenteral antimicrobial therapy: a safe alternative to hospitalization only with proper patient selection (August 13, 2026, online, Slovak)
- 12.15** Low back pain and sciatica: diagnosis, treatment, and analgesic safety in CKD (August 13, 2026, online, Slovak)
- 12.16** Allopurinol in chronic kidney disease: why randomized trials and retrospective data point in opposite directions (August 23, 2026, online, Slovak)
- 12.17** Gentamicin vestibulotoxicity: recognition, diagnosis, and management (August 24, 2026, online, Slovak)
- 12.18** Inhaled cannabis and cardiac ectopy: what a randomized crossover trial shows (and why it does not yet support a recommendation to "treat" arrhythmias) (September 3, 2026, online, Slovak)
- 12.19** Lithium: seven myths that may stand in the way of its use, from a nephrology perspective (September 3, 2026, online, Slovak)
- 12.20** Pielotax and "kidney peptides": an analysis of composition, efficacy, and safety from a nephrologist's perspective (September 5, 2026, online, Slovak)
- 12.21** Ursodeoxycholic acid and ursolic acid: similar names, fundamentally different clinical significance (September 8, 2026, online, Slovak)
- 12.22** Polyneuropathy in chronic kidney disease: diagnosis, treatment, HTEMs, and safe dosing (September 8, 2026, online, Slovak)
- 12.23** Proton pump inhibitors vs. H₂-receptor antagonists in patients with CKD taking nonsteroidal anti-inflammatory drugs (September 9, 2026, online, Slovak)
- 12.24** Long-term use of proton pump inhibitors: what we actually know and how to translate it into practice (September 17, 2026, online, Slovak)
- 12.25** Gabapentin and its "dark side": safety in the nephrology setting (CKD, hemodialysis) (October 2, 2026, online, Slovak)
- 12.26** Chronic pain in chronic kidney disease: which analgesic, and at what cost (October 2, 2026, online, Slovak)
- 12.27** Cannabis, polypharmacy, and driving in older patients: what the data show and what to ask (October 2, 2026, online, Slovak)

CHAPTER 13

Nutrition, lifestyle, and the environment

Protein, fiber, plant-based diets, physical activity, heat waves, and popular nutrition trends. What holds up against the evidence and what it means for patients with kidney disease. Number of articles: 30.

The chapter on nutrition, lifestyle, and the environment brings together thirty texts from May to September 2026, and its composition sets it apart from the other chapters of the book. It is dominated by observational cohorts, cross-sectional analyses, conference abstracts, and narrative or journalistic reviews. Randomized evidence is rare and concerns either surrogate endpoints, such as the meta-analysis of the effect of creatine on creatinine (13.15), or small mechanistic studies (13.28, 13.14). The only pragmatic RCT with hard kidney outcomes, ERIKA, was negative for the tested intervention (13.30).

The chapter's main message is therefore methodological rather than therapeutic. The direction of the evidence is consistent: a predominantly plant-based, minimally processed diet is associated with a lower risk of developing CKD and with lower mortality in patients who already have CKD (13.24, 13.5), whereas ultraprocessed foods, additives, and red meat lie at the opposite end (13.4, 13.5, 13.20). Causality does not follow from these data; vegetarians in the UK Biobank were younger, leaner, and less likely to smoke (13.24). The second theme is that recommendations for the healthy population and fitness targets cannot be mechanically transferred to CKD (13.19, 13.21, 13.22, 13.13).

Plant-based diet between association and recommendation

The strongest data in the chapter are prospective but observational. In the UK Biobank cohort of 416,584 participants without kidney disease followed for a mean of 12.9 years, a vegetarian diet was associated with an HR of 0.81 (95% CI, 0.711 to 0.927) for incident CKD, and the association persisted after adjustment for baseline eGFR and ACR as well as for polygenic risk (13.24). A conference abstract presented at the ERA Congress, in a multicohort analysis of 8,165 adults with CKD, linked a one-standard-deviation increase in adherence to the Planetary Health Diet with an HR of 0.83 for all-cause mortality and an HR of 0.76 (95% CI, 0.64 to 0.90) for kidney-related mortality, whereas the Inverted Pyramid Diet was associated with 32% higher kidney-related mortality (13.5). Agreement in direction across two different populations is the most persuasive argument in the chapter.

Data from other areas point the same way. In the NutriNet-Santé cohort, the highest colorant intake was associated with an HR of 1.38 (95% CI, 1.17 to 1.63) for type 2 diabetes and the highest preservative intake with an HR of 1.24 for hypertension, but additives may be only a marker of an ultraprocessed diet (13.4). A cross-sectional NHANES analysis associated red and processed meat with a higher likelihood of hepatic steatosis and fibrosis (13.20). An expert interview with Kim Williams explains the convergence by the substitution effect, but it remains an opinion, not an analysis of the evidence (13.3).

What matters for the nephrologist is that, in the descriptive data, vegetarians in the UK Biobank had a lower intake of both potassium and phosphorus than the group with high meat consumption, so the label “plant-based diet” does not automatically mean a potassium-rich diet (13.24). The review on fats

adds that the correlation between potassium intake and serum potassium is weak and that no higher risk of hyperkalemia has been observed with the Mediterranean or DASH diet (13.2). The PLADO model, with 0.6 to 0.8 g of protein/kg/day, is therefore usable as a planning framework in CKD as well (13.30).

Protein: one number is not enough

The tension between population and nephrology recommendations is sharpest for protein. The US DGA 2025 to 2030 recommend 1.2 to 1.6 g/kg/day, and educational reviews for healthy adults cite the same range (13.20, 13.8), whereas KDIGO 2024 recommends approximately 0.8 g/kg/day in CKD G3 to G5 not on dialysis and advises not exceeding 1.3 g/kg/day in patients at risk of progression (13.30, 13.16). For an 80-kilogram patient, the fitness target of 1.6 g/kg/day means 128 g of protein daily, compared with 64 g under the CKD recommendation (13.21), and according to a meta-analysis, fat-free mass no longer increases above approximately 1.62 g/kg/day (13.19).

For people without CKD, the chapter offers a reassuring observational finding. The Norwegian RENIS cohort of 1,324 participants, with repeated iohexol GFR measurements over a median of 10 years, found no association between higher reported protein intake and faster mGFR decline, although the highest quartile had a mean intake of 1.8 g/kg/day (13.16). Its strength is measured GFR; its limitations are a single questionnaire and missing data on supplements and extreme intake, so the study does not establish a safe upper limit (13.16).

In established CKD, the evidence for restriction is mixed. An umbrella review of 25 meta-analyses showed moderate certainty of benefit of a low-protein diet for serum phosphate and the risk of kidney failure but low certainty for change in GFR, and the pragmatic ERIKA RCT in 223 patients with CKD G4 to G5 showed no benefit of a very low-protein diet with ketoanalogues for patient or kidney survival and revealed poor adherence (13.30). On dialysis the direction reverses: KDOQI 2020 recommends 1.0 to 1.2 g/kg/day, and the main risk is undernutrition and loss of muscle mass (13.20, 13.30, 13.21). The accumulation of kidney-related and gluten-free restrictions carries the same risk, although KDIGO 2025 does not include a gluten-free diet among standard treatments for IgA nephropathy (13.17).

Creatinine as a diagnostic trap

Several texts share the finding that lifestyle changes creatinine without changing filtration. A meta-analysis of 19 RCTs and one crossover study showed that creatine raises serum creatinine by an average of 0.13 mg/dL (95% CI, 0.07 to 0.18) without a significant difference in eGFR or urea, but studies longer than one year are lacking (13.15). High protein intake and greater muscle mass act in the same way (13.21), and a rise in creatinine after a work shift in the heat may reflect hemoconcentration (13.27). The opposite artifact appeared in an RCT of a ketogenic diet, in which eGFR rose only because of a fall in cystatin C, which probably reflects a change in the marker rather than a kidney benefit (13.28).

None of this establishes the safety of creatine in CKD. The most solid data concern creatine monohydrate at doses of 3 to 5 g daily in persons without kidney disease, and an exploratory one-year study in patients on hemodialysis taking 5 g daily increased fat-free mass by bioimpedance but also total and intracellular water, without improving malnutrition and inflammation scores (13.15). Evidence for cognitive use rests on small studies, such as one in 29 sleep-deprived adults or a pilot trial with 20 participants, and has not been confirmed (13.9, 13.8). When creatinine does not match the

clinical picture, the consistent recommendation is a combined creatinine and cystatin C eGFR together with albuminuria and, for major decisions, measured GFR (13.15, 13.16, 13.21).

Heat, work, and volume status

The environmental strand rests on modeled estimates and observational associations, but their magnitude is striking. The Lancet Countdown estimated a total of 62,775 heat-attributable deaths in Europe in 2024, and mortality rose in 820 of 823 regions (13.13). In a US cohort of 945,251 patients on dialysis, extreme humid heat was associated with an HR of 1.18 (95% CI, 1.15 to 1.20), and a post hoc analysis of DAPA-CKD in 4,017 participants associated more days with a heat index above 30 °C with faster eGFR decline (13.13). The journalistic review does not directly address AKI during heat waves, and the kidney-related interpretation in it is labeled as the author's own (13.7).

A California study of 283 agricultural workers found AKI during a single shift in 12.3%, and piece-rate pay, with an OR of 4.24, was a stronger predictor than heat stress itself, with an OR of 1.34 (13.27). Population-level evidence of a CKD epidemic in Texas is lacking (13.27). AKI with rhabdomyolysis can threaten even a healthy athlete (13.10).

The practical conclusion runs against blanket advice. The universal recommendation to drink more is wrong in patients with advanced CKD, heart failure, or anuria and in those on hemodialysis, and KDIGO 2024 allows temporary interruption of RAAS inhibitors, diuretics, SGLT2 inhibitors, metformin, and NSAIDs only during acute dehydrating illness, not on every hot day; a common error is forgetting to restart them (13.13). Diuretics and beta-blockers can worsen circulatory stability in the heat (13.7), and hyponatremia from excessive free water intake is a risk in sports as well (13.10).

Beverages, sweeteners, and social media trends

On coffee, the chapter counsels calm. In the UK Biobank, an intake of at least 5 cups daily was associated with an HR of 0.68 (95% CI, 0.58 to 0.79) for cirrhosis and an HR of 0.53 for hepatocellular carcinoma, but kidney outcomes were not assessed, and the authors' recommendation to drink coffee for prevention is worded more strongly than the observational design can support (13.23). Microbiome studies of coffee did not include advanced CKD, and their clinical significance is unknown (13.12). Alcohol, by contrast, has no net protective effect, is carcinogenic, and in CKD its indirect risks predominate, particularly volume depletion and AKI (13.11).

Sweeteners are an area of weak signals. The kidney associations are observational: in the UK Biobank, an HR of 1.26 (95% CI, 1.12 to 1.43) for more than one serving of artificially sweetened beverages daily, and in a meta-analysis an RR of 1.40 with an interval of 0.65 to 3.02 (13.26). The signal for xylitol comes from a non-peer-reviewed conference abstract from ESC 2026 with HRs of 1.47 and 1.18; blood levels, not intake, were measured (13.29).

Online trends usually have a true core that is overgeneralized. Chia seeds are a source of fiber, but a cleansing effect has not been demonstrated (13.6). A series of meta-analyses in the Lancet associated 25 to 29 g of fiber daily with a 15% to 30% lower risk of mortality and chronic disease, but it excluded persons with chronic disease, so its application to CKD is an extrapolation (13.22). Colon cleansing has no evidence of benefit, and sodium phosphate caused acute phosphate nephropathy in 21 patients (13.19); blinded challenge confirmed subjective gluten sensitivity in only 16% (13.17); and the rigid gutmaxxing protocol may conceal an eating disorder (13.21). Physical activity shows favorable associations from as few as 7,000 steps daily, with an HR of 0.53 compared with 2,000 steps, yet patients with CKD take an average of about 4,640 steps (13.25), and in atrial fibrillation, physical activity

does not replace anticoagulation (13.18). The microbiome remains a field for research, not for treatment rules (13.1).

What is proven, what is probable, and what remains open

At the level of RCTs and meta-analyses, it is proven that creatine raises creatinine without changing eGFR (13.15), that a very low-protein diet with ketoanalogues brought no survival benefit in a pragmatic RCT and had poor adherence (13.30), that muscle mass no longer increases above 1.62 g of protein/kg/day (13.19), and that most gluten sensitivity is not specific to gluten (13.17). Probable, but based on observational data, are the protective effect of a predominantly plant-based, minimally processed diet (13.24, 13.5), the neutral effect of higher protein intake on healthy kidneys (13.16), the benefit of dietary fiber (13.22), and the harm of heat stress to patients on dialysis (13.13).

Almost everything related to causality and intervention remains open: whether a plant-based diet slows CKD progression in a randomized design (13.24), how safe creatine is in advanced CKD and after transplantation (13.15), whether fiber improves hard kidney outcomes by changing fermentation (13.22), whether microbiome changes precede kidney damage or result from it (13.1), and whether the signal for xylitol and erythritol reflects intake or endogenous production in hyperglycemia (13.29, 13.26).

Implications for practice

For a Slovak outpatient clinic and dialysis center, the chapter yields a simple framework. Dietary counseling should aim for a predominantly plant-based, minimally processed diet with a numerical protein target by stage: approximately 0.8 g/kg/day in CKD G3 to G5 not on dialysis, 1.0 to 1.2 g/kg/day on dialysis, and, in a patient who is losing weight or strength, a looser rather than a stricter plan (13.30, 13.20). Potassium and phosphorus should be managed according to serum potassium, serum phosphate, and the share of additives, not by a blanket ban on fruit, legumes, and coffee (13.22, 13.23, 13.2). The history should include an active question about supplements and regimens, and a rise in creatinine after them should be verified with cystatin C and albuminuria (13.21, 13.15).

Before summer, an at-risk patient, that is, one with CKD G4 to G5, after an episode of AKI, on dialysis, or frail, should have a written individual fluid and medication plan with an escalation contact and monitoring of weight, BP, and urine output (13.13, 13.7). For those working in hot workplaces, a targeted occupational history is appropriate (13.27). The physical activity target should be set from a measured baseline and increased in increments of 500 to 1,000 steps daily, combined with resistance training (13.25, 13.18). With accumulating restrictions, in older age, and during incretin therapy, malnutrition and sarcopenia should be kept in mind and a dietitian involved (13.2, 13.17, 13.26).

What to watch next

Attention should go to peer-reviewed publications of the conference abstracts on the Planetary Health Diet and xylitol (13.5, 13.29), to the prospective longitudinal microbiome cohorts called for by the authors of the largest cross-sectional study (13.1), to long-term RCTs of creatine stratified by G category with measured GFR and cystatin C (13.15), and to fiber studies with hard kidney outcomes (13.22). Data of practical importance will show whether sick day rules during heat waves bring benefit, because the evidence is limited (13.13), and whether CKDnt will be confirmed at the population level outside the traditional regions (13.27). The reassessment of the safety of selected additives called for

by the NutriNet-Santé authors may change the regulatory environment rather than clinical recommendations (13.4).

Articles in this chapter

- 13.1** The gut microbiome and kidney health: what the largest study to date brings (May 24, 2026, online, Slovak)
- 13.2** Diet, fats, and cardiovascular risk in chronic kidney disease (CKD): what makes sense in practice (May 24, 2026, online, Slovak)
- 13.3** How “plant-based” is the current direction of dietary recommendations, and why it makes sense (May 27, 2026, online, Slovak)
- 13.4** Food additives under the microscope: new data link colorants and preservatives with a higher risk of chronic diseases (June 10, 2026, online, Slovak)
- 13.5** Plant-based diet and lower mortality in patients with CKD: what to use in practice (June 16, 2026, online, Slovak)
- 13.6** The “chia seed internal shower”: what is true and what is not from a medical perspective (June 17, 2026, online, Slovak)
- 13.7** Extreme heat and underestimated health risks: what it also means for nephrology (June 26, 2026, online, Slovak)
- 13.8** Protein and creatine are no longer exclusively a “fitness topic”: what follows for nephrology (June 27, 2026, online, Slovak)
- 13.9** Creatine and brain health: is it worth a try, or is it just another supplement? (July 1, 2026, online, Slovak)
- 13.10** Hydration breaks in soccer: a marketing move or a medical necessity? (July 10, 2026, online, Slovak)
- 13.11** Alcohol as a “healthy drink”: why medical recommendations are changing (July 10, 2026, online, Slovak)
- 13.12** Coffee and the gut microbiome: the effect goes beyond caffeine (July 10, 2026, online, Slovak)
- 13.13** Extreme heat: a growing risk for patients with kidney disease (July 17, 2026, online, Slovak)
- 13.14** Can the smell of chocolate improve performance in strength training? (July 24, 2026, online, Slovak)
- 13.15** Creatine in kidney disease: harm, benefit, or diagnostic trap? (August 4, 2026, online, Slovak)
- 13.16** Higher protein intake and kidney function: a ten-year cohort found no faster decline in measured GFR (August 14, 2026, online, Slovak)
- 13.17** Non-celiac wheat sensitivity: gluten may not be the only trigger (August 14, 2026, online, Slovak)
- 13.18** Physical activity in atrial fibrillation: lower risk of stroke and death, not a substitute for anticoagulation (August 14, 2026, online, Slovak)
- 13.19** Diet and gut health according to influencers: where the most common errors arise and what they mean for nephrology practice (August 20, 2026, online, Slovak)
- 13.20** US Dietary Guidelines 2025 to 2030: the risk of misinterpretation in MASLD and CKD (August 22, 2026, online, Slovak)
- 13.21** “Gutmaxxing”: when the pursuit of a healthy gut turns into a rigid protocol, and what it means in the nephrology outpatient clinic (August 28, 2026, online, Slovak)
- 13.22** “Fibermaxxing”: higher fiber intake brings benefit; the question is where, and for whom, the curve levels off (August 28, 2026, online, Slovak)
- 13.23** Coffee and the liver in the UK Biobank: lower risk of cirrhosis and hepatocellular carcinoma, and what applies to the patient with kidney disease (August 28, 2026, online, Slovak)
- 13.24** Vegetarian diet and lower risk of incident chronic kidney disease in the UK Biobank: what applies to practice (August 28, 2026, online, Slovak)
- 13.25** How many steps a day are really enough? What the latest dose-response analysis shows and how to apply it in nephrology (September 3, 2026, online, Slovak)
- 13.26** Sweeteners in MASLD, CKD, and incretin therapy: evidence and uncertainties (September 5, 2026, online, Slovak)
- 13.27** Chronic kidney disease of unclear etiology in heat-exposed workers: is CKDnt arriving in Texas? (September 6, 2026, online, Slovak)
- 13.28** Ketogenic diet and metabolic health: promising results for the liver, not proof of universal superiority (September 11, 2026, online, Slovak)
- 13.29** Xylitol and cardiovascular risk: what the new cohort data really show and what applies in nephrology (September 17, 2026, online, Slovak)
- 13.30** Protein in CKD: the question is not only “how much” but also “for whom” (September 24, 2026, online, Slovak)

CHAPTER 14

Internal medicine and related specialties

Hepatology, rheumatology, psychiatry, dermatology, and other specialties a nephrologist meets in everyday internal medicine practice. Number of articles: 19.

The chapter on internal medicine and related specialties brings together nineteen articles from May to September 2026, united by a single question: what from neighboring specialties does the nephrologist need to know so that a patient with chronic kidney disease (CKD) is treated neither inadequately nor unnecessarily? Thematically, the chapter falls into four strands: metabolic dysfunction-associated steatotic liver disease (MASLD) and its targeted treatment (14.9, 14.17, 14.18, 14.19), psychiatric diagnoses in patients with somatic illness (14.8, 14.10, 14.13, 14.14), new drugs from dermatology and rheumatology that run into kidney-related restrictions (14.5, 14.11, 14.12, 14.15), and diagnostic restraint in infections, inflammatory markers, and functional gastrointestinal disorders (14.4, 14.6, 14.7, 14.16). At the margins, it adds an epidemiologic signal regarding tattoos (14.1), a preclinical concept of a blood test for the glycocalyx (14.3), and information about a hepatology educational event (14.2).

The predominant type of evidence is not the randomized trial but the review, the cohort, and expert opinion. Only a few original RCTs appear in the chapter: ESSENCE in MASH (14.17), UP-AA1 and UP-AA2 in alopecia areata (14.15), SPARTACUS in peripheral spondyloarthritis (14.5), and the Spanish antibiotic trial in cystitis (14.4); none was designed for patients with CKD. The chapter's main message is therefore not that a new nephrology treatment has emerged, but that effective drugs and tools from other specialties are multiplying, that their evidence base in reduced glomerular filtration rate is lacking, and that their dosing should be read from the summaries of product characteristics, not from press releases (14.15, 14.17, 14.9).

Steatotic liver as part of kidney risk

The common starting point of the four articles on MASLD is that the disease shares risk factors with diabetes, obesity, cardiovascular disease, and CKD, and that prognosis is determined by the stage of fibrosis, not by steatosis itself (14.9). The observational NHANES 2007 to 2018 cohort of 14,818 adults showed the worst prognosis with coexisting CKD and advanced fibrosis, but the groups differed markedly in age, diabetes, and obesity, and the cohort does not prove causality (14.18). The proposed cardiovascular, renal, hepatic, and metabolic syndrome is a conceptual proposal from a narrative review, not an accepted classification (14.18).

There is agreement on stepwise fibrosis triage. A FIB-4 below 1.3 indicates low risk (threshold 2.0 above age 65), a value of 1.3 to 2.67 calls for elastography or the ELF test, and a value above 2.67 calls for hepatology assessment; a low FIB-4 may miss about one-tenth of advanced fibrosis cases (14.9, 14.18). The review authors' proposal to test every patient with CKD and one risk factor goes beyond the wording of the 2026 US recommendations and should be labeled as such (14.18). More important for the nephrologist is that aminotransferases tend to be lower in patients on hemodialysis, FIB-4 thresholds are not validated, nonhepatic thrombocytopenia distorts the result, and volume overload

affects liver stiffness; in a pilot study of 41 patients on hemodialysis, approximately 10% exceeded the 8 kPa threshold after dialysis (14.9, 14.18).

The evidence for targeted treatment of MASH varies in strength. The phase 3 ESSENCE RCT showed resolution of steatohepatitis without worsening of fibrosis after 72 weeks in 62.9% of patients with semaglutide 2.4 mg weekly versus 34.3% with placebo (14.17). Tirzepatide has histologic evidence only from phase 2 and has no MASH indication in the EU; retatrutide reduced liver fat on MRI without a histologic endpoint and remains experimental (14.17). Resmetirom and the semaglutide product Kayshild hold conditional marketing authorization in the EU for noncirrhotic MASH with F2 to F3 fibrosis based on surrogate endpoints; they have not been shown to reduce decompensation, transplantation, or mortality, Kayshild is not recommended at an eGFR below 30 mL/min/1.73 m², and resmetirom interacts with statins and CYP2C8 inhibitors (14.9, 14.17, 14.18). Pioglitazone has a histologic signal, but PIVENS, with $p = 0.04$, did not meet the prespecified threshold of $p < 0.025$, fibrosis did not improve, and in IRIS edema (35.6% versus 24.9%) and serious fractures (5.1% versus 3.2%) increased, which outweighs the benefit in CKD (14.19). SGLT2 inhibitors, RAS blockade, and finerenone are not treatments for MASH (14.9, 14.18, 14.19).

The kidney benefit of incretin-based drugs must be separated from their hepatic effect. The FLOW RCT showed that semaglutide 1 mg in 3,533 patients with type 2 diabetes and CKD reduced a composite kidney outcome by 24% (HR 0.76; 95% CI, 0.66 to 0.88); the evidence applies to the specific molecule and dose and cannot be extrapolated to tirzepatide or retatrutide (14.17). The overlap of liver disease with kidney dysfunction and undernutrition is also highlighted by a hepatology livestream that presents no original data (14.2).

Psychiatric diagnosis only after a metabolic cause is excluded

Psychosis is a syndrome, not an etiologic diagnosis, and whether it is substance-induced or primary cannot be determined by a single symptom or a toxicology test (14.8). A meta-analysis of 50 studies with 40,783 individuals showed conversion of substance-induced psychosis to schizophrenia in 25% overall and in 34% after cannabis, but these are heterogeneous observational data (14.8). In patients on dialysis, uremic encephalopathy, disorders of sodium, calcium, and magnesium, hypoglycemia, sepsis, drug toxicity, dialysis disequilibrium syndrome, and thiamine deficiency should be excluded before a psychiatric diagnosis is made; there is no uniform renal dosing for antipsychotics (14.8).

For coexisting psychosis and substance use, the review relies on NICE CG120 and a Cochrane review of 41 randomized trials that found no high-quality evidence of the superiority of any psychosocial intervention; the signal for clozapine comes from low-quality observational data (14.10). For the nephrologist, the physical risks are key: stimulants and synthetic substances can cause hyperthermia, rhabdomyolysis, and acute kidney injury, chronic hyponatremia must not be corrected at an uncontrolled rate, and dehydration, NSAIDs, ACE inhibitors, and ARBs increase the risk of lithium toxicity (14.10).

According to a Hong Kong population-based cohort (3,092 persons versus 902,927 controls), borderline personality disorder carries an HR for all-cause mortality of 5.65, with an estimated loss of 13.01 years of life (14.13). After adjustment for socioeconomic indicators, the evidence for cardiovascular risk weakened to borderline significance; BPD is not a proven independent kidney risk factor, and the main danger remains diagnostic overshadowing (14.13). The same principle applies to suspected ADHD in adults: the six-item ASRS had a specificity of 99.5% but a sensitivity of only 68.7%; in CKD, uremia, anemia, sleep apnea, and polypharmacy contribute to attention impairment; and a meta-ana-

lysis of 22 studies with 46,107 participants showed that methylphenidate increases both heart rate and systolic BP, although the CKD population was not included (14.14).

New drugs from related specialties and kidney limits

Effective treatments from dermatology and rheumatology arrive without kidney data. Two parallel phase 3 RCTs with 1,399 patients showed that upadacitinib achieved the primary endpoint of SALT ≤ 20 at week 24 in 44.6% to 55.0% of patients versus 1.5% to 3.4% with placebo; however, SALT ≤ 20 means at least 80% scalp hair coverage, not complete regrowth, and the media figure of 23% for SALT 0 was not confirmed in the primary summary (14.15). The time horizon was only 24 weeks, data on older patients and kidney outcomes are lacking, and the drug is not recommended at an eGFR below 15 mL/min/1.73 m² (14.15).

The SPARTACUS RCT, presented at EULAR 2026, supported immediate golimumab in peripheral spondyloarthritis: remission at 24 weeks was achieved in 60% of patients versus 33% with a step-up strategy with methotrexate (OR 3.1; $p = 0.006$) (14.5). However, this is a conference abstract with 97 patients that does not address long-term remission or radiographic progression, and TNF inhibitors are not approved for this diagnosis (14.5). In axial spondyloarthritis, the revised ASAS-SPARTAN 2025 criteria remain a research tool; for the nephrologist, what matters is that NSAIDs as the first pharmacologic choice can lead to a decline in glomerular filtration rate, hyperkalemia, and acute kidney injury, particularly in combination with RAS blockade and a diuretic, and that in CKD the argument for earlier targeted therapy is stronger (14.12). Topical finasteride has the weakest support: a retrospective cohort of 123 men without a control group or blinding does not prove equivalence with oral treatment or interchangeability of formulations, which differ in applied dose approximately 5.5-fold, and studies in impaired kidney function are lacking (14.11).

Infections, markers, and diagnostic restraint

A Spanish randomized trial in 768 women showed clinical resolution of symptoms by day 7 in 74% with five-day nitrofurantoin versus 59% with single-dose fosfomycin, although the sample may not have been large enough to distinguish smaller differences (14.4). For recurrent infections in older women, expert opinion emphasizes that some complaints merely mimic cystitis, that a negative urine test without nitrites and leukocytes has a high negative predictive value, and that the main pillar of prevention in genitourinary syndrome of menopause is local vaginal estrogen; older patients with CKD are more vulnerable to the adverse effects of antibiotics (14.6).

CRP, ESR, and plasma viscosity are sensitive but poorly specific; a CRP above 10 mg/L almost always indicates acute disease and an ESR above 100 mm/h has a 97% predictive value for underlying disease, but a mild rise in CRP also accompanies obesity and diabetes (14.7). In irritable bowel syndrome, the Rome criteria allow a positive diagnosis, yet a cross-sectional study of 238 medical students showed that 76% rated a typical presentation as functional and 70% ordered tests anyway; in CKD, anemia, bleeding, or nocturnal diarrhea should be investigated, not attributed to the kidneys (14.16). The tattoo signal calls for the same caution: a Swedish study found a 21% higher, statistically nonsignificant risk of lymphoma, and a Danish twin study found a 62% higher risk of skin cancer; these are observational associations subject to confounding (14.1). The concept of a blood test of the erythrocyte glycocalyx rests only on animal models (14.3).

What is proven, what is probable, and what remains open

Proven at the RCT level are the histologic improvement of MASH with semaglutide 2.4 mg and the kidney benefit of semaglutide 1 mg in diabetes and CKD (14.17), the efficacy of upadacitinib for SALT ≤ 20 (14.15), the short-term advantage of immediate golimumab (14.5), and the superiority of five-day nitrofurantoin over single-dose fosfomycin (14.4). Probable, but based on observational data, are worse survival with coexisting CKD and advanced fibrosis (14.18), higher mortality in borderline personality disorder (14.13), and the transition of substance-induced psychosis to schizophrenia in one-quarter of patients (14.8). Fibrosis triage in hemodialysis (14.9), the diagnosis of recurrent infections (14.6), and the interpretation of inflammatory markers (14.7) remain expert opinion.

It remains open whether any targeted treatment of MASH will reduce decompensation or mortality (14.9, 14.17), whether the histologic signal of pioglitazone outweighs its fluid-related and skeletal risks (14.19), what the long-term risks of upadacitinib and golimumab are in the populations excluded from the trials (14.15, 14.5), whether tattoos are causally related to lymphoma (14.1), and whether the glycocalyx test will reach patients (14.3). In dialysis, neither FIB-4 nor elastography thresholds are validated (14.9, 14.18), and no psychiatric diagnosis in the chapter has a proven independent kidney mechanism (14.13, 14.14).

Implications for practice

In patients with CKD and type 2 diabetes or abdominal obesity, it makes sense to calculate FIB-4 in a targeted way and, at a value of 1.3 to 2.67, to add elastography or ELF; in hemodialysis, however, measurement should be done at a standardized time and with awareness that lower aminotransferases and volume status distort the result (14.9, 14.18). Targeted treatment of MASH should be chosen according to the strength of the evidence and kidney-related limits: semaglutide 2.4 mg has phase 3 evidence, but Kayshild is not recommended at an eGFR below 30 mL/min/1.73 m², resmetirom interacts with statins, and pioglitazone is suitable only for selected patients without congestion or fracture risk (14.17, 14.9, 14.19). With incretin-based drugs, hydration, glycemia during concomitant insulin therapy, and protein intake in advanced CKD should be monitored (14.17).

Before a psychiatric diagnosis is made in a patient on dialysis, the first priority is to exclude a metabolic, infectious, or drug-related cause (14.8, 14.14), and in a patient with a known psychiatric diagnosis, vigilance is needed against diagnostic overshadowing of an acute coronary syndrome, embolism, or acute kidney injury (14.13). With NSAIDs, lithium, methylphenidate, and new immunomodulatory drugs, BP, creatinine, eGFR, potassium, and interactions should be checked (14.12, 14.10, 14.14, 14.15). Antibiotics for recurrent urinary tract infections should follow only after accurate confirmation of infection (14.6, 14.4), and inflammatory markers, tests for functional bowel complaints, and media signals should be read in clinical context without escalating investigations (14.7, 14.16, 14.1).

What to watch next

In the near term, it is worth following the final ESSENCE results and data on clinical liver events with semaglutide and resmetirom, which alone can convert conditional marketing authorization into full authorization (14.17, 14.9); phase 3 trials of tirzepatide and retatrutide in MASH (14.17); validation of FIB-4 and elastography in patients on dialysis beyond the pilot study of 41 persons (14.18); long-term safety data for upadacitinib in reduced eGFR, together with the reimbursement decision in Slovakia (14.15); full publication of SPARTACUS with data on radiographic progression (14.5); the first human studies of the glycocalyx test (14.3); prospective verification of the tattoo and lymphoma signal (14.1); and whether systematic exclusion of metabolic and drug-related causes before a psychiatric diagnosis

is becomes established in centers treating patients with CKD and psychiatric comorbidity (14.8, 14.10, 14.13).

Articles in this chapter

- 14.1** Tattoos and the risk of lymphoma or skin cancer: a signal that deserves attention (May 17, 2026, online, Slovak)
- 14.2** News in hepatology: what the STREAMED UP livestream will offer (May 19, 2026, online, Slovak)
- 14.3** A new simple blood test could detect vascular damage, and thus heart and kidney risk, earlier (May 24, 2026, online, Slovak)
- 14.4** Nitrofurantoin vs. fosfomycin: what a new Spanish study shows in uncomplicated cystitis in women (May 27, 2026, online, Slovak)
- 14.5** Early TNF inhibitor treatment in peripheral spondyloarthritis led to higher remission rates than step-up csDMARD therapy (June 10, 2026, online, Slovak)
- 14.6** Recurrent urinary tract infections in older women and the link to genitourinary syndrome of menopause (GSM): diagnosis, antibiotics, and prevention without them (July 1, 2026, online, Slovak)
- 14.7** How to interpret inflammatory markers in primary care: CRP, ESR, and plasma viscosity (PV) in practice (July 1, 2026, online, Slovak)
- 14.8** Primary or substance-induced psychosis? Differential diagnosis and principles of acute management (August 5, 2026, online, Slovak)
- 14.9** Metabolic dysfunction-associated steatotic liver disease: diagnosis, fibrosis assessment, and relevance for nephrology practice (August 13, 2026, online, Slovak)
- 14.10** Psychosis and coexisting substance use: integrated assessment, treatment, and protection of physical health (August 13, 2026, online, Slovak)
- 14.11** Topical finasteride in male androgenetic alopecia: efficacy at 52 weeks and the limits of the evidence (August 14, 2026, online, Slovak)
- 14.12** Axial spondyloarthritis: early diagnosis, imaging, and individualized treatment (August 15, 2026, online, Slovak)
- 14.13** Borderline personality disorder and physical health: higher morbidity cannot be explained by the psyche alone (August 15, 2026, online, Slovak)
- 14.14** Diagnosing ADHD in adults: what it takes to avoid mistaking distractibility for a diagnosis (August 20, 2026, online, Slovak)
- 14.15** The JAK1 inhibitor upadacitinib in severe alopecia areata: what the new phase 3 trials show and how to read media claims correctly (September 3, 2026, online, Slovak)
- 14.16** Rome criteria for disorders of gut-brain interaction (DGBI) and when to “test further” in IBS: lessons from a study of medical students (September 3, 2026, online, Slovak)
- 14.17** Incretin-based agonists in MASLD and MASH: liver protection and relevance for the nephrologist (September 13, 2026, online, Slovak)
- 14.18** Steatotic liver disease: why it belongs in the risk assessment of patients with chronic kidney disease (September 13, 2026, online, Slovak)
- 14.19** Pioglitazone in MASLD and MASH: histologic benefit vs. cardiorenal risk (September 24, 2026, online, Slovak)

CHAPTER 15

Artificial intelligence, digital medicine, and experimental research

Artificial intelligence, wearable sensors, regulation of digital tools, and experimental research from organoids to regenerative medicine. Number of articles: 19.

The chapter on artificial intelligence, digital medicine, and experimental research is the most heterogeneous in this volume. It brings together nineteen texts that range from economic scenarios for the development of artificial intelligence up to 2030 (15.18), through legal and regulatory reviews (15.10, 15.16), to preclinical work on autophagy in tubular cells (15.3), kidney organoids (15.11, 15.17), and wound healing in a jellyfish (15.12).

The type of evidence defines this chapter. Review articles, commentaries on news and professional sources, and opinion pieces predominate; randomized data appear only indirectly, for example in the form of a trial of 6,030 patients that showed no benefit of AKI alerts in the electronic health record (15.7). As of October 2026, the chapter's main message is therefore sobering: the technical performance of algorithms is well documented, evidence of improved patient outcomes is rare, and most recommendations for practice have the character of expert opinion rather than conclusions from interventional studies.

Predictive models and the gap between performance and benefit

Predictive models form the most solid part of the evidence base. A review of machine learning in nephrology reports that models can predict AKI up to 48 hours before the need for dialysis, that extreme gradient boosting outperformed traditional models in a retrospective analysis of 2,010 patients after vascular and thoracic procedures, and that a CKD progression model achieved C-statistics of approximately 0.84, 0.81, and 0.79 at horizons of 1, 2, and 5 years, compared with approximately 0.77 for KidneyIntelX (15.7). The same sources, however, show the other side: a model trained on 703,782 patients had a higher false-positive rate and questionable transferability because men predominated in the dataset, and a randomized trial of 6,030 patients did not show that EHR alerts shortened hospital stay or reduced mortality (15.7). Good discrimination is therefore not the same as clinical benefit, which is probably the most important methodological lesson of the entire chapter.

In ADPKD the picture is similar but more concrete. Automated deep-learning segmentation of the kidneys on MRI or CT agrees closely with expert segmentation, shortens analysis, and improves reproducibility, but performance is degraded by differing protocols, very large kidneys, exophytic cysts, or prior procedures (15.13). The review authors therefore insist on visual verification of every TKV and point out that KDIGO 2025 recommends the Mayo Imaging Classification based on htTKV and age, with subclasses 1C to 1E among the criteria for rapid progression when tolvaptan is being considered; however, the MIC applies only to typical morphology and should not be used for a pathogenic variant outside PKD1 and PKD2 (15.13).

In transplantation, the iBox system has advanced furthest: it predicted the long-term risk of graft failure better than nephrologists and has been externally validated in the United States and Europe

(15.7). By contrast, the reflection that a nephrologist who can program may build custom predictive tools for a dialysis center is an opinion piece without trial data, and it merely opens the question of real-world use in Slovakia (15.1).

Generative AI in the outpatient clinic and the physician's responsibility

The second strand concerns generative language models and their place in clinical work. The most valuable empirical observation, although not a validation study, is the experience from a hands-on workshop in New York in March 2026: identical prompts led to markedly different recommendations and to a different logical structure of reasoning in ChatGPT, Gemini, and Claude (15.8). In assessing photographs of meals and of the catheter exit site in peritoneal dialysis, AI sometimes achieved high agreement with an expert but was sensitive to image quality (15.8). For the clinician, inconsistency between models is a more fundamental problem than hallucinations themselves, because it undermines the notion that the answer of a single tool can be used as a reference.

The commentary on the legal risks of AI scribes carries this uncertainty over into documentation. The risks center on three points: the vendor's compliance in storing and accessing records, documented patient consent, and the physician's full responsibility for the accuracy of the record (15.10). In nephrology this is sensitive for drug doses, creatinine and eGFR, proteinuria, electrolytes, and dialysis treatment records, so the author proposes an approved vendor selection, a standard consent script, and a rule that an unverified record does not enter the documentation without correction (15.10).

The third text in this strand addresses the situation in which a patient arrives with a diagnosis from a chatbot. The author points out that KDIGO defines CKD as an abnormality of kidney structure or function lasting at least three months, so a conclusion of kidney failure based on a single eGFR value of 52 mL/min/1.73 m² is unfounded, and proposes a six-step approach ranging from exclusion of an urgent condition to a concrete plan, including which medications remain unchanged (15.14). An important caution is that both patient and physician are subject to anchoring and that a hypothesis should be judged by the evidence, not by its origin; names, birth numbers (Slovak personal identification numbers), and complete medical reports do not belong in unverified chatbots (15.14).

Regulation, equity of access, and new devices

The regulatory review brings to the chapter a framework that most commentaries on specific tools lack. According to AMA data, more than 80% of physicians use AI professionally, compared with 38% in 2023, and the August 2026 FDA CDRH discussion paper on generative AI devices is neither a draft nor binding guidance (15.16). The key observation is that most AI devices have been cleared through the 510(k) pathway based on equivalence to an existing device, so regulatory clearance need not rest on prospective studies, and good discrimination does not mean lower mortality (15.16). Liability remains contested: a proposal by the Washington Medical Commission placed it entirely on the physician, whereas the AMA chief executive officer proposed assigning it according to who best understands the risk of the tool (15.16).

A 2022 report by the WHO Regional Office for Europe adds the dimension of equity. A scoping review of 22 reviews and meta-analyses showed consistently higher use of digital health in cities and among people who are better educated, economically better off, younger, and members of the majority language group; the quality of the included studies was low, and the search ended in May 2022, before generative AI became widespread (15.15). For nephrology, the key warning is that algorithms trained on historical data may misinterpret low use of care as low need, so a CKD progression model may en-

trench inequity and lend it an appearance of objectivity (15.15). These are associations from observational reviews, not proof of causality, but the trend is consistent across sources.

Wearable devices and AI-assisted POCUS also belong to this strand. The review of wearable devices reports the strongest benefit in diabetes, heart failure, and arrhythmias, but in CKD a smartwatch measures neither eGFR nor albuminuria, and in patients on dialysis, interdialytic trends are only a supplement to, not a substitute for, assessment of volume status; meanwhile, the devices can generate more than 5 GB of data per patient per week (15.6). AI-assisted point-of-care ultrasound for nurse practitioners and physician assistants is described as a triage tool, not a substitute for formal echocardiography; its transfer to nephrology is the author's reflection, and Robert Klempfner, cited in the source, is a cofounder and the chief medical officer of AISAP (15.9). The declared conflict of interest should be weighed as carefully as the claims themselves.

The relational and ethical dimension of technology

Three opinion pieces frame technology in terms of the physician-patient relationship. The interview with Ann O'Hare challenges the assumption that time saved by AI will automatically bring more humane care and warns that an AI scribe that deletes the nonessential parts of a conversation may strip the record of its human dimension; her conclusion that the future of nephrology depends mainly on what kind of physicians we choose to be is an opinion without empirical data (15.4). The commentary on the emotional persuasiveness of chatbots shifts attention from hallucinations to the risk that an AI that appears empathetic and is always available will create intimacy and reinforce distorted beliefs; according to the author, those most at risk are adolescents, people with psychotic spectrum disorders, and the socially isolated, and motivational interviewing is recommended instead of moralizing (15.5). The economic scenarios through 2030 from The Anthropic Institute are not a peer-reviewed publication; in the extreme scenario, GDP is 32.4% above the trajectory without AI, but the authors explicitly do not regard the scenarios as forecasts and expect a mixed model for healthcare (15.18).

Experimental research as a biological framework

The preclinical section of the chapter is the cleanest methodologically but the most distant clinically. The review of selective autophagy describes mitophagy, lipophagy, and ER-phagy as pathways that determine both repair and maladaptation after kidney injury, with insufficient, excessive, or poorly timed flux potentially promoting ferroptosis, senescence, and fibrosis during the transition from AKI to CKD (15.3). The authors reject the crude approach of simply enhancing or blocking autophagy and call for cell-specific modulation with stratification by biomarkers of autophagic flux, which are not yet available (15.3).

Kidney models are improving but remain immature. Synthetic Wnt-secreting organizers strengthened distal nephron programs and oriented nephrons toward the signal source at the WNT11 and WNT9B boundary, but the organoids still lack vascularization, a drainage system, and physiologic function (15.11). The review of kidney-on-a-chip models shows that shear stress influences polarity, cilia, and transporter expression, but higher flow is not automatically more physiologic, and permeability in a glomerulus-on-a-chip is not equivalent to glomerular filtration; the FDA Modernization Act 2.0 supports these models as an alternative to animal testing, but it does not abolish animal studies (15.17).

Two further studies are only indirectly relevant to nephrology. The wound-healing model in the jellyfish *Clytia hemisphaerica* describes scarless wound closure by lamellipodial crawling, actomyosin contraction, and collective migration, the choice among which is governed by the availability of the

basement membrane; the absence of a scar results from the lack of a vascular network and an inflammatory phase, so direct transfer to human skin is not possible (15.12). Inhibition of SagA with the compound pghi-4 reduced the vancomycin MIC in the ERV165 strain by up to eightfold and, in a murine experiment with six animals per group and an unblinded analysis, reduced the bacterial burden in the liver and spleen; renal elimination, removal by dialysis, and interactions with immunosuppressants remain unclear (15.19). The commentary on the shift of clinical research toward precisely defined diseases, biomarkers in phase 2, and the Asia-Pacific region is a review of trends from a sponsored source, and the author himself states that it does not yet change anything in routine care (15.2).

What is proven, what is probable, and what remains open

At the level of randomized data, practically the only proven finding in this chapter is a negative one: AKI alerts in the electronic health record in 6,030 patients did not improve length of hospital stay or mortality (15.7). It is probable, though based on observational data, reviews, and experience, that inconsistency of outputs between language models is a systemic property, not a matter of chance (15.8), that digital tools without targeted design deepen inequities (15.15), and that regulatory clearance says nothing about clinical benefit (15.16).

Almost everything related to patient outcomes remains open: whether AI scribes will improve or worsen the quality of documentation and the relationship with the patient (15.4, 15.10), how often patients arrive with a diagnosis from AI and how accurate these suggestions are (15.14), whether predictive models for ADPKD will hold up in other populations (15.13), who bears liability for harm (15.16), and whether preclinical findings on autophagy, organoids, and SagA inhibition will ever lead to biomarkers or drugs (15.3, 15.11, 15.19).

Implications for practice

For a Slovak outpatient clinic, hospital ward, or dialysis center, the chapter yields a framework that is organizational rather than clinical. Every AI output, whether a TKV, a predictive score, or a scribe's text, must be verified by a physician, and an unverified record does not belong in the documentation (15.10, 15.13). When selecting a tool, the vendor should be asked for validation data in a similar population, and deployment should start in silent mode with an audit before clinical use; regulatory clearance is not a sufficient argument (15.16). For digital services, particularly home dialysis and transplant follow-up, differences between subgroups should be measured, telephone and in-person channels should be preserved, and nonuse should not be taken as lack of interest (15.15).

In conversation with the patient, information from a chatbot should be assessed according to the evidence, facts should be separated from hypotheses, the CKD duration criterion should be verified, and the visit should end with a concrete plan (15.14); vulnerable patients should be asked sensitively about emotional use of AI (15.5), and when scribes are introduced, informed consent should be secured and it should be clear where the data are stored (15.10). Preclinical work on autophagy, organoids, wound healing, and antibiotic potentiation should be followed as a direction of development, not as a basis for treatment decisions (15.3, 15.11, 15.12, 15.19).

What to watch next

In the near term, it is worth following whether the FDA CDRH discussion paper progresses to draft guidance and how the question of liability is resolved (15.16), whether prospective studies with clinical outcomes appear for the prediction of AKI and CKD progression, not merely new C-statistics (15.7), whether an externally validated model of response to tolvaptan in ADPKD emerges (15.13), and wheth-

er data on renal elimination and toxicity are added for pghi-4 (15.19). In the Slovak setting, how these tools will actually be used remains a separate question (15.1).

Articles in this chapter

- 15.1** When nephrology meets code (April 25, 2026, online, Slovak)
- 15.2** Kidney research is changing: from large trials to more precise and targeted nephrology (May 19, 2026, online, Slovak)
- 15.3** Selective autophagy in the kidney: when cellular recycling determines both injury and repair (May 19, 2026, online, Slovak)
- 15.4** Will artificial intelligence make medicine more human again? (May 19, 2026, online, Slovak)
- 15.5** "The most dangerous thing" about AI: not hallucinations, but how AI can become emotionally persuasive (May 27, 2026, online, Slovak)
- 15.6** Wearables in chronic disease: more data is not enough; protocols and clinical purpose are what matter (June 10, 2026, online, Slovak)
- 15.7** Artificial intelligence in nephrology: what we already know, where the limits lie, and where it is heading (June 17, 2026, online, Slovak)
- 15.8** AI in nephrology in practice: what the "Hands-On Primer" offers for clinical reasoning and safe integration (June 21, 2026, online, Slovak)
- 15.9** AI-assisted POCUS speeds up cardiac decisions by NPs and PAs, and what it could mean for nephrology (July 1, 2026, online, Slovak)
- 15.10** Legal pitfalls of using AI scribes in the outpatient clinic and what they mean for nephrology practice (July 2, 2026, online, Slovak)
- 15.11** Synthetic Wnt-secreting organizers improve the spatial organization of human kidney organoids (July 5, 2026, online, Slovak)
- 15.12** A jellyfish that heals wounds without scars: what a simple model can teach regenerative medicine (July 7, 2026, online, Slovak)
- 15.13** Artificial intelligence in ADPKD: current possibilities and limits of clinical use (July 13, 2026, online, Slovak)
- 15.14** When the patient arrives with a diagnosis from AI: what it changes in the nephrology outpatient clinic (July 15, 2026, online, Slovak)
- 15.15** Digital health can also deepen inequities: what the evidence base says and what it means for nephrology practice (August 19, 2026, online, Slovak)
- 15.16** Regulation of medical artificial intelligence: what physicians should know and what it means for nephrology (September 9, 2026, online, Slovak)
- 15.17** Kidney-on-chip and kidney organoids: mechanical forces open new possibilities for research into kidney disease (September 9, 2026, online, Slovak)
- 15.18** Artificial intelligence through 2030: what different economic scenarios may mean for wages, work, and healthcare (September 9, 2026, online, Slovak)
- 15.19** SagA inhibition may enhance the effect of vancomycin against resistant *Enterococcus faecium* (September 12, 2026, online, Slovak)

CHAPTER 16

Congresses, the profession, and medicine in society

Congress reviews, research methodology, physicians' working conditions, and the ethical and social context of medicine. Number of articles: 9.

The chapter on congresses, the profession, and medicine in society stands apart from the clinical core of the book. Its nine texts, published from May to August 2026, bring no new randomized trials or registry cohorts; commentaries, congress reports, methodological reviews, and editorial reflections predominate, and several of the texts themselves point out that they are a personal opinion or educational material that does not replace individual clinical decision-making (16.4, 16.5, 16.8). The level of evidence is low in the classic sense of the term, yet the chapter teaches how to handle evidence, how to translate it into decisions, and how to talk with a patient whose laboratory values do not capture their condition.

The main message of the chapter is twofold. On the patient side, the texts converge on the conviction that communication style and attention to the experience of illness are clinically relevant variables, not a courtesy add-on (16.2, 16.4, 16.7, 16.8). On the side of the profession, they show that the quality with which congress findings are transferred, the stability of research funding, and the organization of work determine whether knowledge reaches the bedside at all and who will apply it there (16.5, 16.6, 16.9).

Language that is not understood and language that demeans

A commentary by Aryu Anthony Kamyab opens a problem that nephrology knows from everyday practice: justified simplification imperceptibly turns into infantilization. The author distinguishes health literacy from intelligence; an educated person may not know the term paresthesia, but the explanation “nerve-related tingling or pins and needles” is understandable without being childish (16.2). The opposite pole is explaining chemotherapy as treatment for “naughty cells,” which an adult finds demeaning and which affects both trust and willingness to ask questions (16.2). A distinct form is elderspeak, that is, diminutives, terms of address such as “sweetie,” needlessly raised volume, or the collective plural, as in “are we ready now?”; according to the author, studies suggest it can increase resistance to care (16.2).

It should be said what this text is and what it is not. It is a professional reflection combined with patient experience, not a quantitative study, and the claim about an effect on cooperation with treatment has no numerical support in the source (16.2). Its value lies in naming an error that physicians rarely notice in themselves and in offering a simple approach: begin by asking what the patient would like to know and whether they want a brief or a detailed explanation, use the technical term and explain it immediately, and verify comprehension as the conversation proceeds (16.2).

Disease, illness, and what the numbers do not show

A reflection on the difference between disease as a nameable disorder of organ function and illness as a disruption of ordinary functioning, routine, and one's relationship to one's own body adds content to the communication thread (16.4). A patient who says they are ill is usually describing the second dimension, and the recognition of illness often precedes the diagnostic process (16.4). For nephrology, the key paradox is that marked distress need not be accompanied by a clear-cut finding, and a patient with CKD or on dialysis may experience substantial functional limitation despite relatively stable laboratory values (16.4). The author concludes that the patient's account should be taken as a clinically relevant signal and that sleep quality, treatment tolerance, and functional capacity should be assessed in a targeted way, as these are factors that cannot be reduced to a single parameter (16.4).

A methodological review of qualitative research gives this intuition a research framework. Qualitative research answers a different type of question than an experiment or a registry: it explores meanings, experiences, and the context of decision-making, for example treatment burden, the choice of dialysis modality, acceptance of vascular access, or decisions about conservative care (16.7). The review also teaches critical reading: sample size is judged not by a power calculation but by the information power of the data, and claims of effectiveness or prevalence derived from themes are warning signs (16.7).

The example of a secondary analysis of semi-structured interviews with 20 US nephrologists is instructive precisely because of its limitations: participants often saw dialysis and hospice as mutually exclusive models and were unsure who should lead the conversation, but they came from a single city, 85% were classified as white, and 75% were men, so the result cannot be converted into population prevalence (16.7). Moreover, palliative care is not a synonym for hospice and can be provided concurrently with dialysis (16.7). The evidence on narrative medicine, which is meant to train this competence, is cautiously encouraging, not definitive: a 2019 review of 36 publications found measurable changes in attitudes, knowledge, and skills but no clear evidence of lasting behavior change, and a critical review from 2026 with 17 studies and 835 participants evaluated narrative medicine in only four of them, with high heterogeneity precluding meta-analysis (16.8).

Congresses and the discipline of knowledge transfer

Two congress reports illustrate two different approaches. The report on ECO 2026 in Istanbul is a pre-congress, thematic one: it recalls that obesity contributes to the development of hypertension, type 2 diabetes, albuminuria, glomerular hyperfiltration, and obesity-related glomerulopathy, and that in advanced kidney disease it affects transplant preparation, dialysis treatment, and nutritional status (16.1). A practical caveat is the warning that body mass index in patients on dialysis may not reliably reflect nutritional status, sarcopenia, or inflammatory burden, and that with incretin-based therapy, questions remain open about safety at reduced glomerular filtration rate and about the effect on albuminuria (16.1).

The post-congress analysis of ERA 2026 takes the opposite route and deliberately cites no specific trials or doses (16.6). Instead of news, it offers a working protocol: define the cardiovascular-dominant patient with CKD by history of events, blood pressure, albuminuria, and metabolic factors, introduce a CKD-CV check at every visit, and eliminate inconsistent parallel plans among nephrology, diabetology, and cardiology (16.6). In transplantation, it recommends not settling for the creatinine trend and setting decision points for escalation; in dialysis, monitoring stability, safety, and long-term goals; and in CKD monitoring, avoiding the error of measuring everything without knowing what it changes

(16.6). This is an expert recommendation of an organizational nature; the author himself asks readers to verify specific claims in the congress recordings (16.6).

Both texts share one methodological requirement with the overview of books: to distinguish biological plausibility, an experimental result, an observational association, and evidence of clinical benefit, because an improvement in a biomarker need not mean lower mortality, fewer hospitalizations, or better quality of life (16.8).

A profession under pressure

The Medscape survey, according to which 59% of physicians surveyed would consider a lower salary in exchange for better work-life balance, compared with 63% in the previous survey, reflects a stated preference, not a decision (16.9). The figure of 46% combines burnout, depression, or both, that is, distinct constructs, so it does not indicate the proportion of physicians with diagnosed depression (16.9). Its limitations are self-selection, self-assessment, a hypothetical question with no specified size of the salary reduction, and the limited transferability of US data (16.9).

More valuable than the survey are the meta-analytic data the text adds, and their partial contradiction. A meta-analysis by Panagioti and colleagues with 20 comparisons from 19 studies and 1,550 physicians showed a small but significant reduction in burnout (standardized mean difference, -0.29), with organization-directed interventions achieving -0.45 (95% CI, -0.62 to -0.28) and individual interventions only -0.18 (16.9). However, a more recent meta-analysis among physicians in specialty training, with 33 studies and 2,536 participants, found a small effect of individual interventions on emotional exhaustion (Cohen's d , -0.25) and depersonalization (-0.17), whereas organizational interventions showed no significant effect and made up less than a quarter of the studies (16.9). The contradiction more likely reflects different populations and the small number of organizational studies in residents than evidence that the organization of work does not matter. For nephrology, limited substitutability is decisive: in a center with two nephrologists, protected time off for one means shifting the burden onto the other (16.9).

Two texts are a reminder that medicine does not exist outside politics and violence. A commentary on events at the ADA links the confiscation of attendees' badges to the distribution of an editorial about cuts to NIH research and concludes that fewer grants may slow research on biomarkers and mechanisms of progression of diabetic kidney disease (16.5). This is a personal opinion, and contested claims should be verified in primary materials (16.5). A BBC report on more than 300 kidnapped migrants in Libya who were tortured and intimidated with threats of kidney removal, with ransom demands of US\$5,000, relies on testimonies and photographs, but it was unable to fully verify the removal of organs (16.3).

What is proven, what is probable, and what remains open

Little in this chapter is proven in the sense of controlled data: meta-analyses of interventions against burnout show small effects and a predominance of organizational measures in practicing physicians, while in residents the picture is inconsistent (16.9). Probable, though resting only on reflection, qualitative data, and reviews of limited quality, is that infantilizing language weakens trust, that the experience of illness despite stable laboratory values is clinically relevant, and that narrative competence improves attitudes and skills without evidence of lasting behavior change (16.2, 16.4, 16.7, 16.8).

What remains open is the safety of incretin-based therapy at reduced glomerular filtration rate and its effect on albuminuria (16.1), the actual impact of research cuts on nephrology innovation (16.5), the

extent of organ removal in Libyan captivity networks (16.3), and whether physicians' stated willingness to trade salary for time will survive contact with a real offer (16.9).

Implications for practice

For a Slovak outpatient clinic and dialysis center, the chapter yields a framework, not a protocol. Speak clearly without infantilizing, explain a technical term immediately, verify comprehension as the conversation proceeds, ask what is hardest for the patient today and what they fear most about treatment, and reflect the answer in the plan (16.2, 16.4, 16.7). Conversations about the choice of modality and about withdrawal of dialysis require narrative competence, which literature can support but which by itself will not make anyone a better physician (16.8).

It is reasonable to translate congress news into predefined decision points shared by the nephrology and dialysis team and to verify them in primary recordings (16.6, 16.5). In patients on dialysis with obesity, the assessment of nutritional status should not be based on body mass index (16.1). In refugees and victims of violence presenting with oliguria, changes in creatinine, hematuria, or hypertension, consider urgent nephrology evaluation combined with forensic documentation and psychosocial support (16.3). At the workplace level, prioritize teams with mutual cover, staffing standards, predictable scheduling, and limits on after-hours administrative work over individual advice on resilience (16.9).

What to watch next

In the near term, it is worth watching whether primary recordings emerge from ECO 2026 and ERA 2026 that confirm or refine the framework recommendations on incretin therapy at reduced filtration and on the cardiovascular-dominant patient with CKD (16.1, 16.6); whether qualitative and mixed-methods studies on decision-making about conservative care in the European context, with transparent reporting according to COREQ and SRQR, will be added (16.7); whether the evidence on narrative medicine will move from attitudes to behavior (16.8); how the grant environment for research on diabetic kidney disease will evolve (16.5); and whether organizational interventions against burnout tested in small specialized teams, reflecting the reality of Slovak nephrology, will emerge (16.9).

Articles in this chapter

- 16.1** ECO 2026: the European Congress on Obesity will be held in Istanbul (May 12, 2026, online, Slovak)
- 16.2** A condescending tone in medicine: when plain language stops being respectful (June 5, 2026, online, Slovak)
- 16.3** BBC: kidnappings of migrants in Libya and threats of kidney removal (June 16, 2026, online, Slovak)
- 16.4** When we feel ill: what medicine misses and why it matters in nephrology too (June 26, 2026, online, Slovak)
- 16.5** The new ADA as a prompt for reflection: research grants, political interference, and the impact on nephrology (June 27, 2026, online, Slovak)
- 16.6** Post-congress analysis of ERA 2026 for nephrology: how to translate the highlights into practice in CKD, dialysis, transplantation, and cardiovascular risk (July 1, 2026, online, Slovak)
- 16.7** Qualitative research in nephrology: what numbers do not reveal about decision-making in CKD (July 15, 2026, online, Slovak)
- 16.8** Twelve books that can help physicians better understand illness, the patient, and themselves (August 2, 2026, online, Slovak)
- 16.9** Physicians are not just asking for higher pay. Increasingly, they want control over their own time back (August 5, 2026, online, Slovak)

APPENDIX

Cheat sheets, checklists, and algorithms for practice

Concise overviews, checklists, and algorithms for quick orientation at the bedside and in the outpatient clinic. Number of articles: 11.

The appendix collects eleven tools from June 2026: an algorithm and three checklists on anemia of CKD (A.1, A.2, A.3, A.4), a one-page algorithm for the management of IgA nephropathy (A.5), four cheat sheets on fluid, electrolyte, and acid-base balance (A.6, A.7, A.8, A.9), and two cheat sheets on immune-mediated glomerulopathies (A.10, A.11). They do not offer new evidence; rather, they translate existing guidelines and textbook physiology into a form usable at the bedside.

The type of evidence is therefore derivative: the tools rest on the KDIGO 2026 guideline for anemia and the KDOQI US Commentary (A.1, A.2), on KDIGO 2025 for IgA nephropathy (A.5), on KDIGO risk stratification in membranous nephropathy (A.11), and, for intravenous fluids, on the SMART, BaSICS, and PLUS trials (A.9). The main message: a cheat sheet is only as good as its source, and its value lies in consistent decision-making and documentation.

Anemia of CKD as a model of structured decision-making

The four anemia tools (A.1, A.2, A.3, A.4) repeat the same logic: confirm anemia by hemoglobin and its trend, add a blood count with MCV, RDW, and reticulocytes, interpret ferritin and TSAT together with inflammatory status, and only then decide on iron (A.2, A.3). The thresholds are consistent: in hemodialysis, start intravenous iron at ferritin ≤ 500 ng/mL and TSAT $\leq 30\%$, and withhold routine administration at ferritin > 700 ng/mL or TSAT $\geq 40\%$ (A.1, A.3, A.4). ESA or HIF-PHI come only after correctable causes have been addressed, in CKD G5D typically at hemoglobin of approximately ≤ 9.0 to 10.0 g/dL, with maintenance below approximately 11.5 g/dL (A.1, A.2, A.4).

The emphasis on phosphate with ferric carboxymaltose deserves attention: the monitoring plan should be recorded in advance, especially with repeated doses, and phosphate should be checked immediately if weakness or bone pain develops after intravenous iron (A.1, A.2, A.4). The second strong element is the documentation section, which requires recording values, interpretation, the rationale for the approach, and the date of follow-up so that a retrospective audit is possible (A.2, A.3, A.4). The limitation is obvious: the ferritin and TSAT thresholds are consensus cutoffs from a guideline commentary, not results of randomized trials, and the tools themselves note that hemoglobin is not the sole decision criterion and that the choice outside hemodialysis is individualized (A.1, A.4).

Algorithms for glomerulopathies

The IgA nephropathy algorithm (A.5) translates KDIGO 2025 into a practical sequence: biopsy at proteinuria ≥ 0.5 g/day; stratification by eGFR, proteinuria, and the Oxford classification; a proteinuria target of below 0.5 g/day at minimum and ideally below 0.3 g/day; foundational kidney protection with sodium below 2 g/day, BP of approximately $\leq 120/70$ mm Hg, RAS blockade, and consideration of

SGLT2 inhibition; and only then specific therapy (A.5). The author explicitly notes that this is a framework approach and, where Nefecon is not available, offers a reduced-dose systemic glucocorticoid regimen (A.5). This is a concession to availability, not evidence of equivalence.

The cheat sheets on complement (A.10) and membranous nephropathy (A.11) are condensed pathophysiology and diagnostics. The C3 and C4 pattern as a first orientation between the alternative and classical pathways (A.10), the proportions of target antigens, with PLA2R accounting for approximately 70% of primary forms, and the importance of the anti-PLA2R titer both for diagnosis without biopsy and for estimating the risk of relapse (A.11) are useful, but they are summaries, not validated decision rules. The reminder about meningococcal vaccination before C5 inhibition (A.10) and thromboprophylaxis when serum albumin is below 25 g/L (A.11) are among the safety points worth having in a cheat sheet.

Cheat sheets for fluid, electrolyte, and acid-base balance

The four cheat sheets on acid-base balance, diuretics, electrolytes, and intravenous fluids (A.6, A.7, A.8, A.9) rests on physiological formulas and reference ranges. Winter's formula, albumin correction of the anion gap, and the delta ratio (A.6), dose equivalence of loop diuretics and sequential nephron blockade in resistance (A.7), the serum sodium correction limit of ≤ 8 to 10 mmol/L per 24 h, and the rule of correcting magnesium first (A.8) are standard tools gathered in one place. The only point at which the appendix explicitly refers to randomized evidence is the comparison of balanced crystalloids with 0.9% NaCl based on the SMART, BaSICS, and PLUS trials (A.9); even here, however, the cheat sheet gives only the direction of the result, not the effect size. The cheat sheets themselves note that clinical context takes precedence over numbers (A.6).

What is proven, what is probable, and what remains open

What is proven is what the tools have taken from guidelines and physiology: the approach to anemia according to KDIGO 2026 (A.1, A.2), the proteinuria target in IgA nephropathy according to KDIGO 2025 (A.5), the rules of compensation and correction (A.6, A.8), and the preference for balanced crystalloids in resuscitation (A.9). The specific thresholds taken from commentaries and practice are probable, for example ferritin of 500 and 700 ng/mL or a monitoring interval of 1 to 3 months in hemodialysis (A.2, A.4). What remains open is whether a structured checklist with documentation actually changes patient outcomes; none of the tools claims this, and the appendix contains no validation data.

Implications for practice

For a Slovak dialysis center, the most usable items are the pair of anemia forms with a documentation section (A.3, A.4), which allow uniform recording of thresholds, the formulation, and the follow-up date and, with ferric carboxymaltose, also the phosphate monitoring plan (A.2). In the outpatient clinic, the practical items are the requirement to consider intravenous iron if oral iron is ineffective or not tolerated within 1 to 3 months (A.4) and the IgA nephropathy algorithm with its documentation sentence (A.5). The cheat sheets on fluid, electrolyte, and acid-base balance (A.6, A.7, A.8, A.9) are intended for the hospital ward and the admissions unit. Local protocols and individual assessment take precedence, as the tools themselves state (A.4, A.5).

What to watch next

It is worth watching whether future versions add the strength of recommendations and effect sizes, particularly for the iron thresholds (A.1) and intravenous fluids (A.9), whether data emerge on adherence and outcomes when checklists are used in practice, and how the IgA nephropathy algorithm (A.5) will be adapted to the availability of treatment in Slovakia.

Articles in this chapter

- A.1** Anemia of CKD 2026 in practice: an algorithm from diagnosis to ESA and HIF-PHI according to the KDOQI US Commentary (June 19, 2026, online, Slovak)
- A.2** Checklist for practice: anemia of CKD according to KDIGO 2026 and the KDOQI US Commentary (June 19, 2026, online, Slovak)
- A.3** Checklist for practice (one A4 page): anemia of CKD (HD and non-HD) (June 19, 2026, online, Slovak)
- A.4** Checklist for practice: anemia of CKD for dialysis and the outpatient clinic (June 19, 2026, online, Slovak)
- A.5** Practical appendix: an algorithm for the management of IgA nephropathy according to KDIGO 2025, taking into account the KDOQI US Commentary (June 19, 2026, online, Slovak)
- A.6** Acid-base: cheat sheet (June 23, 2026, online, Slovak)
- A.7** Diuretics: cheat sheet (June 23, 2026, online, Slovak)
- A.8** Electrolytes: cheat sheet (June 23, 2026, online, Slovak)
- A.9** Intravenous fluids: cheat sheet (June 23, 2026, online, Slovak)
- A.10** Complement and kidney disease: cheat sheet (June 24, 2026, online, Slovak)
- A.11** Membranous nephropathy: cheat sheet (June 24, 2026, online, Slovak)

List of abbreviations

A selection of the most common abbreviations. Abbreviations specific to individual trials are explained in the full text of the article.

ABI	ankle-brachial index
ACC	American College of Cardiology
ACE	angiotensin-converting enzyme
ACEi	angiotensin-converting enzyme inhibitor
ACR	American College of Rheumatology
ADA	American Diabetes Association
ADPKD	autosomal dominant polycystic kidney disease
AHA	American Heart Association
AI	artificial intelligence
AKI	acute kidney injury
ANCA	antineutrophil cytoplasmic antibodies
APOL1	apolipoprotein L1
ARB	angiotensin receptor blocker
ASCVD	atherosclerotic cardiovascular disease
ASN	American Society of Nephrology
AVF	arteriovenous fistula
AVG	arteriovenous graft
BMI	body mass index
BNP	B-type natriuretic peptide
CGM	continuous glucose monitoring
CI	confidence interval
CKD	chronic kidney disease
CKD-MBD	chronic kidney disease-mineral and bone disorder
CKM	cardiovascular-kidney-metabolic (syndrome)
CRP	C-reactive protein
CT	computed tomography
CVC	central venous catheter
CVOT	cardiovascular outcome trial
DKA	diabetic ketoacidosis
DKD	diabetic kidney disease
DOI	digital object identifier
EASD	European Association for the Study of Diabetes

ECG	electrocardiogram
eGFR	estimated glomerular filtration rate
EMA	European Medicines Agency
ERA	European Renal Association
ERBP	European Renal Best Practice
ESA	erythropoiesis-stimulating agent
ESC	European Society of Cardiology
ESKD	end-stage kidney disease
FDA	US Food and Drug Administration
FSGS	focal segmental glomerulosclerosis
GDMT	guideline-directed medical therapy
GFR	glomerular filtration rate
GIP	glucose-dependent insulintropic polypeptide
GLP-1	glucagon-like peptide 1
GLP-1 RA	GLP-1 receptor agonist
HbA1c	glycated hemoglobin
HD	hemodialysis
HDF	hemodiafiltration
HF	heart failure
HFpEF	heart failure with preserved ejection fraction
HIF-PHI	hypoxia-inducible factor prolyl hydroxylase inhibitor
HLA	human leukocyte antigen
HR	hazard ratio
IgAN	IgA nephropathy
KDIGO	Kidney Disease: Improving Global Outcomes
KDOQI	Kidney Disease Outcomes Quality Initiative
KRT	kidney replacement therapy
LDL	low-density lipoprotein
LN	lupus nephritis
MACE	major adverse cardiovascular events
MASH	metabolic dysfunction-associated steatohepatitis
MASLD	metabolic dysfunction-associated steatotic liver disease
MCO	medium cut-off (dialysis membrane)
MGRS	monoclonal gammopathy of renal significance
MRA	mineralocorticoid receptor antagonist
NNT	number needed to treat
NSAID	nonsteroidal anti-inflammatory drug
nsMRA	nonsteroidal mineralocorticoid receptor antagonist
OL-HDF	online hemodiafiltration

OR	odds ratio
PD	peritoneal dialysis
PMID	PubMed identifier
POCUS	point-of-care ultrasound
PPI	proton pump inhibitor
PTH	parathyroid hormone
RAAS	renin-angiotensin-aldosterone system
RAASi	renin-angiotensin-aldosterone system inhibitor(s)
RCT	randomized controlled trial
RR	relative risk
SGLT2	sodium-glucose cotransporter 2
SmPC	summary of product characteristics
TMA	thrombotic microangiopathy
TMAO	trimethylamine N-oxide
TSAT	transferrin saturation
TTP	thrombotic thrombocytopenic purpura
UACR	urine albumin-to-creatinine ratio
UPCR	urine protein-to-creatinine ratio
UTI	urinary tract infection
WHO	World Health Organization
ŠÚKL	State Institute for Drug Control (Slovakia)

About the author

Lubomír Polaščin, MD, is a physician specializing in nephrology and internal medicine. He has worked in dialysis and nephrology since 1995 and from 2013 to 2022 headed two dialysis centers in Bratislava, Slovakia.

Alongside clinical work, he writes specialist and literary prose, translates medical texts between Slovak and English, lectures, consults, and develops digital medical tools. He is the author of the nefro.polascin.net portal, which besides specialist articles offers clinical calculators, diagnostic tools, and cheat sheets for nephrology practice.

More information: portfolio.polascin.net · books.polascin.net

Where nephrology and internal medicine stand as of October 2026, according to 387 specialist articles from the nefro.polascin.net portal.

Extended editorials for each of the 16 chapters and for the appendix: the main trends in the evidence, agreements and contradictions, the line between what is proven, what is probable, and what remains open, implications for practice, and what to watch next.

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